

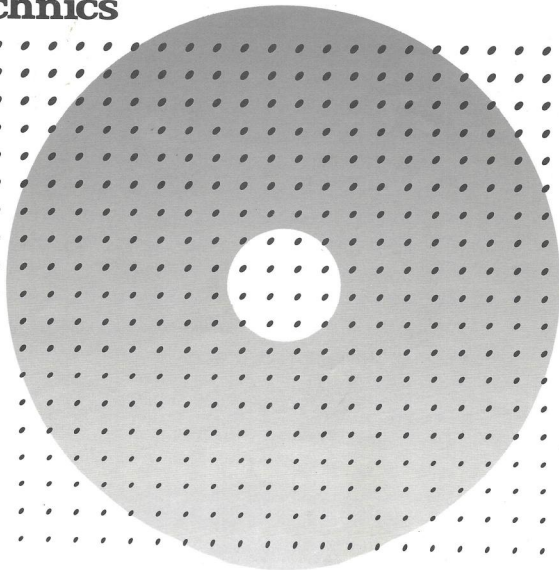
Technical Guide

COMPACT
disc
DIGITAL AUDIO

Vol. 25

Auto CD Technology

**Panasonic
Technics**



AUTO PRODUCTS DIVISION

Matsushita Communication Industrial Co., Ltd.

APPLICABLE MODELS

TYPE	MODELS	
	NORTH AMERICA	EUROPE AND OTHERS
A	CQ-E850EU	CQ-DP25EN
	CQ-DP25EC	CQ-DP26EG
B	CX-DP11EU/EC	CX-DP11EN
	CX-DP10EU/EC	CX-DP10EN
C	CQ-DP5EU/EC	CQ-DP5EG
	CQ-E800EU	CX-DP3EN

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1. Traverse Mechanism And Optical Pick Up

1-1 Traverse Mechanism

The traverse mechanism (Fig. 1-1) drives the optical pickup base by the rack gear which is driven by the motor via the worm gear, gear A, and gear B. The gear A and gear B are double clutch gears. The inner **RESET** switch is turned on by the optical pickup base when the optical pickup is driven inner-most position by the traverse mechanism.

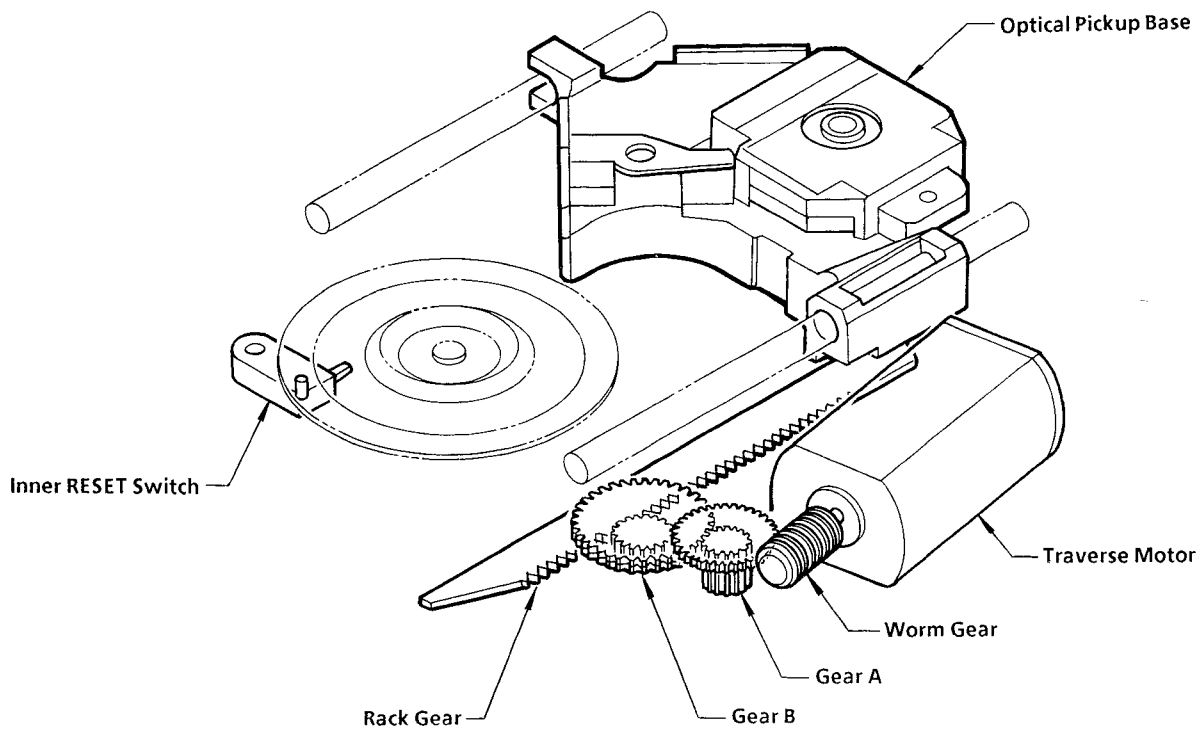


Fig. 1-1 Traverse Mechanism

1-2 3-Beam Type vs. Single Beam Type

- 1-2-1 The largest difference in optical structure between the 3-beam type and single beam type is that single beam type has no diffraction grating. Therefore, the single beam type has only one beam on optical path, and its photo-detector is divided into four instead of six (four plus two).
- 1-2-2 The two sub-beams of the 3-beam type have the function of generating tracking errors signal. The process of generating tracking error signal by the single beam type that is without sub-beams is described in Section 1-4. Focus error signal generated by the conventional astigma method.
- 1-2-3 3-Beam type has a small RF detector area ($150\text{ }\mu\text{m}$ squared). Therefore, it has a disadvantage when playing discs with fingerprints.
- 1-2-4 1-Beam type (4-segments) overcomes the fingerprint playability but it has a disadvantage when playing scratched discs. The area of the 4-segment photodetector is large ($1000\text{ }\mu\text{m}$ squared). Therefore, the error generated by fingerprints is included and corrected. Since the focus and tracking detectors are included as part of the RF detector, the spike is detected as an error signal.
- 1-2-5 1-Beam type (6-segments) has a reduced size RF detector ($250\text{ }\mu\text{m}$ squared) and two large and separate detectors have been added. Any spikes generated by a scratched disc do not appear as a large spikes to the tracking and focus detectors. The total area of the photodetector is now $1400\text{ }\mu\text{m}$ squared.

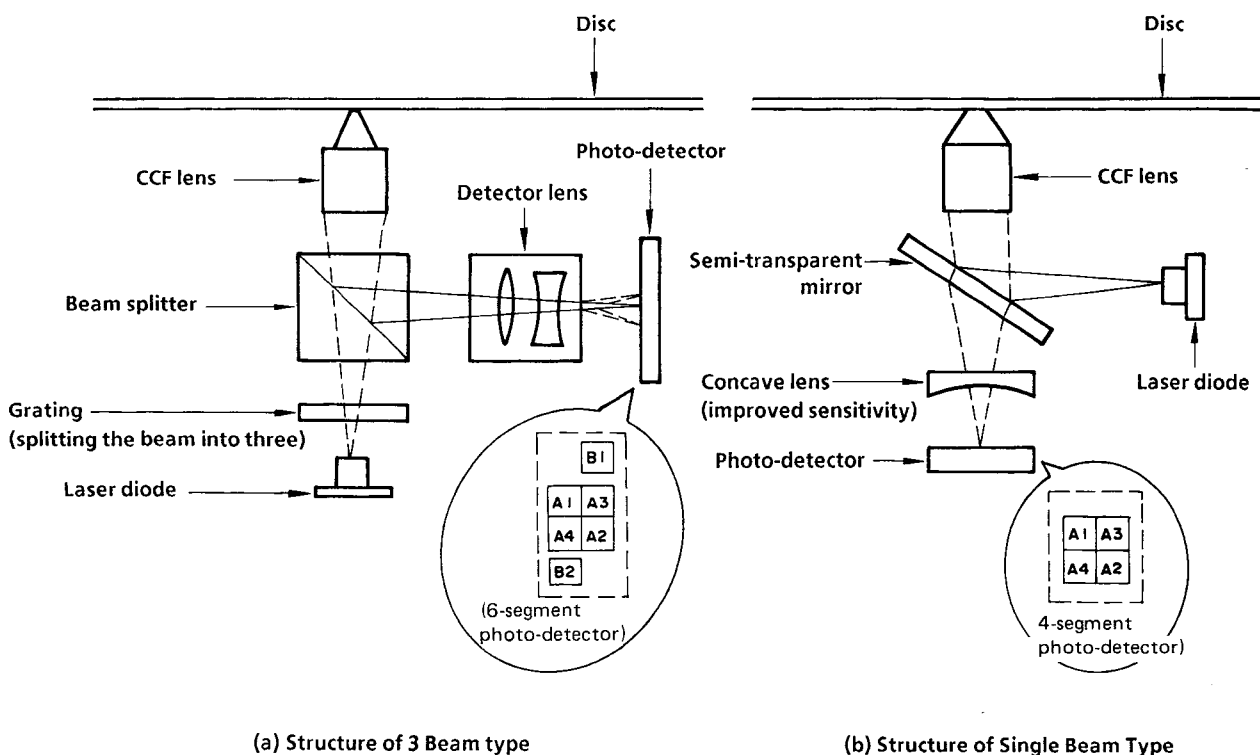


Fig. 1-2 Structure of Optical Pickup

1-3 Photo Detector

The photo Detector is probably the single most important component in the compact disc player. The reason for this is that all intelligence needed to operate the unit is developed from the returning laser beam by the photo detector. This will include focus error signal, tracking error signal as well as the intelligence signal.

The single beam type (Fig. 1-3b) has a large PD surface area to prevent RF signal level reduction. However, if the beam irradiated at a scratch or other defects on the disc surface, the servo errors occur and decrease playability.

In order to compensate such divergent conditions, the photo-detector structure shown in Figure 1-3c is used, the four parts in the center are used for error detection, and the total sum including the periphery (B1 + B2) is the RF signal.

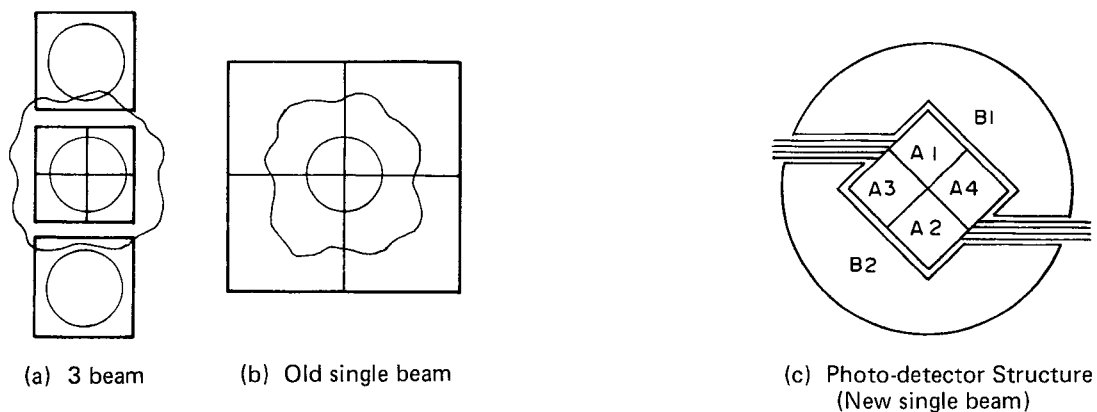


Fig. 1-3

1-4 Tracking Error Signal Generation

Fig. 1-4 shows the basic configuration of the single beam type (phase difference detection method).

The output of the optical pickup is the sum of each diagonal element of the 4-segment photo-detector.

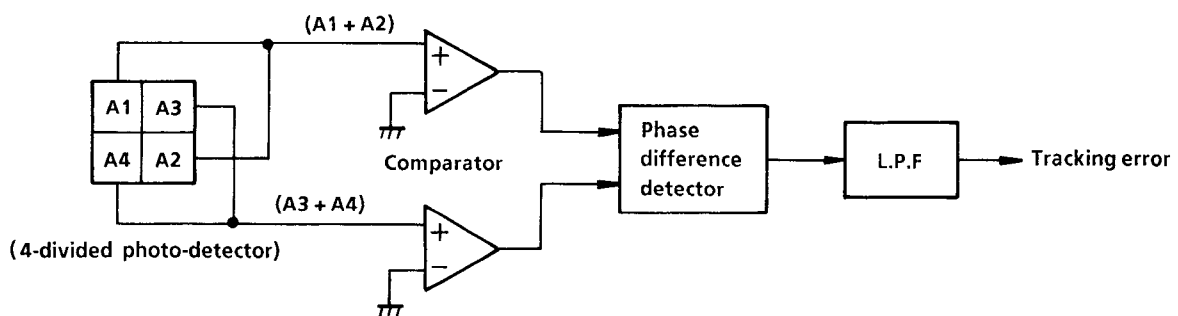
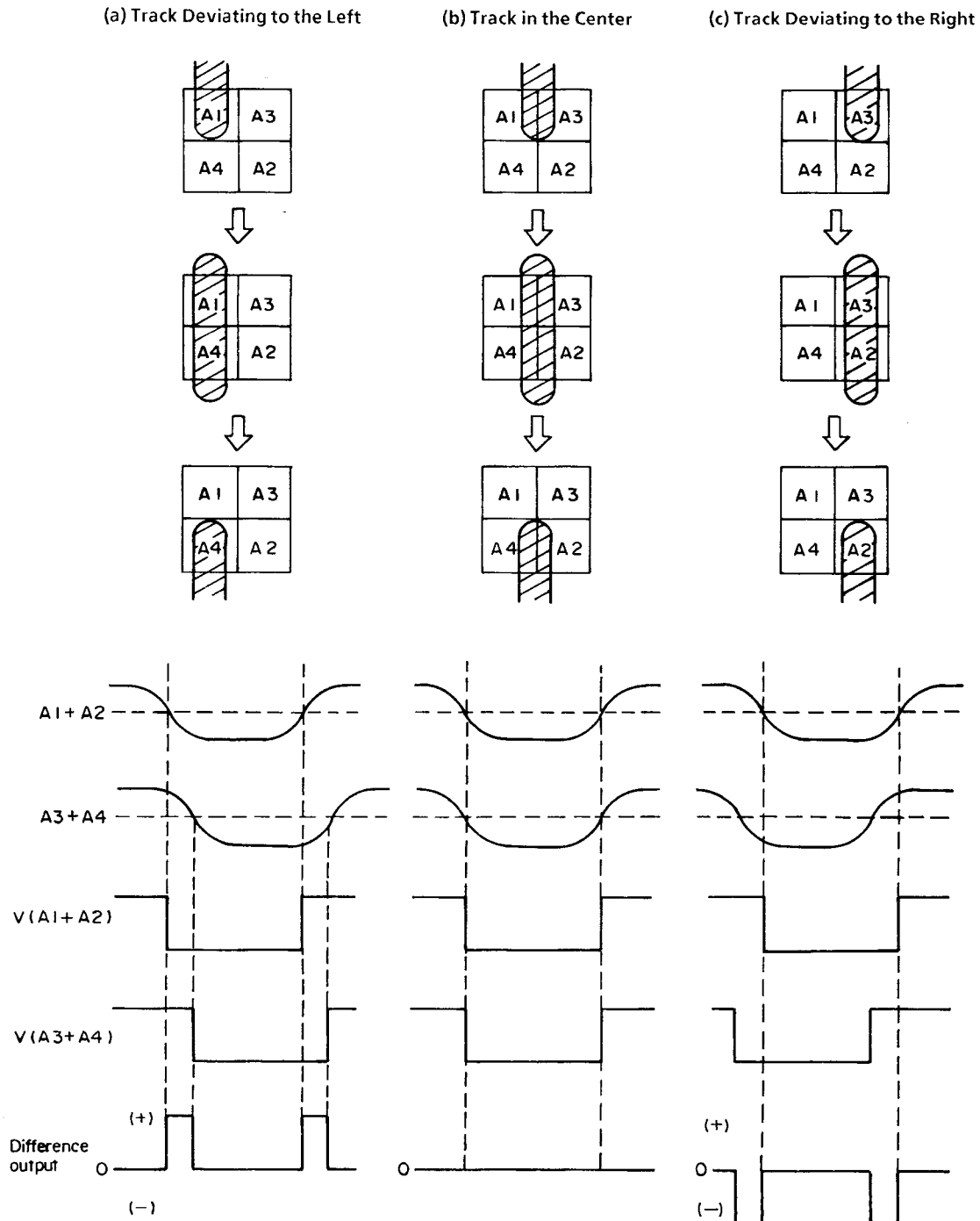


Fig. 1-4 Basic Configuration

If a track pit image runs over the 4-segment photo-detector and deviates to the left or the right, a phase difference corresponding to the track deviation is produced between the output signals ($A1 + A2$) and ($A3 + A4$). This phase difference is detected by the phase detector, and passed through the lowpass filter (L.P.F.) to generate a tracking error signal.

Fig. 1-5 shows a tracking error detection diagram.

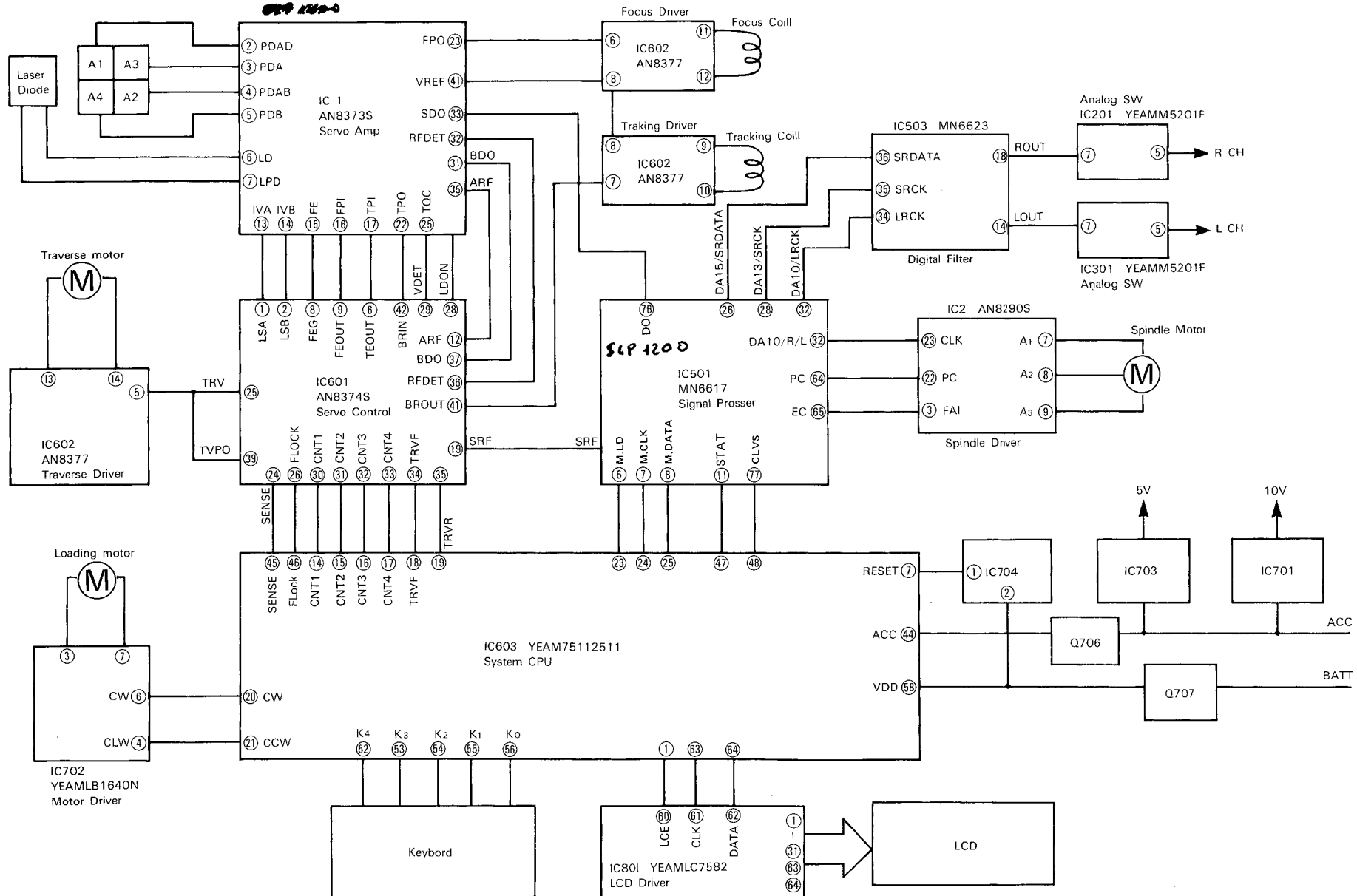


- A positive pulse is generated if $V(A1 + A2)$ leads $V(A3 + A4)$, and a negative pulse if vice versa.

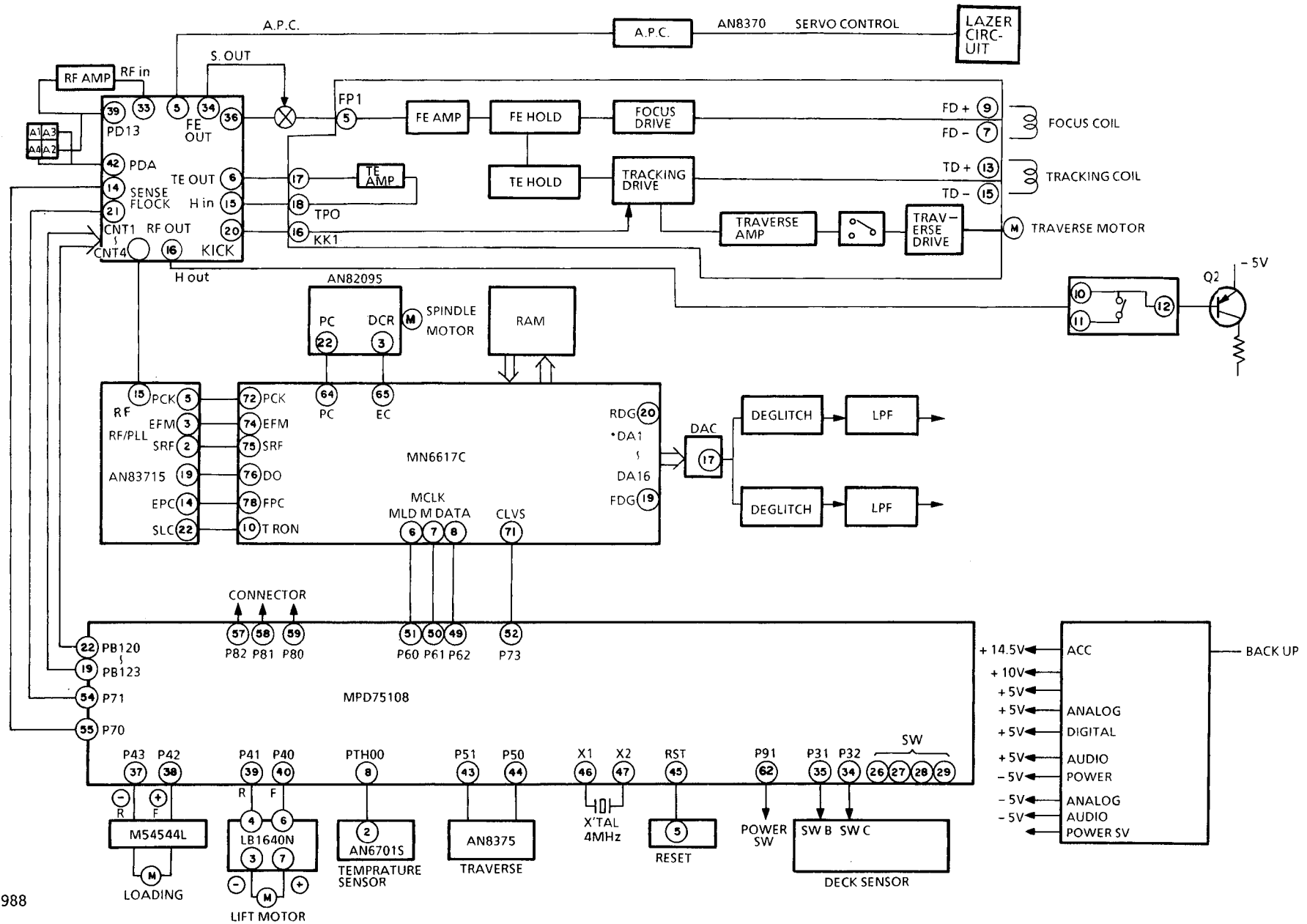
Fig.1-5 Tracking Error Detection Diagram

2. SYSTEM BLOCK DIAGRAM

2-1 SYSTEM BLOCK DIAGRAM (Type A)



2-2 BLOCK DIAGRAM (Type B)



2-3 BLOCK DIAGRAM (Type C)

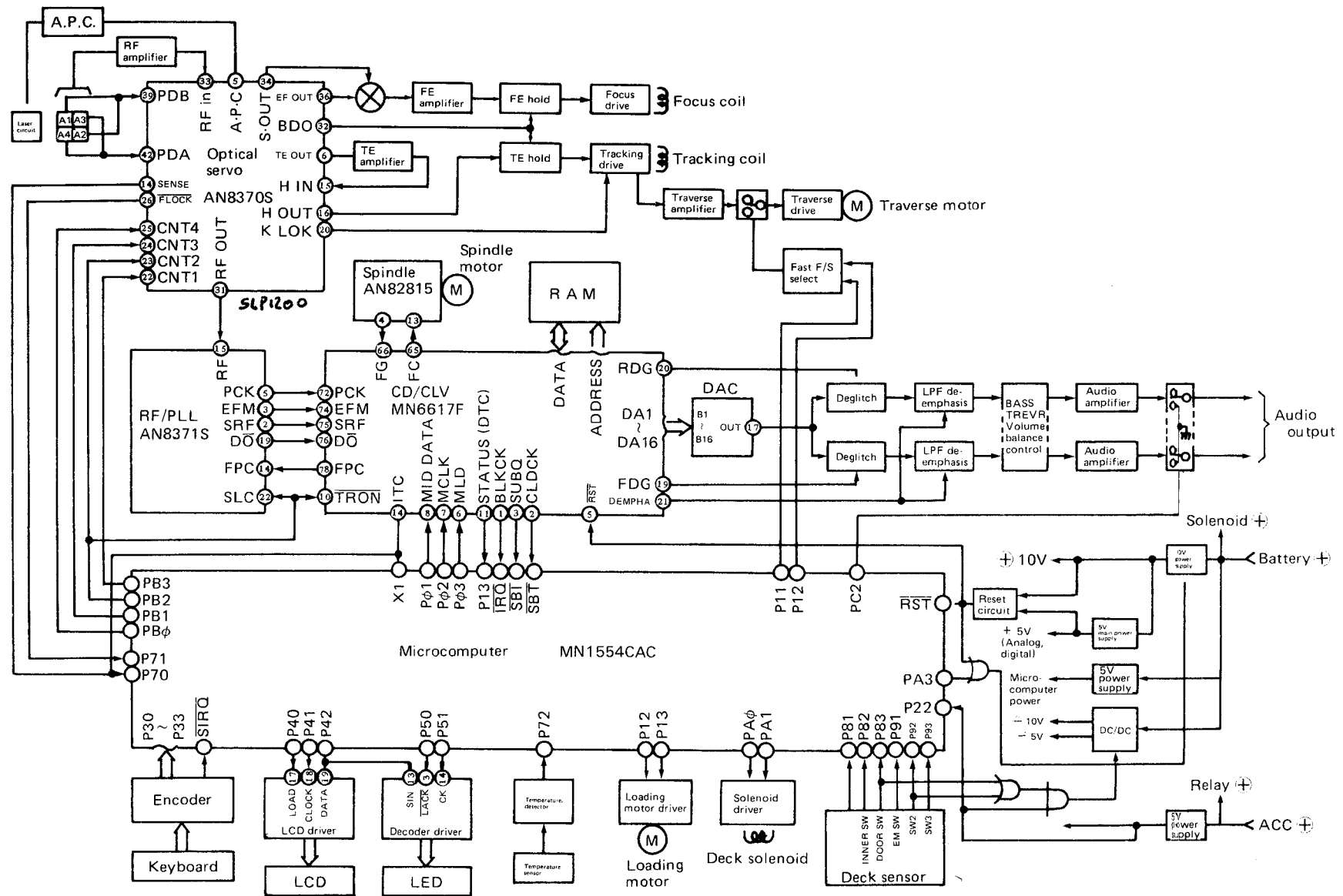
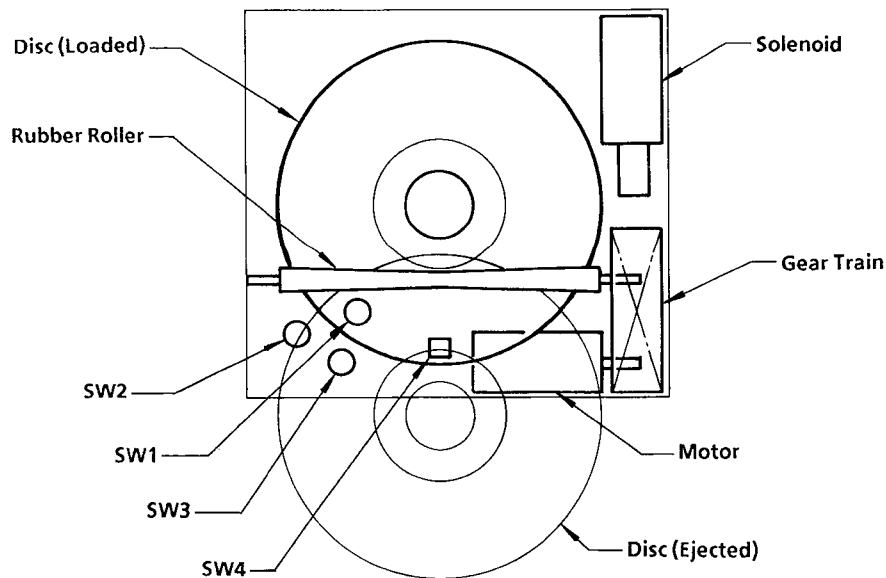


Fig. 2-1 Block Diagram

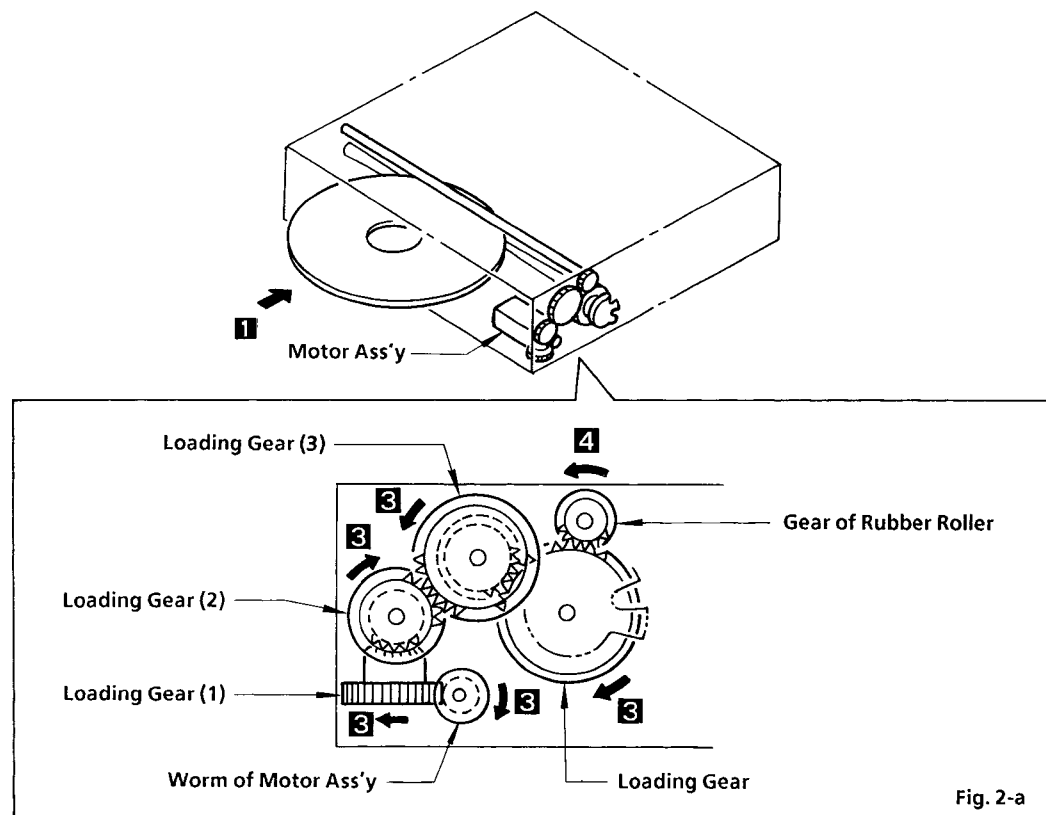
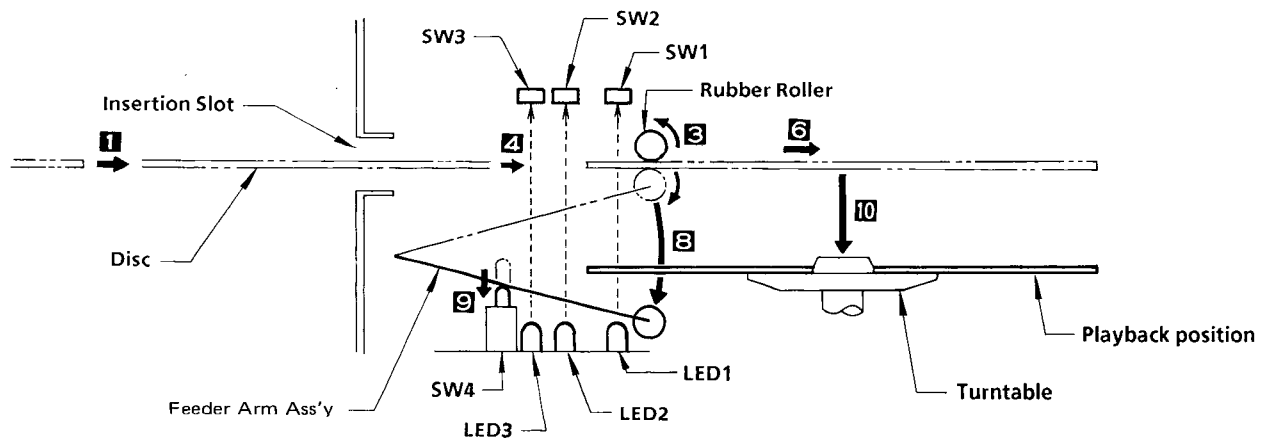
3. Loading Mechanism And It's Operation (TYPE A)

A3-1 Disk Loading Mechanism



Name	Function	Description	Remarks
SW1	Disk in detection	To detect whether or not a disk is inserted.	Photosensor
SW2	Disk position detection	Detects the timing for Switching from horizontal movement to vertical movement. Reference point for determining how far to eject the disk.	Photosensor
SW3	Disk loaded or unloaded detection	To detect whether the disk has been properly loaded. To detect whether the disk has been properly unloaded.	Photosensor
SW4	Mechanism status detection	To detect whether the disk loading mechanism is in play mode.	Contact Switch (Micro Switch)
MOTOR	Disk loading motor	Motor for loading and unloading the disk.	Direction of motor Loading: CCW Unloading: CW
SOL	Clutch Switing	Switches disk movement from vertical to horizontal.	When solenoid on Horizontal movement
Rubber Roller	To move Disk in and out	Converts motor rotary motion into liner motion.	
Gear train			

A3-2 Loading Operation



Loading

* SW1, SW2 and SW3 are ON when No Disc.

- 1 Insert the disc into the player.
- 2 When SW3 turns OFF, the solenoid is activated the lock lever assembly slides, and its hook secures the cam gear.
- 3 SW1 turns OFF. The motor assembly and the loading gears rotate.
- 4 The rubber roller rotates.
- 5 The disc will graspe by the two rubber rollers and load into the player.

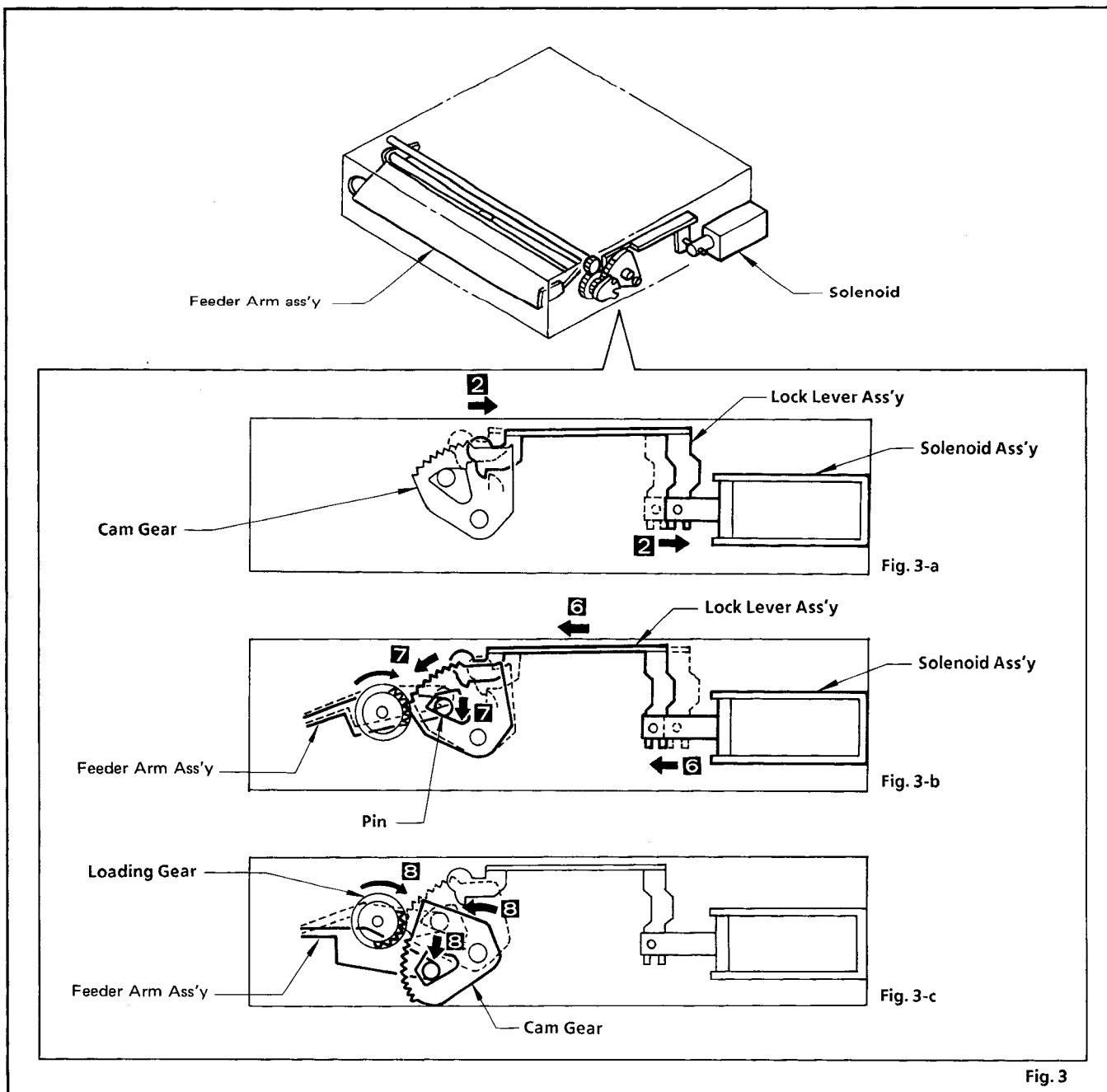


Fig. 3

- 6 When the disc moves beyond SW2 (SW2: ON), the solenoid is turned off, and the hook is released.
- 7 The weight of the disc lowers the feeder arm assembly slightly. Its lowering force rotates the cam gear slightly counterclockwise, then the cam gear and loading gear are engaged.
- 8 The loading gear rotates clockwise and the cam gear rotates counterclockwise.
- 9 The rotation of the cam gear lowers the feeder arm assembly and turns on SW4.
- 10 The disc also lowers and is set on the turntable of optical pickup.

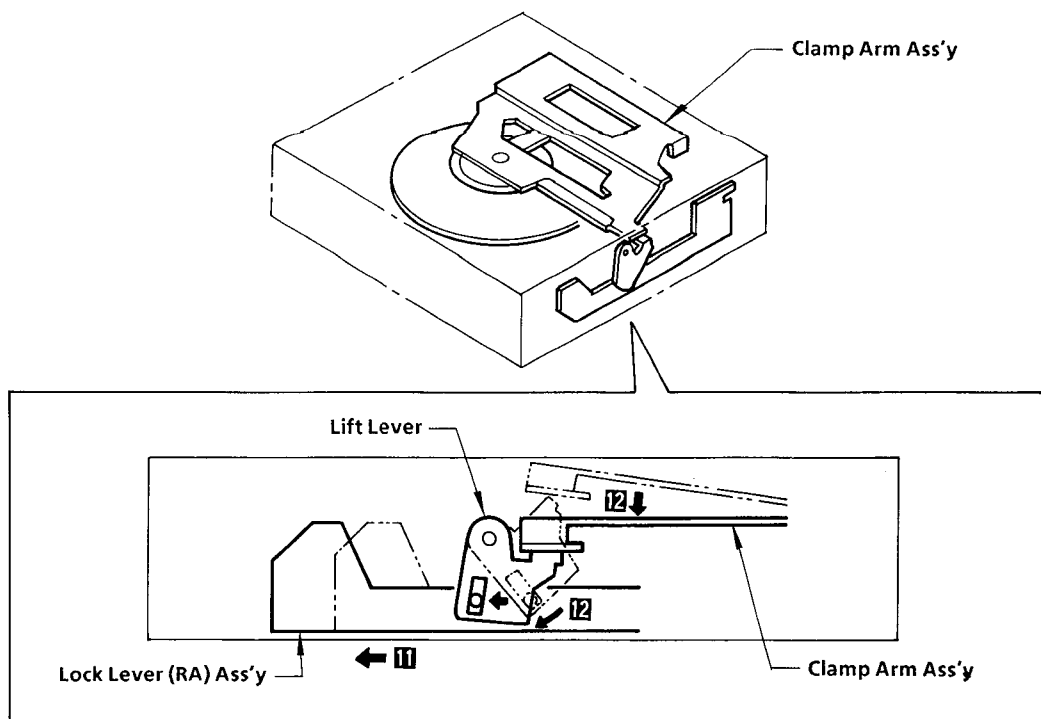


Fig. 4

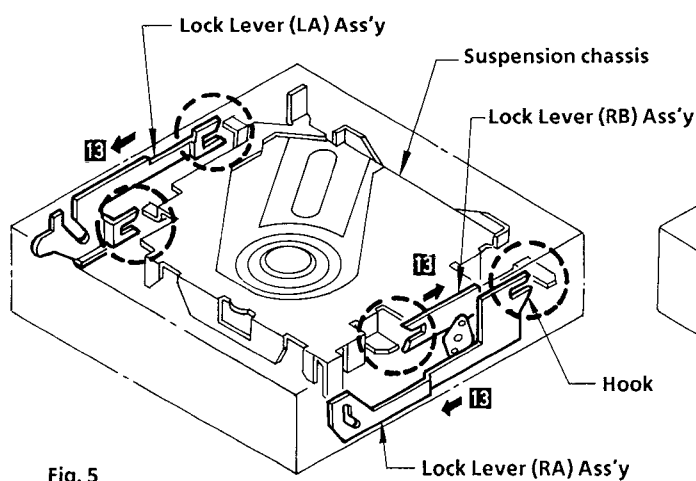


Fig. 5

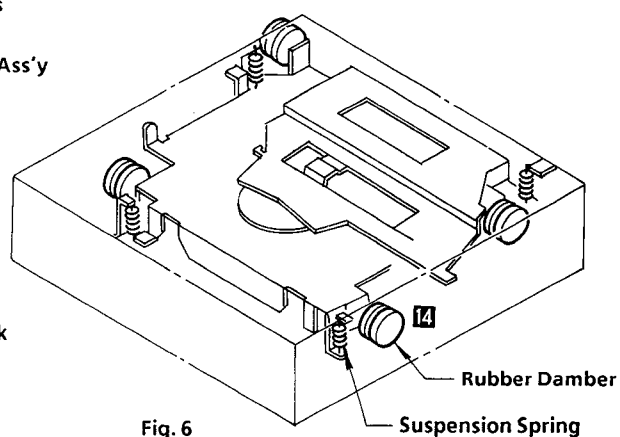


Fig. 6

- 11 When the feeder arm assembly lowers, the lock lever (RA) assembly slides forward.
- 12 The lift lever rotates clockwise and lowers the clamp arm assembly.
The disc is secured in the position for playback by the force of the magnet.
- 13 When the lock lever (RA) assembly slides forward, the lock lever (RB) assembly and the lock lever (LA) assembly also slide, and the secured suspension chassis are released.
- 14 The suspension chassis is then supported by the suspension springs and rubber dampers.

A3-3 Unloading/Reloading Operation

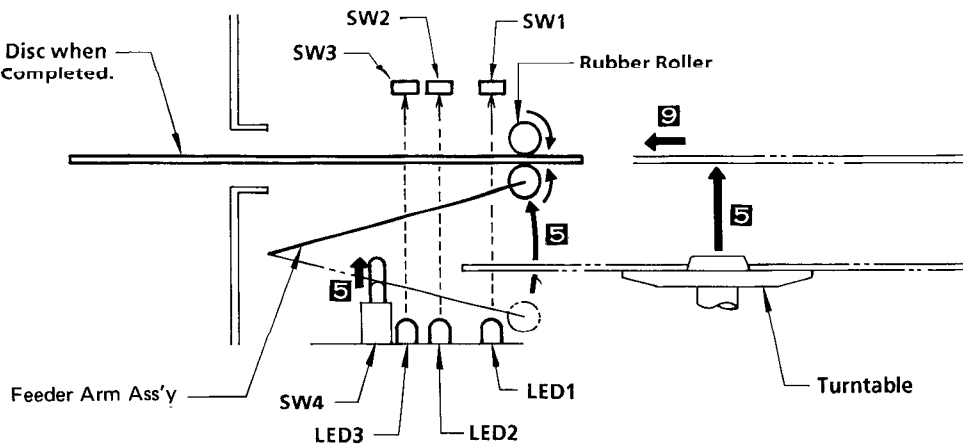


Fig. 7

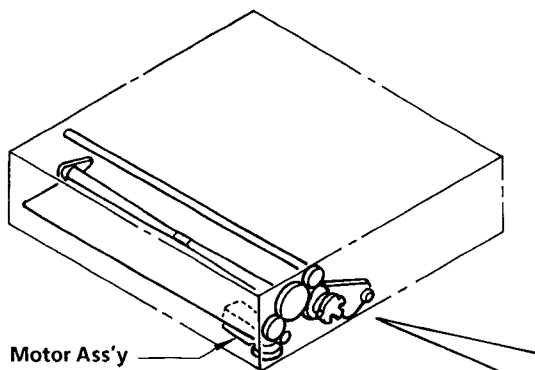


Fig. 8

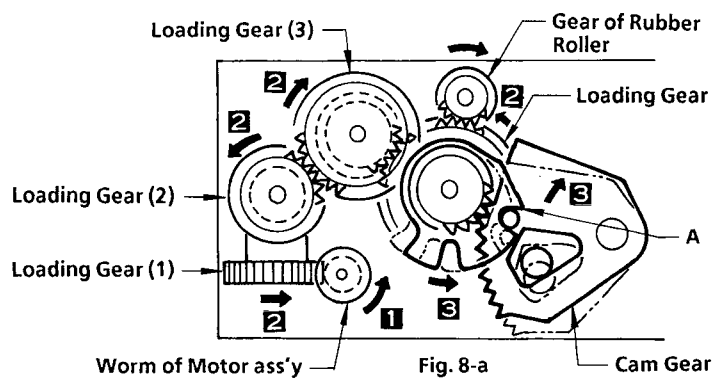


Fig. 8-a

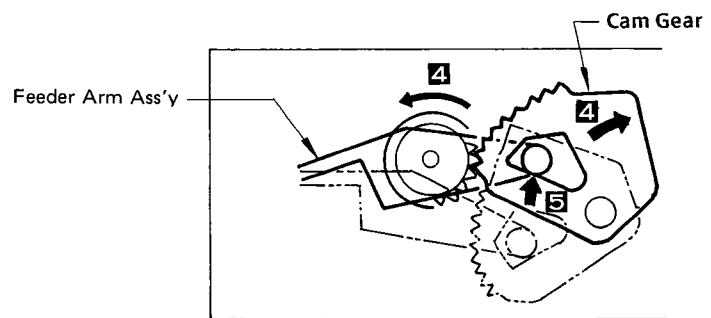


Fig. 8-b

Unloading

- 1 When the eject signal is received, the motor assembly rotates in the direction opposite that of loading.
- 2 The loading gears rotate.
- 3 Part A on the loading gear rotates the cam gear slightly clockwise, then the cam gear and loading gear are engaged.
- 4 The loading gear rotates the cam gear clockwise.
- 5 The cam gear lifts the feeder arm assembly. (SW4 : Off)

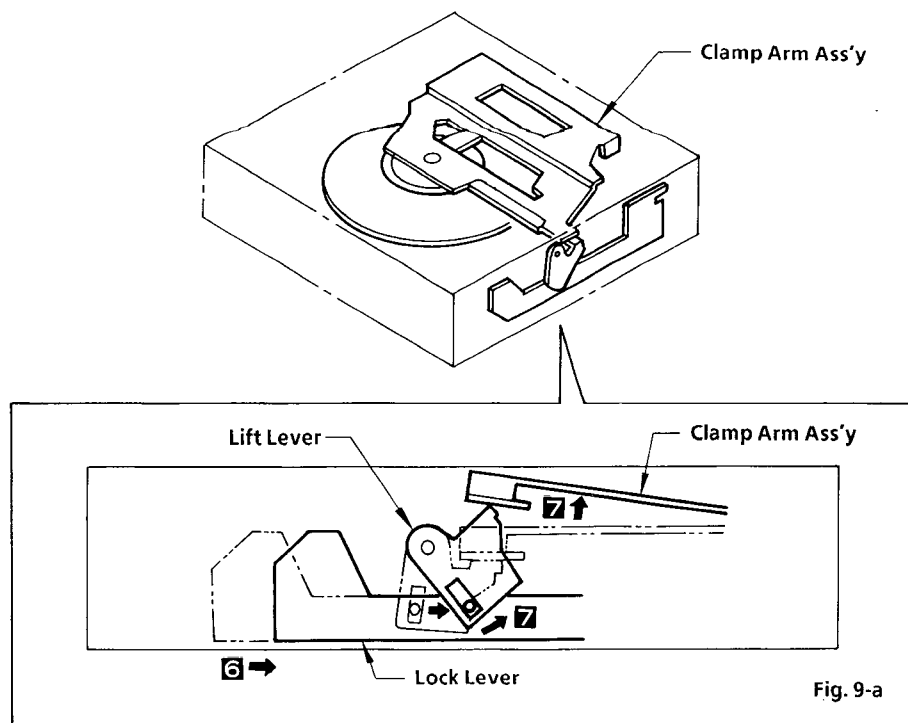


Fig. 9

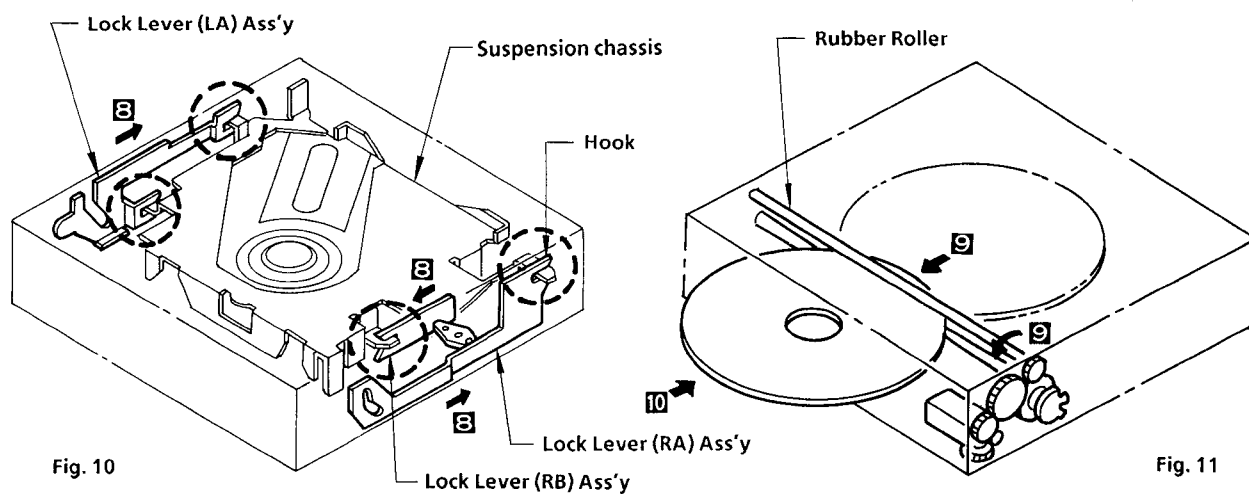


Fig. 10

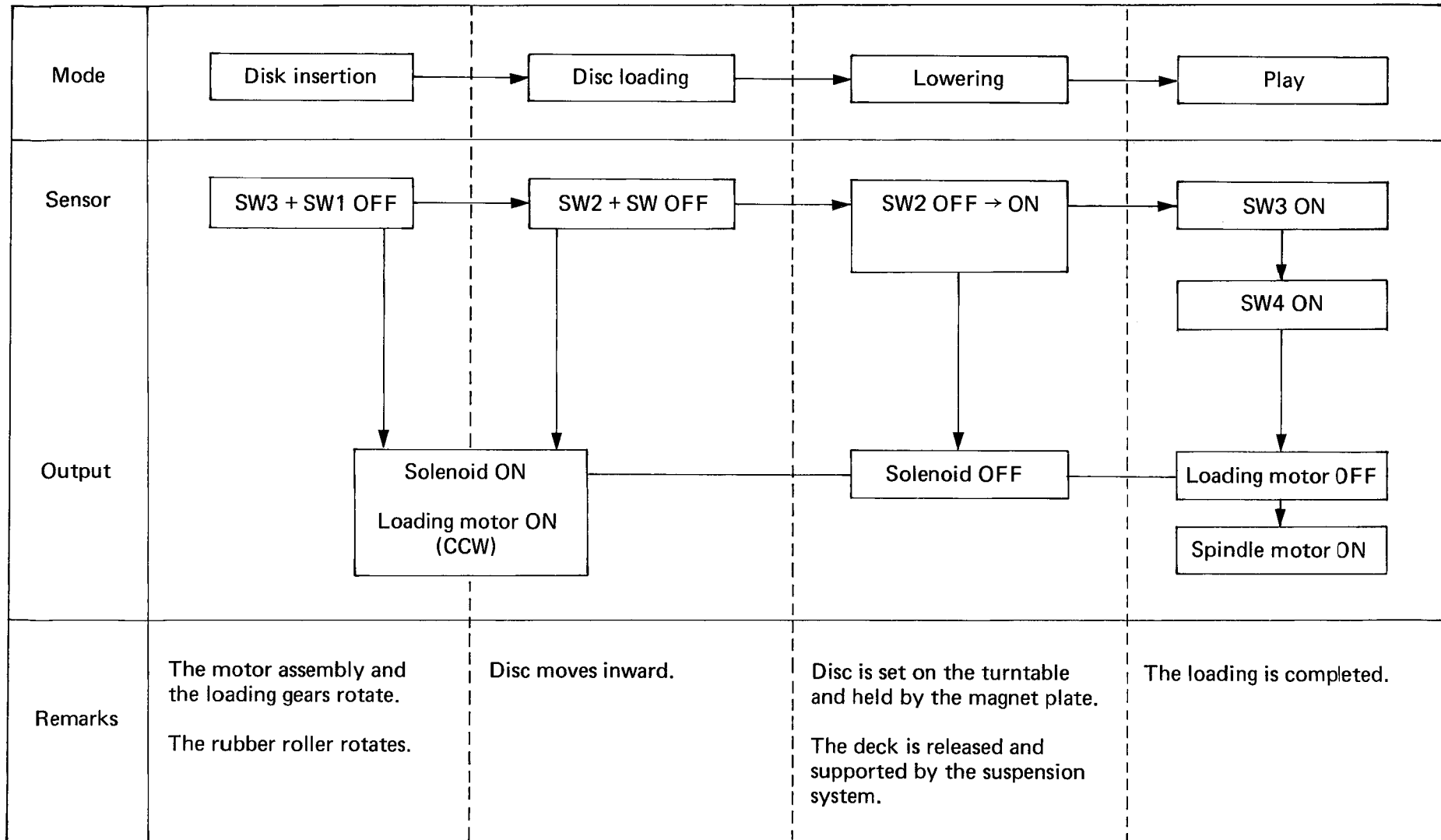
Fig. 11

- 6 When the feeder arm assembly lifts, the lock lever (RA) assembly slides backward.
- 7 The lift lever rotates counterclockwise and lifts the Clamp Arm assembly.
- 8 When the lock lever (RA) assembly slides, the lock lever (RB) assembly and the lock lever (LA) assembly also slide, and secures the suspension chassis with four hooks.
- 9 With these movements, the disc is grasped by the rubber roller and unloaded from the player. When the disc moves beyond SW2, SW2 turns on, the motor assembly stops rotating, and the ejection is completed.

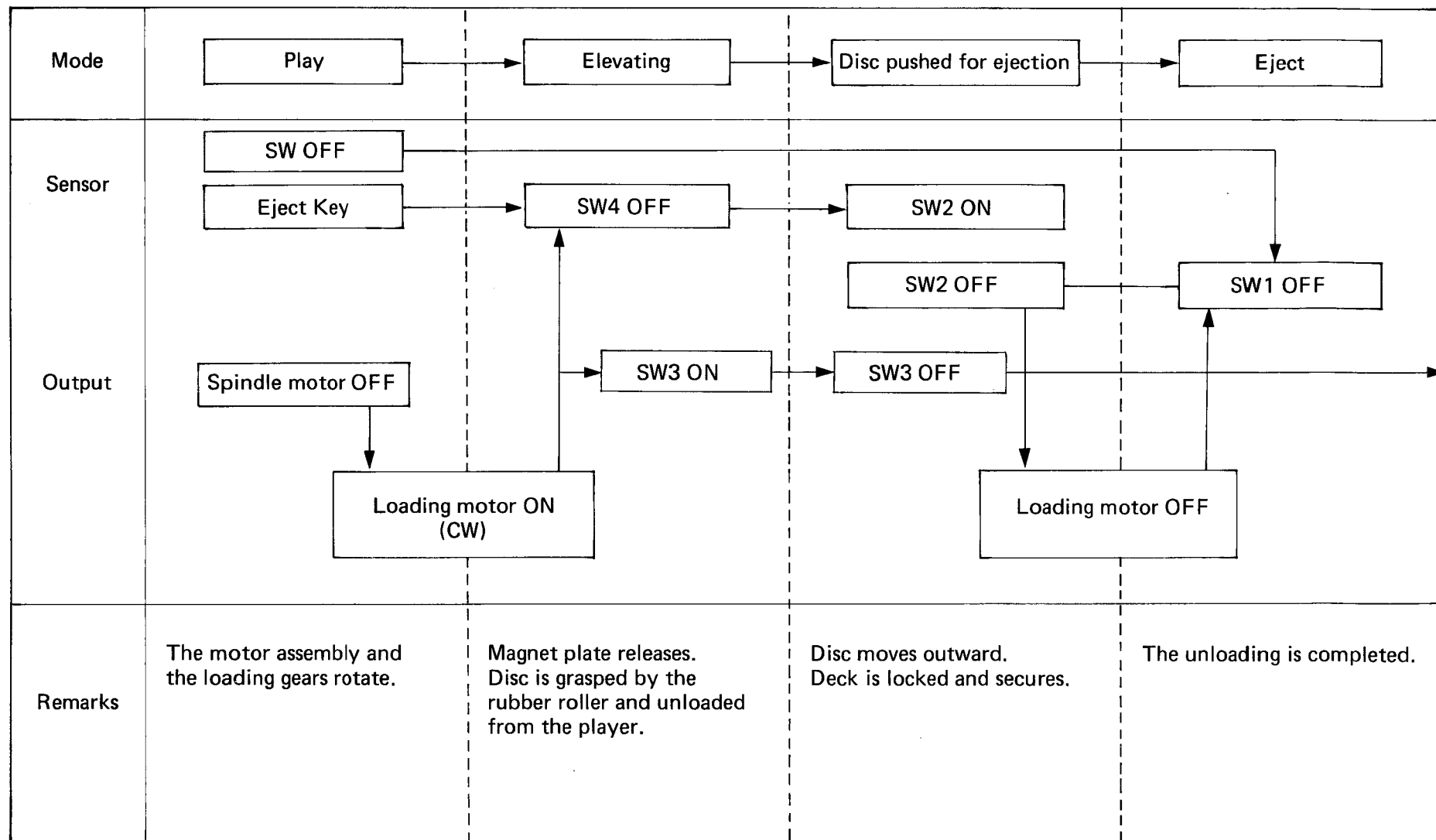
Reloading

- 10 If the disc is pushed inward from the position at which unloading has finished, the operation of loading begins.

3-4 Loading Operation



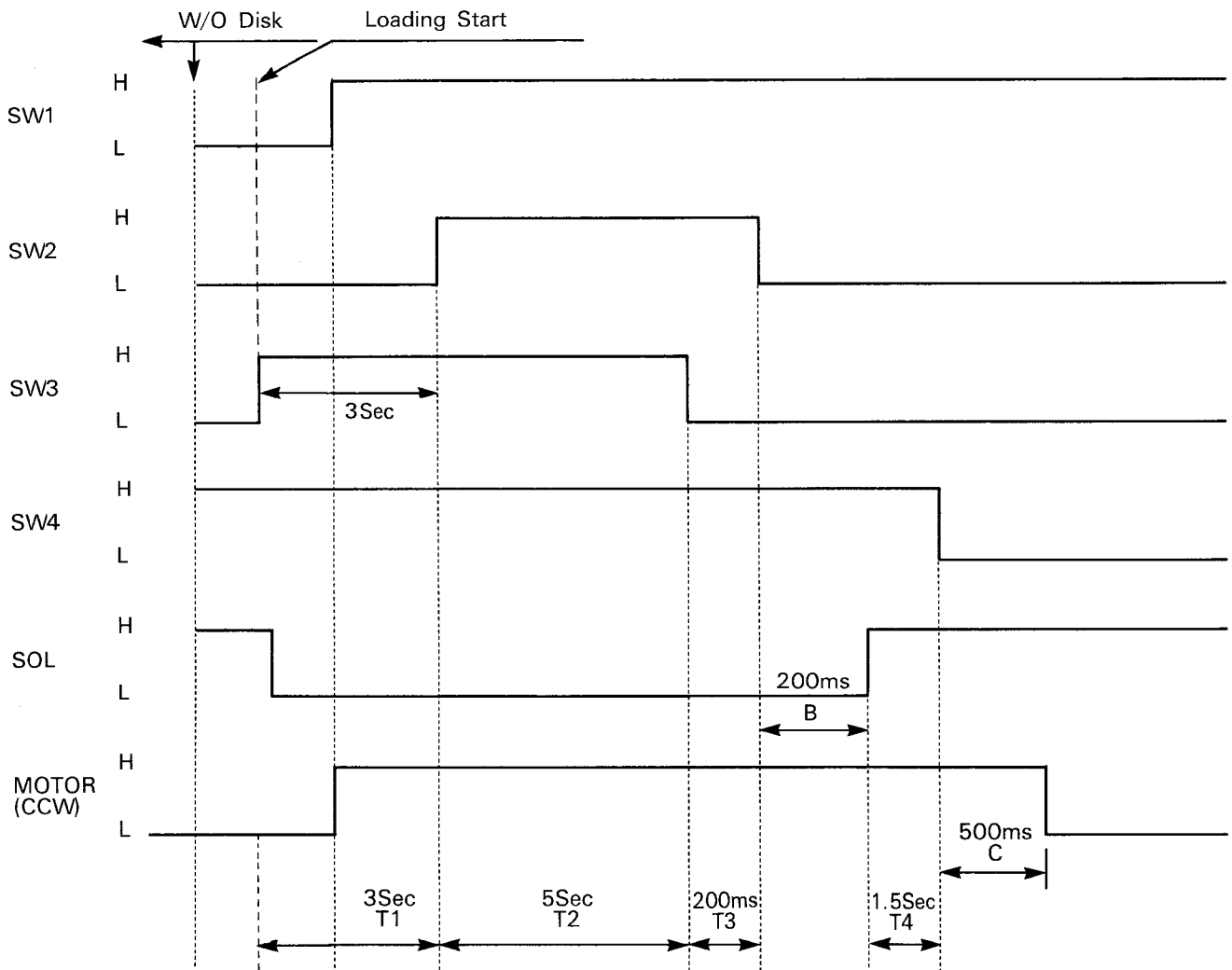
3-5 Eject Operation



A3-6 Timing Chart

LOADING

Active H



B = 200ms

C = 500ms

T1 = 3sec

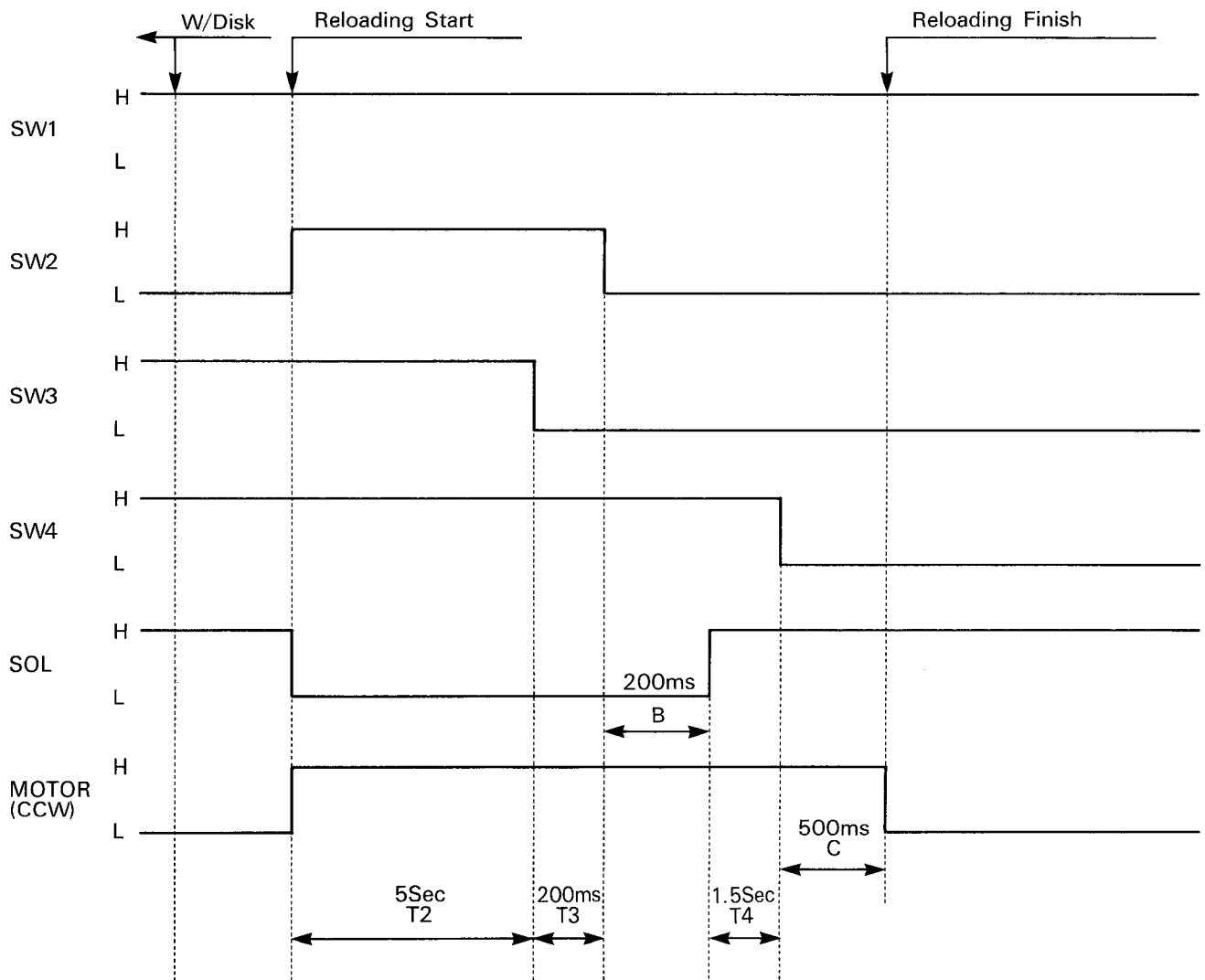
T2 = 5sec

T3 = 200ms

T4 = 1.5sec

RELOADING

Active H



A = 200ms

B = 200ms

C = 500ms

T2 = 5Sec

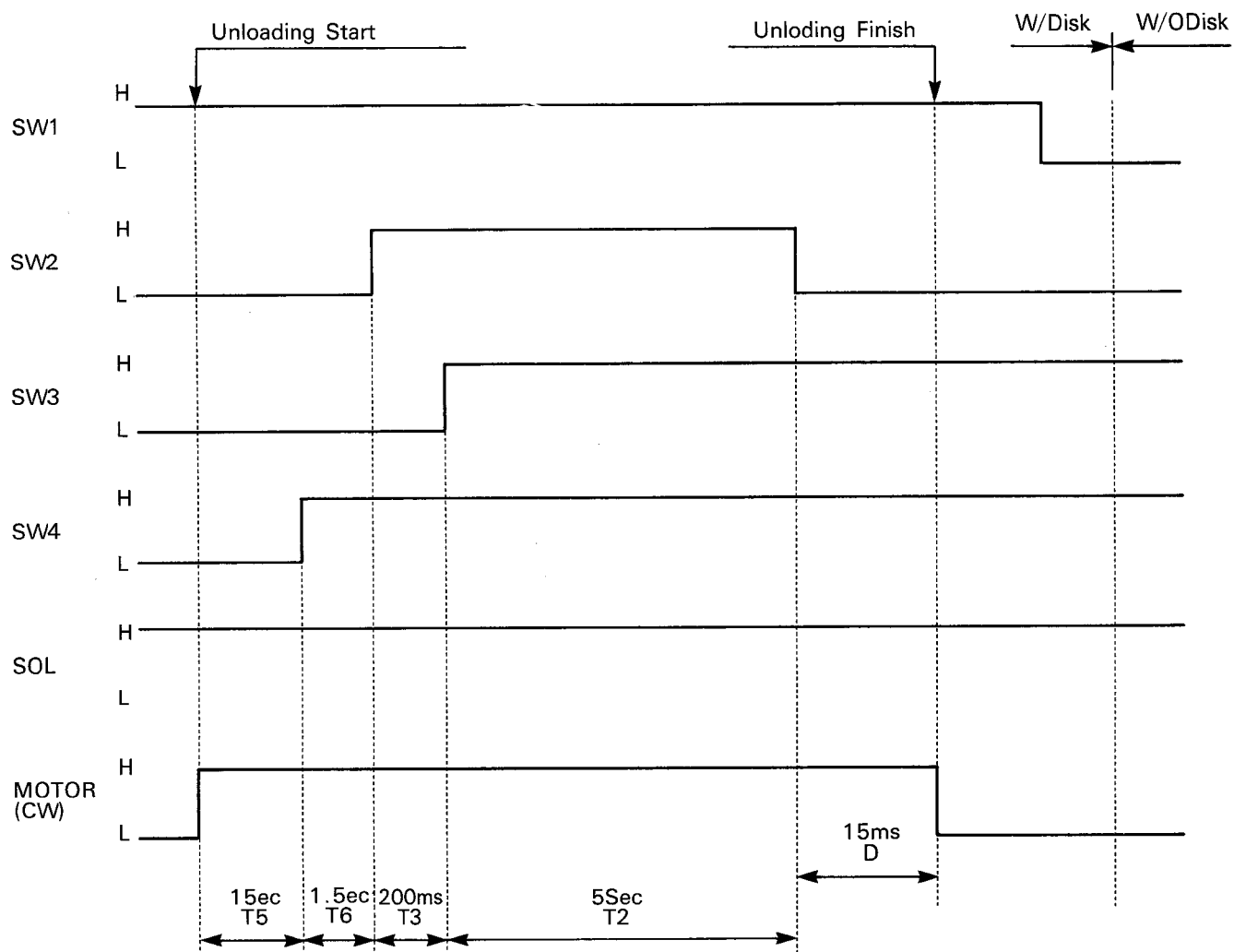
T3 = 200ms

T4 = 1.5Sec

UNLOADING

W/O DUST COVER

Active H

 $D = 15\text{ms}$ $T3 = 200\text{ms}$ $T5 = 1\text{Sec}$ $T6 = 1.5\text{Sec}$

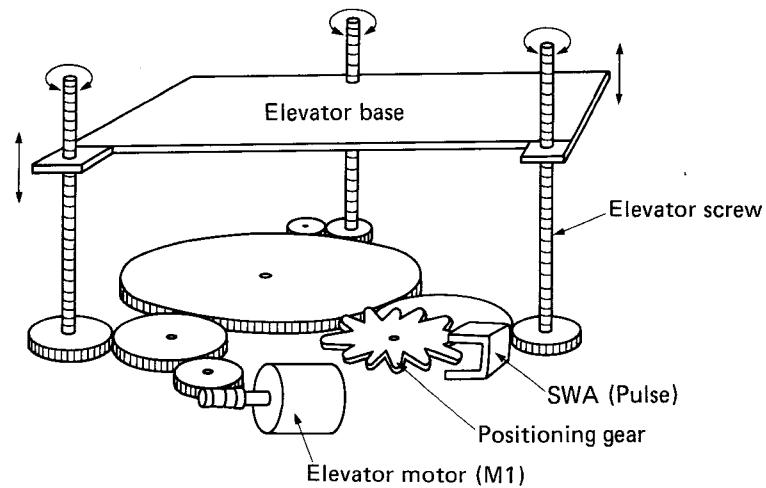
CD CHANGER LOADING MECHANISM (TYPE B)

B3-1 Elevator Motor and Disk Loading Positioning

When the magazine placed on the elevator base with cover closed, the elevator motor (M1) runs, and the elevator base moves downward until it hits SW2 and stop.

The elevating process is disk positioning.

- 1 The stop position at the SW2 is the initial position of Disc Loading.
- 2 For moving the magazine to the correct loading position, it is detected by the pulse of 70 interval.



6 Fig. Elevator Mechanism

CD CHANGER LOADING MECHANISM (TYPE B)

B3-2 Operation of Loading Motor and Solenoid

(1) Disk loading/unloading

Disk loading to and unloading from the optical pick deck can be done by the loading motor (M2) and roller assembly.

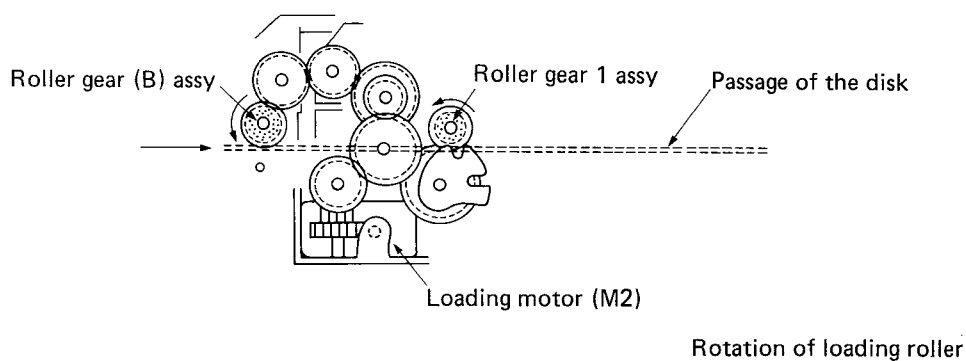


Fig. Right Side View of the Deck

(2) Unloading the disk from the magazine

When the loading motor rotates to the loading direction, the solenoid turns on, the gear lock lever goes down and engages with the cam gear. Then the cam gear pushes the disk ejecting lever once. As a result, the magazine lever is pushed, and kicks the disk out to the loading mechanism.

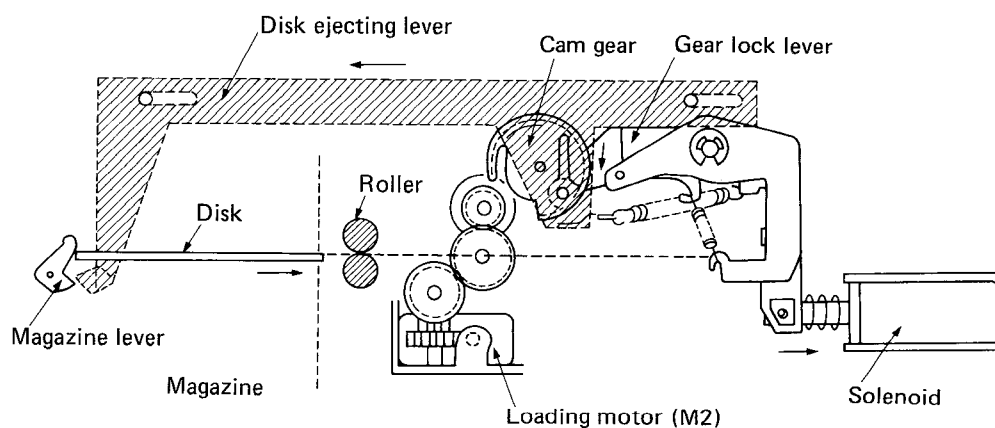
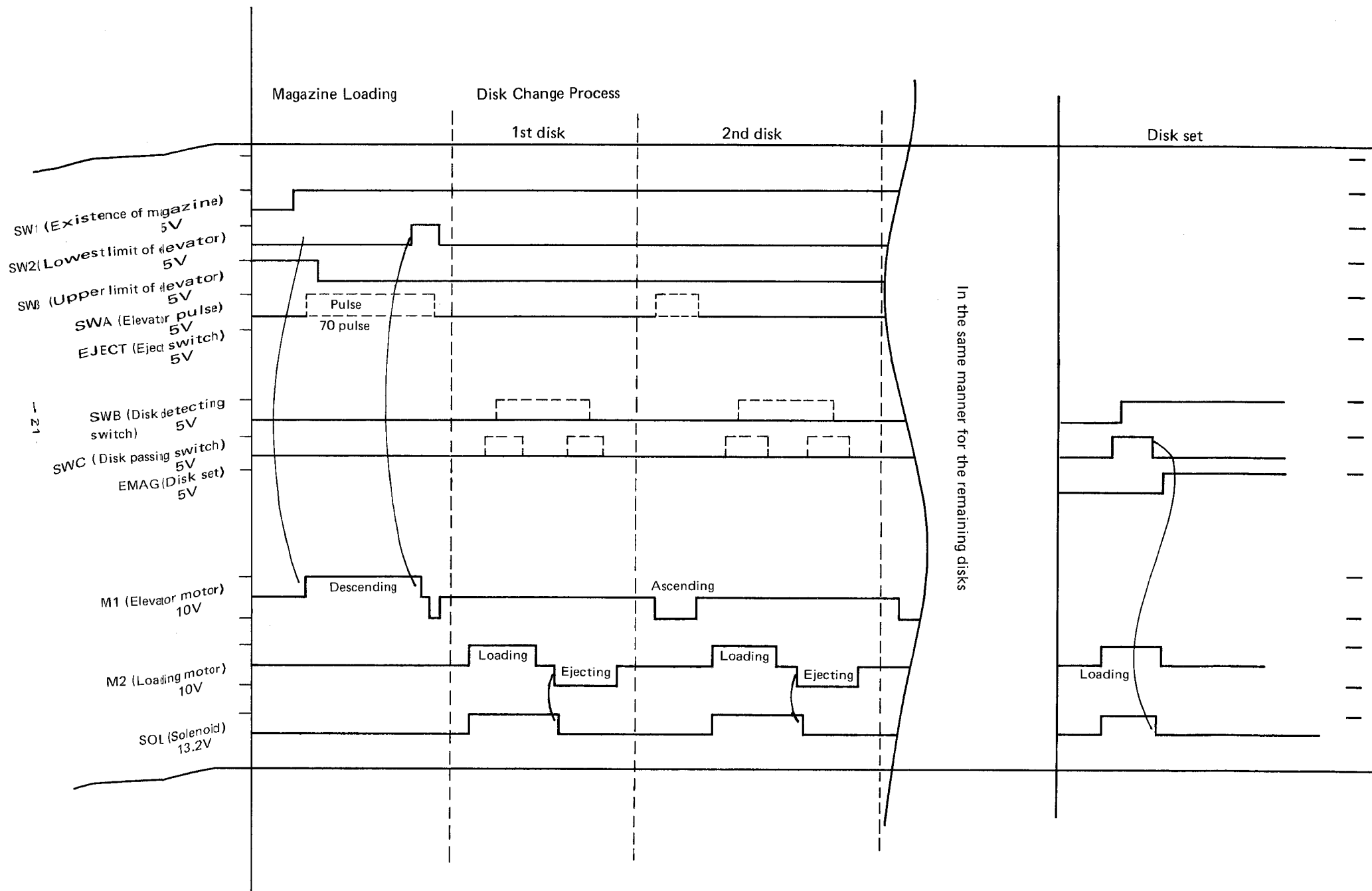


Fig. Right Side View of the Deck

After this, the loading operation is the same as Type A.

B3-3 CD Auto-Changer Loading Timing Chart



3. LOADING MECHANISM and OPERATION (TYPE C)

C3-1 Disk Loading Mechanism

The following switches (sensors) detect the disk position.

- Door switch
- SW2
- SW3

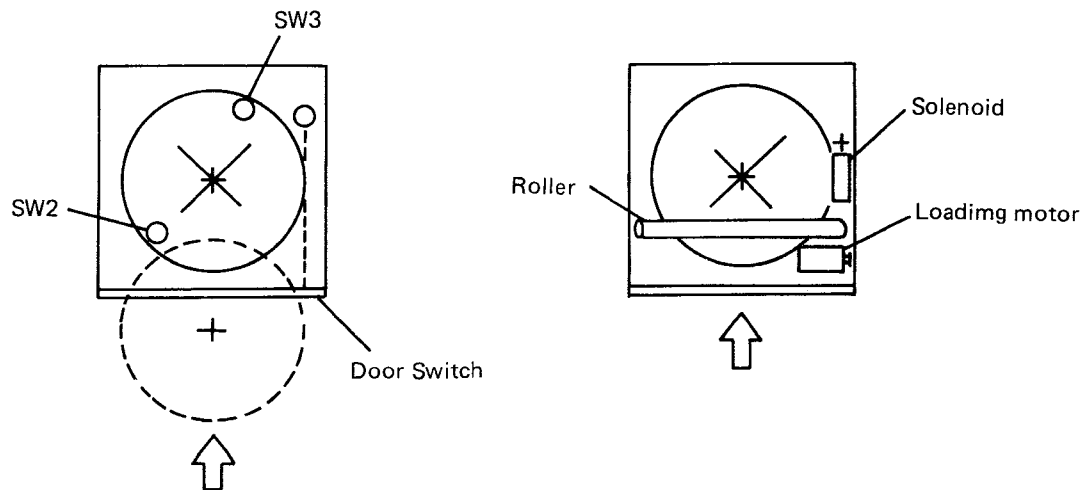


Fig. 3-1 Loading Mechanism Sensor Locations

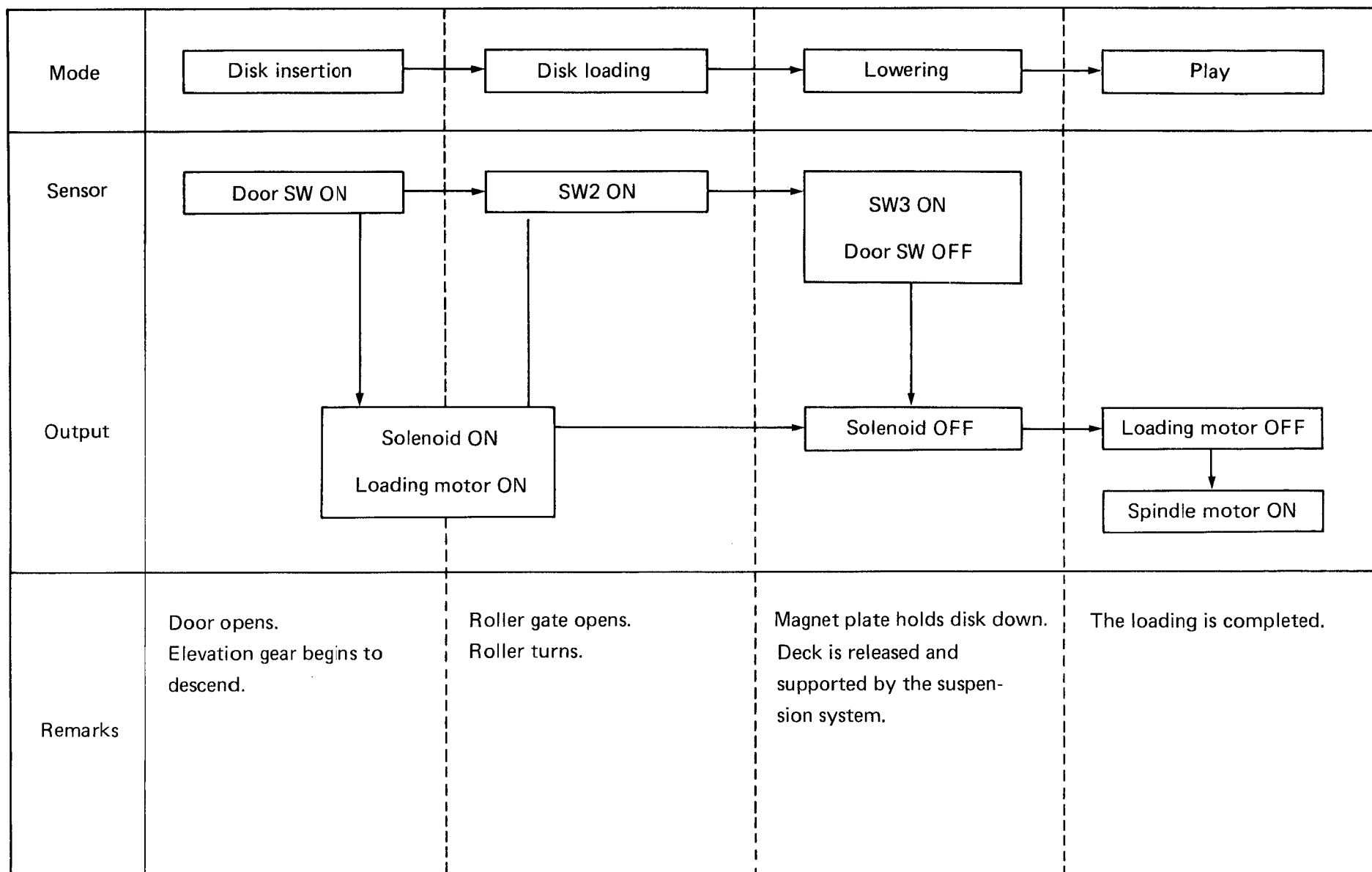
The following devices load and unload the disk:

- Loading motor : Bidirection motor to load, eject, raise, and lower the disks.
- Solenoid : Opens and closes the door, and suspension system operation.

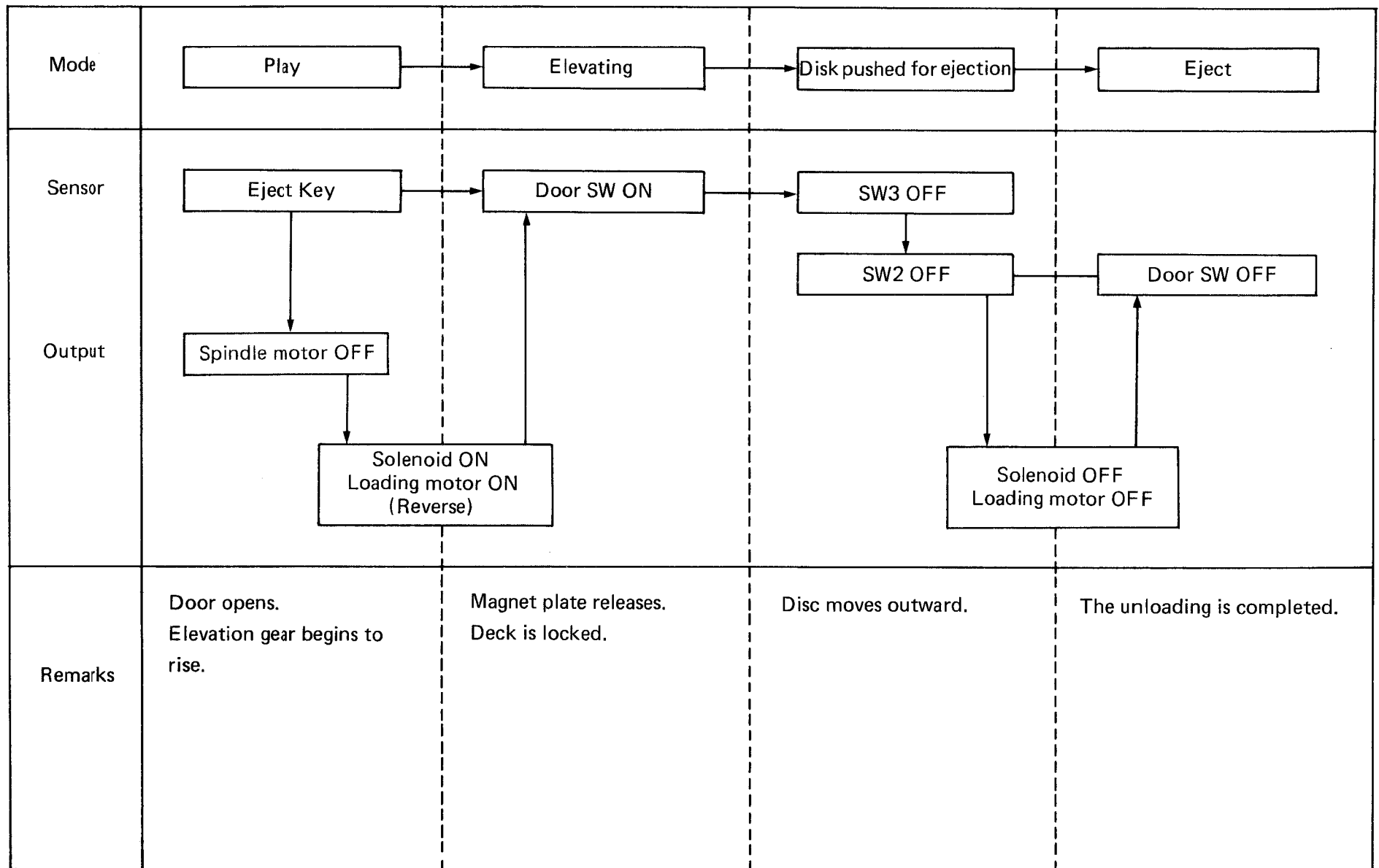
In the play mode, the deck sit on a cushion base with the disk hold down by the magnet plate.

* The drop detection switch is a standby switch that is used to release the magnet plate by means of EJECT, should it drop when there is no disk loaded.

C3-2 Loading Operation



C3-3 Eject Operation



C3-4 Disk Loading Circuit

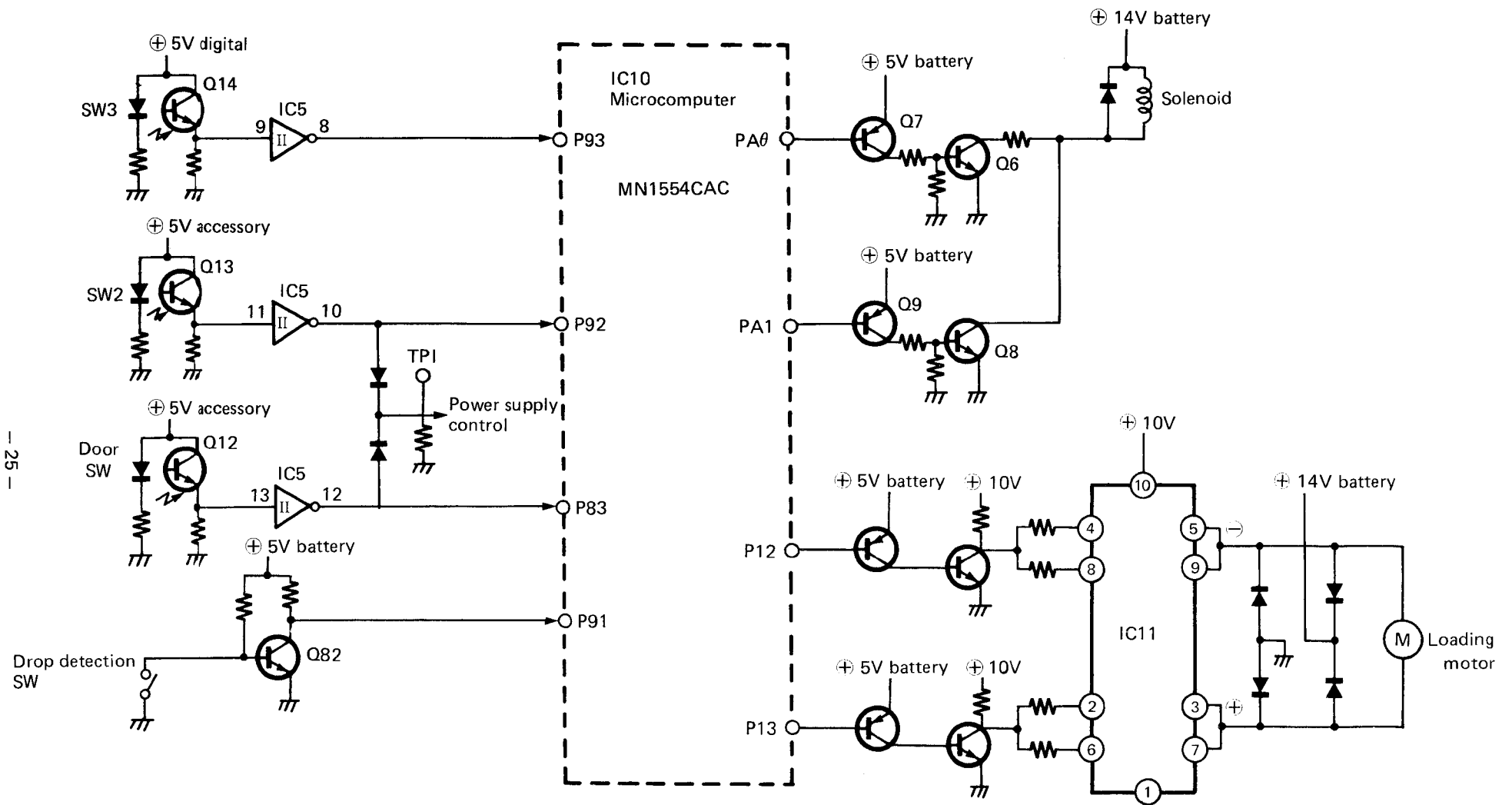


Fig. 3-2 Disk Loading Circuit

(1) Sensor input

During disk loading or ejection operation, the microcomputer controls the loading motor and solenoid with four sensor inputs.

When a disk is set, the sensor information input to the microcomputer is as follows.

SW3	(P93)		0V
SW2	(P92)		0V
Door SW	(P83)		5V
Drop detection SW	(P91)		5V

Loading starts when door SW or SW2 switches on the following:

- Power supply
- Solenoid
- Loading motor

Eject starts when EJECT SW is pressed.

(2) Solenoid control

The solenoid is controlled by $PA\phi$ and PA1 outputs.

PA1 is for starting the solenoid.

$PA\phi$ is for holding the solenoid.

The solenoid operates when the disk goes in or out. (0V during operation)

(3) Loading motor control

There are three loading motor control modes.

- Forward operation for loading
- Reverse operation for eject
- Stop

Pin \ Mode		Loading	Eject	Stop
IC10	P ⑫	5V	0	5V
	P ⑬	0	5V	5V
IC11	P ⑤ ⑨	0	10V	0
	P ③ ⑦	10V	0	0

IC11 is a complementary transistor driver, whose equivalent circuit is shown in Figure 3-3.

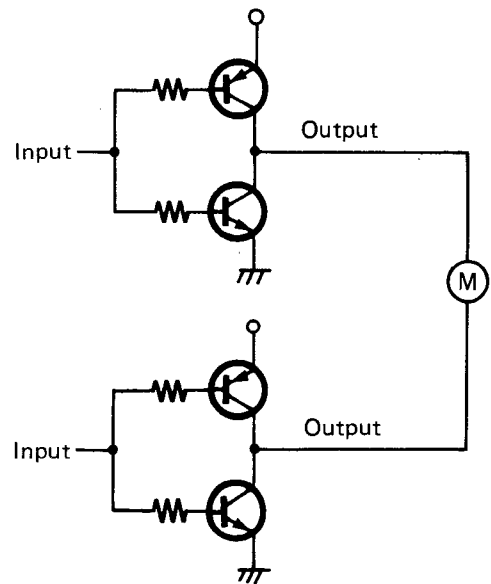


Fig. 3-3 Loading Motor Driver Circuit

4. CIRCUIT OPERATIONS (TYPE A)

4-1 Traverse Control Circuit (Fig. 4-1)

To the optical pickup at the starting position, RF must be present to operate the traverse motor. The IN SW changes (H) → (L) → (H) by the movement of optical pickup block.

When SW4 becomes (L), and IN SW is (H). TRV R (L) and TRV F (H) are output from IC601, the traverse motor drives the optical pick from outside to inside.

When IN SW turns (L), TRV R turns (H) and TRV F turns (L), and the optical pick moves from inside to outside. When IN SW is (H) again, the optical pick up is at the starting position, the traverse motor has ended the starting position setting operation, i.e. both TRV R and TRV F have become (L).

During the setting operation CNT 1, 2, 3, and 4 are all at (H); but just as soon as completion of the operation, CNT 1 remains at (H), while CNT 2, 3, and 4 change to (L).

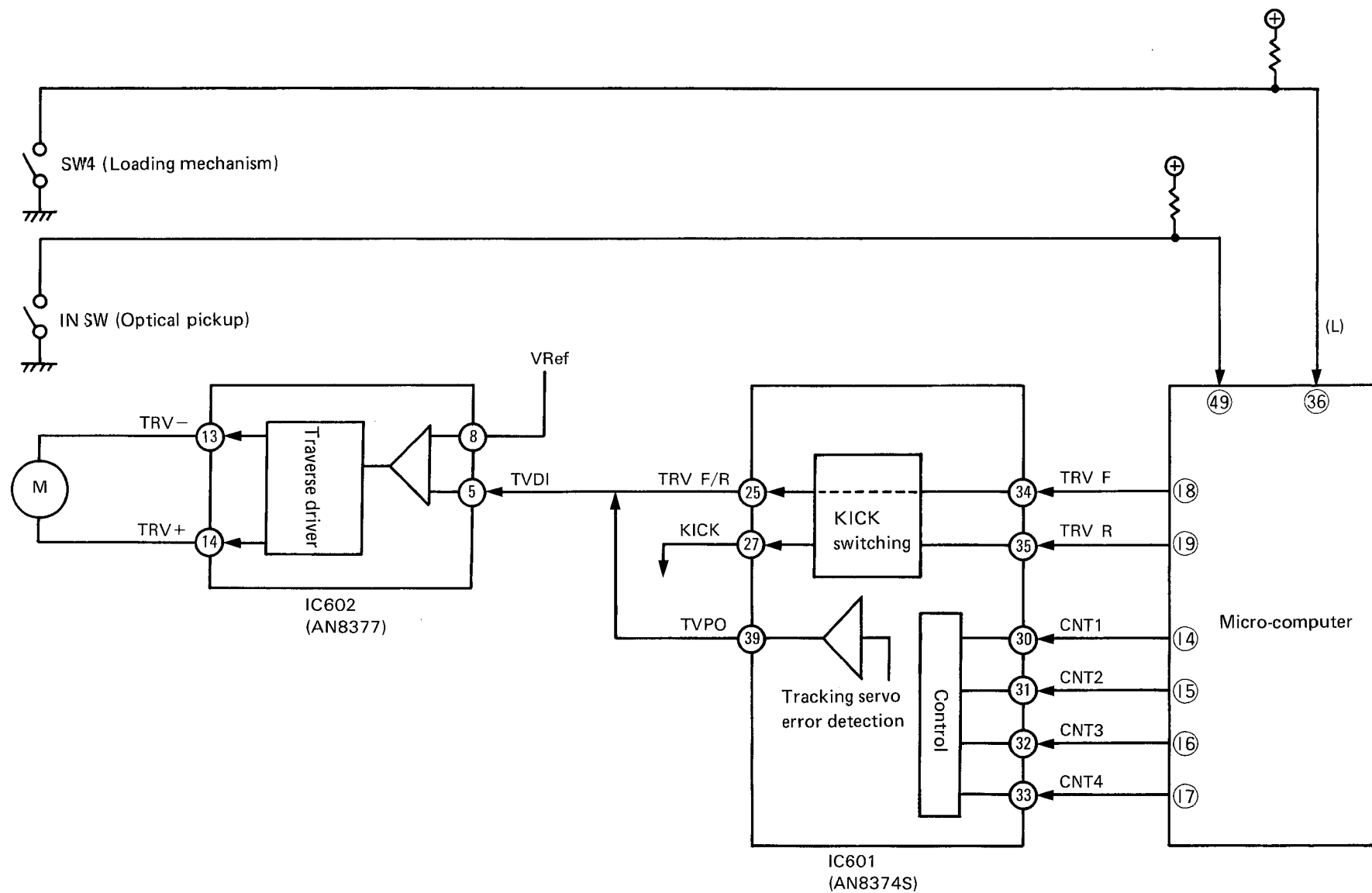
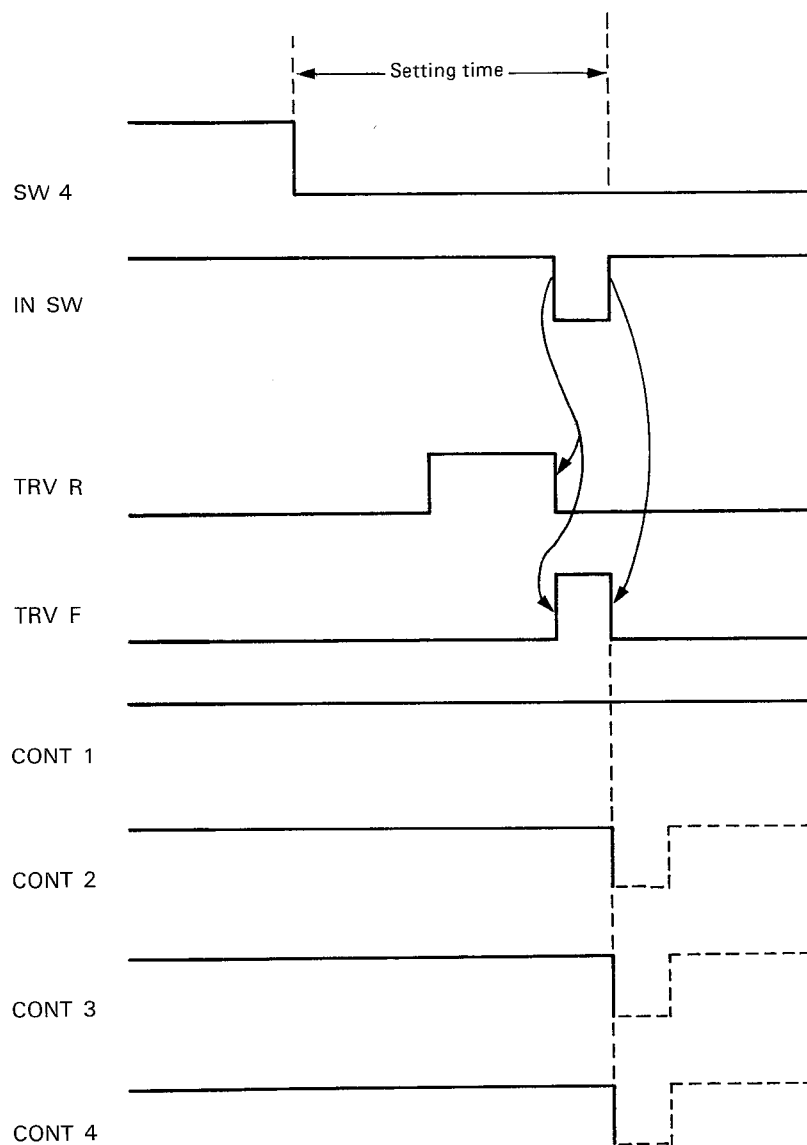


Fig. 4-1 Traverse Control Circuit (Optional pickup starting position setting)



Traverse Controlling Logic

	TRV R	TRV F
OPEN	0	0
FORWARD	0	1
REVERSE	1	0
STOP	1	1

Fig. 4-2 Traverse Timing Chart (At setting of optical pickup starting position)

When the optical pickup starting position is set, this circuit makes focus search and detects the disk. The function is finished by the SENSE pulse (a disk exists) from IC601 pin 24.

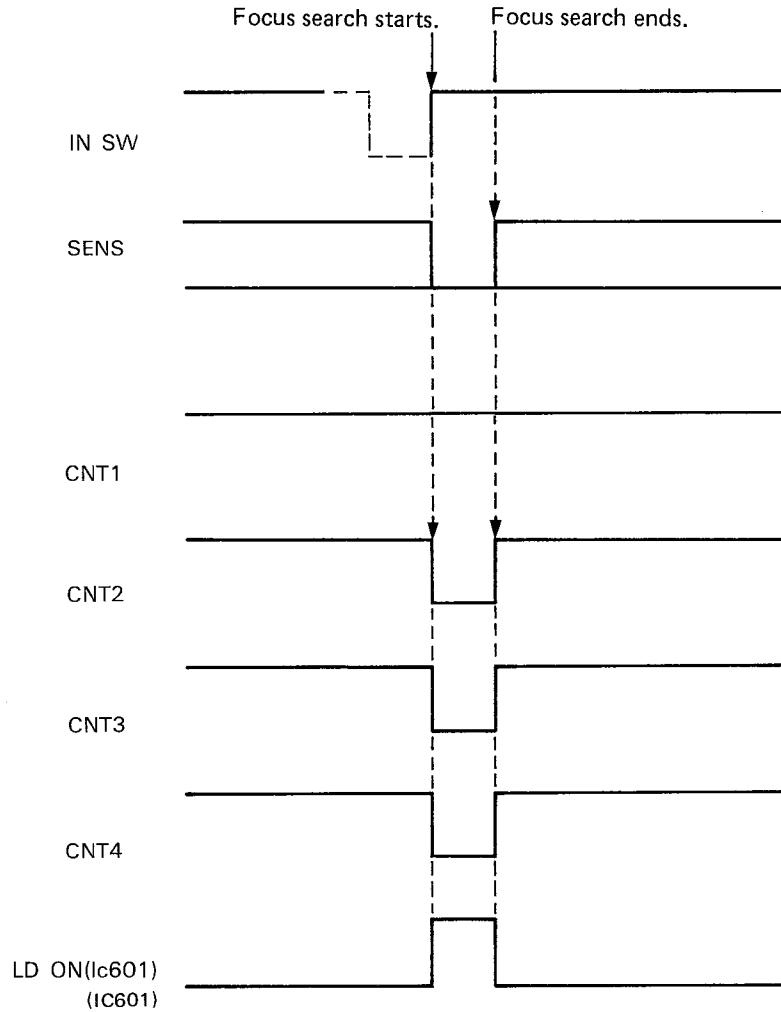
By the riser edge from (L) to (H) of IN SW, disk detecting instruction (CNT 1 (H), 2, 3 and 4 (L)) is output from the micro-computer. Following this instruction IC601 instructs search oscillation and laser emission.

The search oscillation (triangular wave) appears at pin 10, along with FE out at pin 9, feed to IC1 pin 16. The output at pin 23, from the FPO terminal passes through the filter of IC1, drives the focus coil in IC602. As a result, the optical pickup moves up and down.

Meantime, the laser emission instruction LD ON at pin 28 of IC601 joins the LD APC circuit at pin 31 of IC1 LD ON. The laser power supplies at pin 7 of IC1, the laser emits.

The focus control circuit detects this state with the photo-detector and joins the FE signal at pin 15 of IC1 into IC601. And it detects the "S-curve" when the proper disk exists, and search oscillation as well as laser emission are normal. The detecting operation ends when (L) → (H) pulse from the SENSE applied to the micro-computer.

(2) Focus Search Timing Chart



(1 "H", 0 "L")

HEX No.	Mode	Logic input				Logic control signal output							Remarks
		CNT1	CNT2	CNT3	CNT4	LD ON	AF ON	TR ON	HOLD	BR ON	KICK	SENSE	
0	PLAY	0	0	0	0	ON	ON	ON	OFF	OFF	OFF	Off track	Play mode
3	BRAKE	0	0	1	1	ON	ON	ON	OFF	ON	OFF	Off track	Brake mode
4	KICK CROSS	0	1	0	0	ON	ON	OFF	ON	OFF	OFF	CROSS	Kick mode
5	KICK REV	0	1	0	1	ON	ON	OFF	ON	OFF	ON (REV)	Shaping CROSS	
6	KICK FWD	0	1	1	0	ON	ON	OFF	ON	OFF	ON (FWD)	Shaping CROSS	
7	ACCESS	0	1	1	1	ON	ON	OFF	OFF	OFF	OFF	Shaping CROSS	Coarse feed
8	DISK DETECT	1	0	0	0	ON	ON	OFF	OFF	OFF	OFF	S latch	Disk detection (Search Oscillation)
F	STANDBY	1	1	1	1	OFF	OFF	OFF	OFF	OFF	OFF	Shaping CROSS	Forced return

Control Logic Table

A4-3 Spindle Motor Circuit

When disk detection ended, the spindle motor starts to rotate. The focus servo and the traverse servo are commanding the spindle motor operation; and after reading the TOC, play starts.

If there is any trouble in focus or traverse servo, it will become EJECT mode.

When the SENSE pulse appears, CNT 1, 2, 3, and 4 are (H). MLD, MCLK, and the MDATA are output from the micro-computer. The spindle motor starts to run by outputting the CLV data at pin 64 PC (L) of IC501 to IC2.

When the number of revolution increased, LD ON and AF ON signals (CNT 1, (L), 2, 3, and 4 (H)) are output from the micro-computer.

When the micro-computer receives the FLOCK (L) signal from IC601, the TRON signal (CNT 1, 2 (L), 3, and 4 (H)) is output from the micro-computer.

The TRON signal forms tracking servo while applying brake, then to the CLV control along with the rough servo by SRF signal of IC601 which is obtained from the RF signal.

When this circuit changes from the rough servo to the refined servo. CLVS of IC501 turns (H); and the PLAY signal (CNT 1, 2, 3, and 4 (H)) is output from CPU. At this time, reading of the TOC starts.

1. At the head of lead-in track, it positions by kick pulse while reading the data.
2. It reads TOC in the micro-computer.
(If there is any abnormality, it will reject at the place where abnormal function is affirmed.)
3. Play starts from the track 1.

Fig. 4-4 Spindle Motor CLF-Control

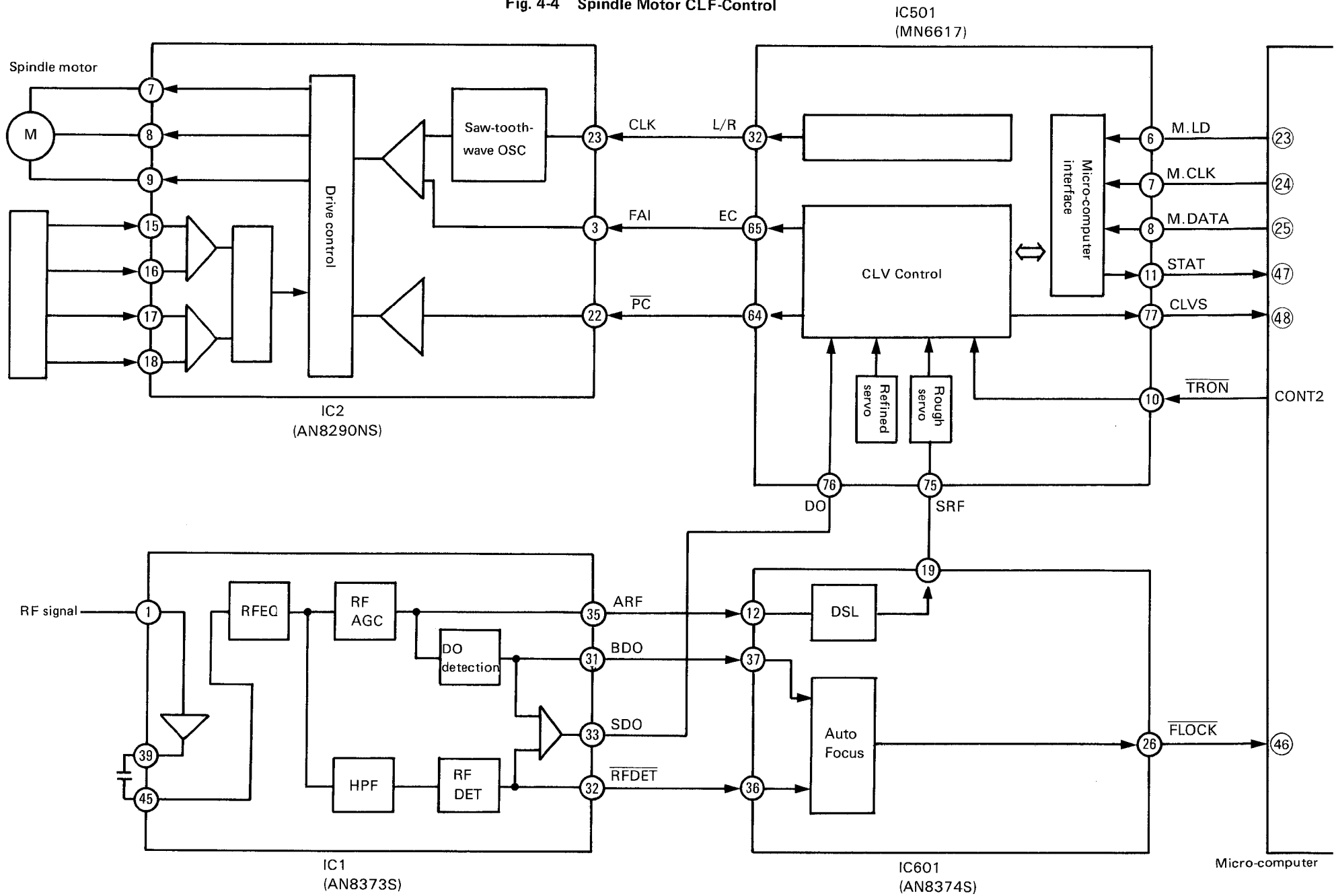
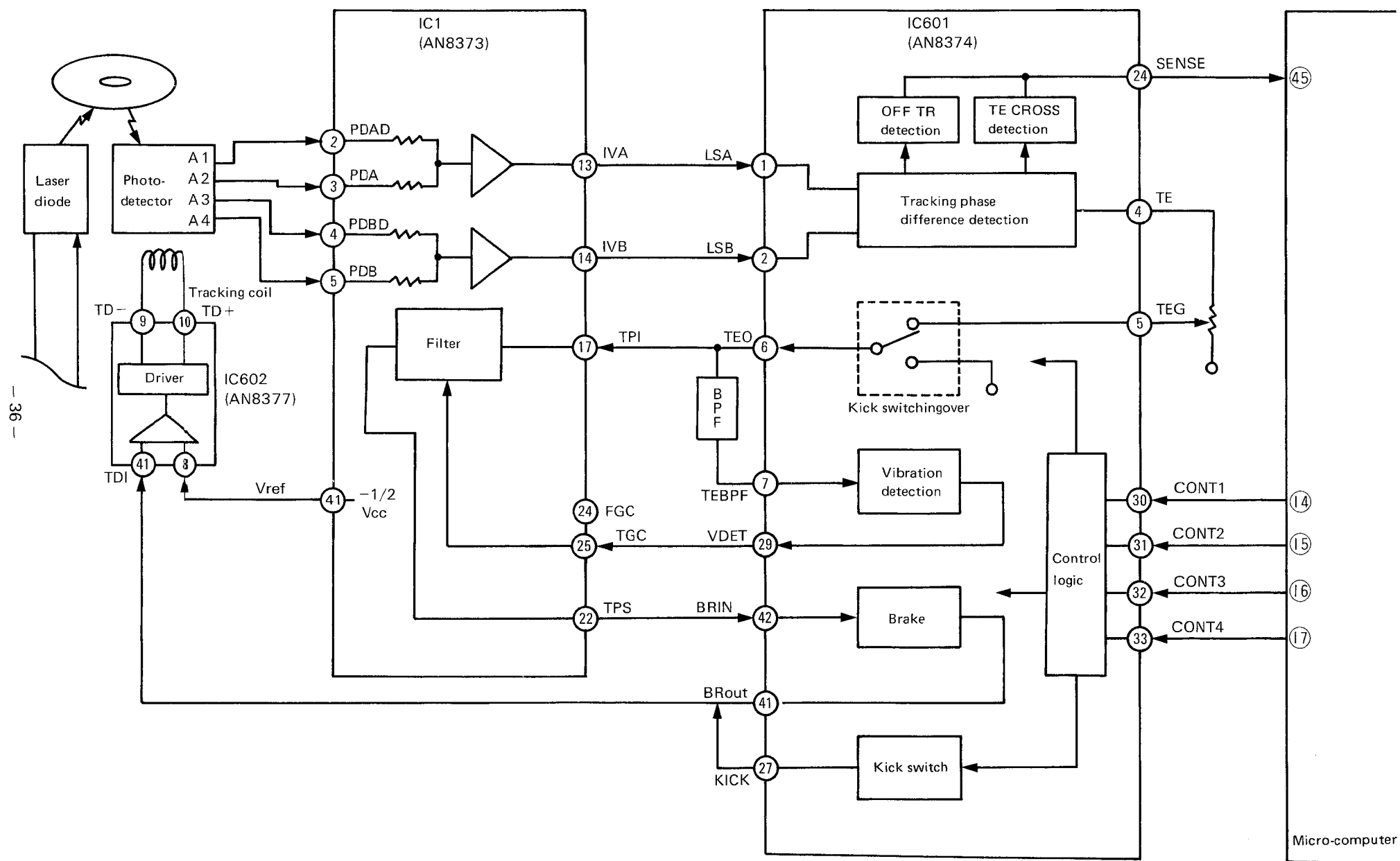


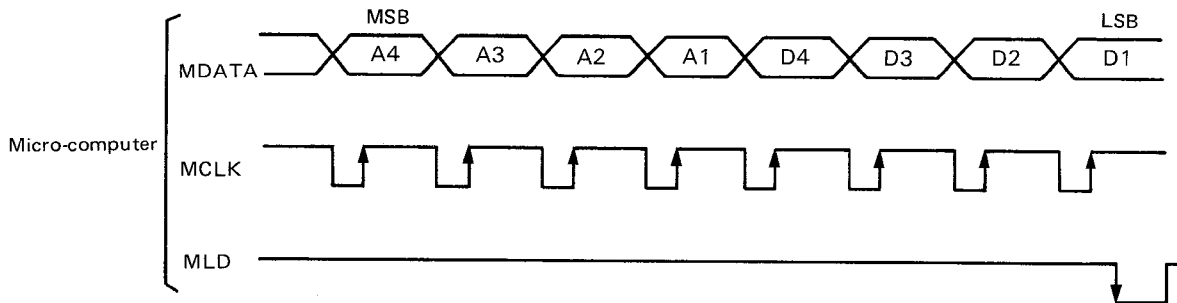
Fig. 4-6 Tracking Servo Circuit



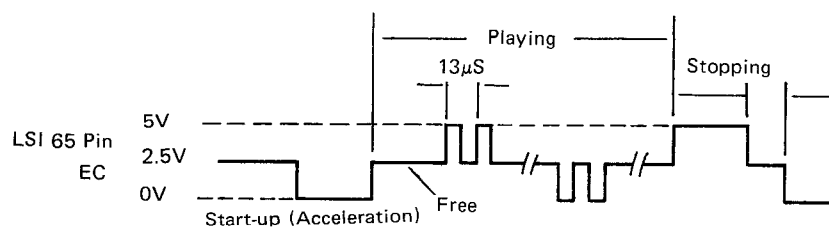
(4) Timing Chart

No.	←(MSB) 8 bits (LSB)→								Remarks
	A4	A3	A2	A1	D4	D3	D2	D1	
①	0	0	0	1	1 (with FG)	1 (TT ON)	0 (STOP)	0	CLV mode
					0 (without FG)	0 (TT OFF)	0 (ACC)	1	
							1 (BRAKE)	0	
							1 (PLAY)	1	
②	0	0	1	0	—	—	1 (ATT-12dB)	1 (MUTE ON)	Interpolation circuit
							0 (ATT OFF)	0 (MUTE OFF)	
③	0	1	1	1	—	—	0 (CRC)	0	Status output switching
							1 (CLVOK)	0	
							1 (TT STOP)	1	

Contents of M. DATA



Transferring M. DATA (Micro-computer → LSI)



CLV Control Output of EC Terminal

A4-4 Description of New Servo Circuit (MN8373S and MN8374S)

1. General

The servo circuit in CQ-E850, CQ-DP25/26 has been entirely changed with new IC's for performance improvement and compactness.

2. Circuit Configuration

The servo circuit used in the conventional FF-1 CD players comprised a servo IC (AN8370S) and discrete circuit. In contrast, the modified servo circuit consists of three IC chips with on discrete circuit.

- (1). IC-1 (servo amplifier)
=Processes analog servo signals.
- (2). IC-601 (servo controller)
=Processes digital servo signals.
- (3). IC-602 (servo driver)
=Drives the focus, tracking, and traverse coils.

The block diagram of IC-1 (AN8373S) and IC-601 (AN8374S) are shown in figure 4-9.

3. Departures from the Conventional Servo Circuit and some Cautions

While the new servo circuit replaces discrete components with integrated circuits, the basic circuit configurations and operations are almost identical to those of the conventional servo circuit, with the exception of the following details:

- (1). PLL adjustment eliminated
=Servo signals are automatically pulled into the PLL lock range by using the CLK (88.2KHz) signal.
- (2). 2.5V reference for tracking error and focus error signals
=Because the new circuit operates on a signal +5V supply, the reference voltage for the error signals has been changed to 2.5V as shown in figure 4-7.

NOTE

The 2.5V reference is also used for turntable height adjustment.

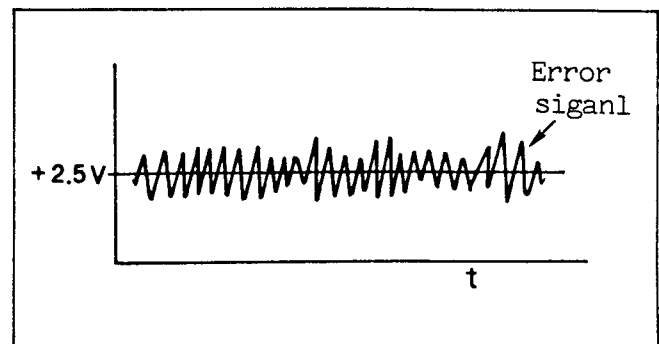


Figure 4-7

Focus and Tracking Error Signal

5. Data Signal Flow

(1). RF Signal Flow

- * The photo detectors A1 - A6 output is applied to IC-101's RF IN INPUT (pin 39) via an RF amplifier within the Optical Pick-up.
- * After being amplified by another RF amplifier within IC-101, the signal passes through an equalizer for 3T compensation, goes into the AGC (Auto Gain Control) circuit where it undergoes automatic gain control, and appears at the ARF (RF signal controlled by auto gain control) output (pin 35) of IC-101.

(2). EFM Data Generation

- * The RF signal applied to IC-102's ARF input (pin 12) is sliced by the DSL (data sliced) into a pulse signal EFM (eight to fourteen bit modulation), and then coupled to the PLL circuit.
- * The PLL is synchronized with the EFM signal and then coupled to the signal processing block.

6. PLL Operations

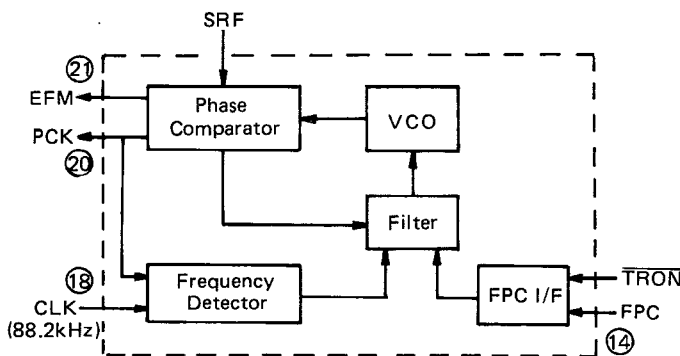


Figure 4-8 PLL Block

- (1). The new PLL circuit has replaced the PLL adjustment with a frequency comparator, one of whose input is clocked by the CLK signal (88.2KHz/11.34us) supplied from the digital signal processor.

The PLL circuit shows figure 4-8.

- (2). If PLL lock is lost, the frequency comparator's error output, this error signal will be controlled VCO's free-running frequency. The large error output of the frequency comparator drives the VCO until range. When the PLL is locked (TRON active), the FPC (frequency comparison error signal) is active and the VCO is controlled by the phase comparator's error output.

7. Control Logic (IC-601)

Commands used to set the servo circuit status are supplied from the system CPU in parallel 4 bit data (CNT1 - CNT4).

SENSE is the status read signal for IC-601, which is transferred to the system CPU. All other control logic operations are identical to those of the conventional control logic.

8. Focus Servo Circuit Operation

- (1). The outputs of the photo detector are coupled to pin 2 - 5 of IC-101, where they are phase corrected by the A1 + A2 and A3 + A4 signals and converted from current to voltage signals. They are transferred to the focus balance circuit and subjected to photo detector position error correction by means of VR101 (best eye adjustment).

- (2). The difference between the A1 + A2 and A3 + A4 signals, produced within the FE mixing circuit, serves as the Focus Error signal which appears at IC-101 pin 15.

The signal is then applied to IC-601 pin 8. VR63 is used for offset voltage injection control.

- (3). Since the Focus switch is closed in the PLAY mode, the FE signal appears at IC-601 pin 9 and couples to the loop filter within IC-101, where it is used to determine the focus servo's frequency response. The output of the loop filter at IC-101 pin 23, is transferred to the focus driving circuit and drives the focus actuator.

4. Servo Circuit Block Diagram

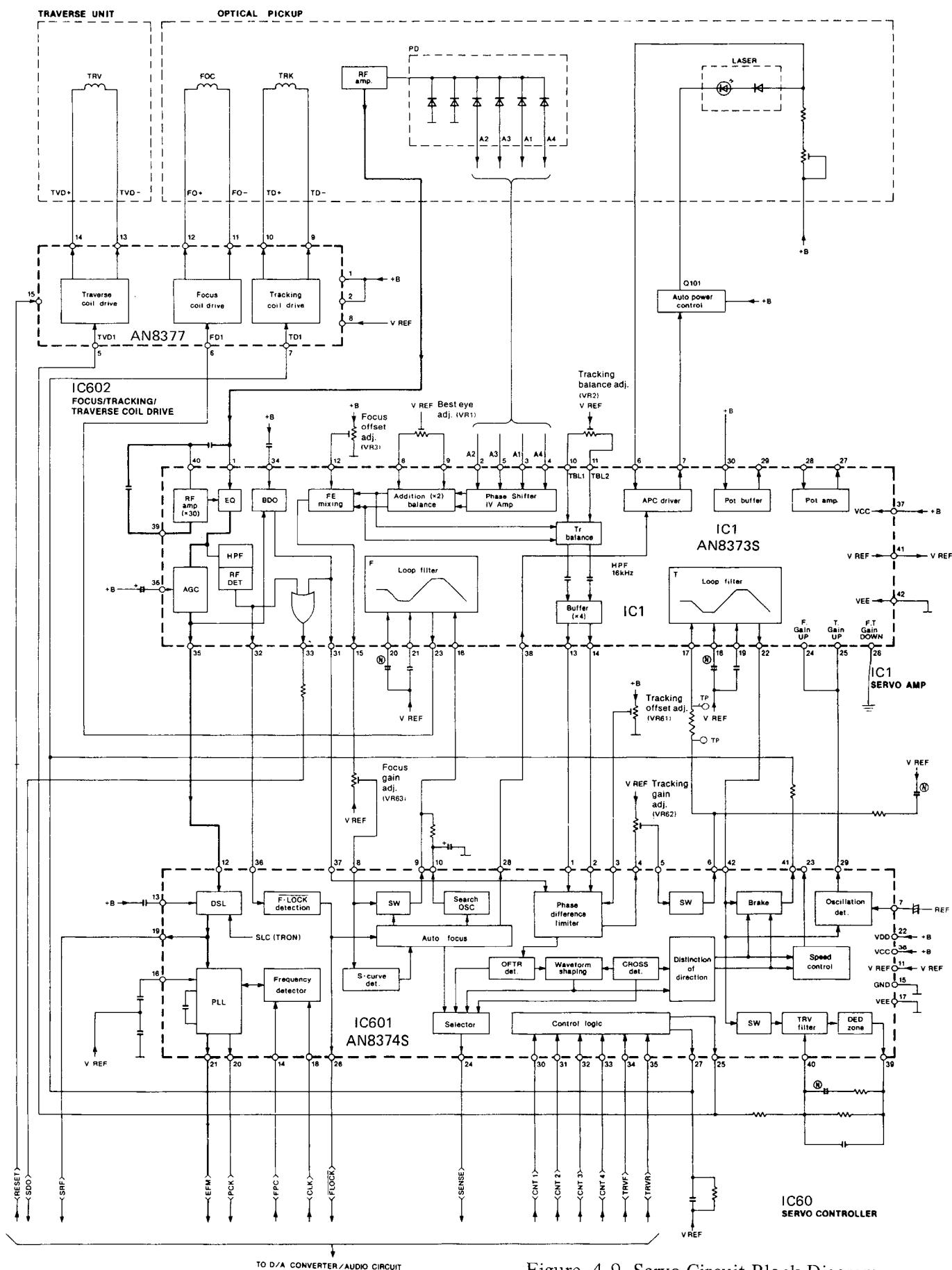


Figure 4-9 Servo Circuit Block Diagram

9. Tracking Servo Circuit Operation

- (1). The focus balance circuit output, $A1 + A2$ and $A3 + A4$, are also transferred to the tracking balance circuit. The output of the tracking balance circuit are applied to the phase difference limiter within IC-601, where they are phase compared to produce a tracking error signal.

The tracking error signal at IC-601 pin 4 is subjected to gain adjustment, then applied to the loop filter, within IC-101 via the tracking switch.

- (2). The loop filter output appearing at IC-601 pin 22, is coupled to pin 42 (BR IN/ tracking error input) of IC-601, where it is fed into a braking circuit. The output of the braking circuit appears at pin 41 (BR OUT/ tracking drive control signal) and couples to the tracking coil driver to drive the tracking actuator's.

10. Traverse Servo Circuit Operation

- (1). PLAY mode:

The tracking error signal applied to IC-601 pin 42 goes through the traverse switch and TRV (traverse circuit) filter (for response compensation) and appears at pin 39. This output is used to control the traverse actuator.

- (2). KICK mode:

The KICK signal decoded from the system CPU command by the control logic appears at pin 27. It is used to control the tracking actuator which KICKs the pick-up by a span of 1, 10, or 100 tracks according to the system CPU command.

- (3). Forced traverse mode:

The TRVF, TRVR signals supplied from system CPU to pins 34 and 35, respectively, are decoded to produce a traverse drive signal. The signal appears at pin 25 and drives the traverse coil.

11. RF Signal Detection and Auto Focus Circuit

The auto focus circuit within IC-601 performs focus search. If it detects an FE signal S-Curve it closes the focus switch contacts to close the focus servo loop. At this point the RF detector within IC-101 detects the RF signal and supplies it to the F.LOCK DET. input at IC-601 pin 36.

The F.LOCK signal is used to indicate to system CPU that focus is "IN". If focus goes "OUT" and the RF signal is no longer detected, the auto focus circuit opens the focus loop switch and starts focus search again.

12. Dropout Detection Circuit

- (1). Black Band Dropout (BDO)

Dropouts are detected from the RF signal within IC-101. In order to stabilize the tracking servo against dropout, the BDO signal is supplied from IC-601 pin 37 to the phase difference limiter and used to hold the tracking servo operation.

- (2). Signal Dropout (SDO)

If an RF signal dropout is detected by the RF detector, or a BDO is detected, the SDO signal is supplied to the signal processor to stabilize the CLV servo operation.

The SDO signal is supplied to the signal processor to stabilize the CLV servo operation. The SDO signal, produced within IC-601, is used to stabilize PLL circuit operation.

13. Braking Circuit

The braking circuit applies a brake to the tracking actuator so as to speed the tracking servo pull-in sequence required after a SKIP or KICK operation.

The direction of actuator movement is detected by the OFTR DET (off track detection), CROSS (tracking cross) and DIRECTION signals from IC-102 and a brake signal is added to the tracking error signal.

16. Pin Description of AN8373S (IC1)

Pin No.	Mark	I/O Division	Function
1	AMP1	I	RF signal input (X30 amp.)
2	PDAD	I	Photo detector current input (A2)
3	PDA	I	Photo detector current input (A1)
4	PDBD	I	Photo detector current input (A4)
5	PDB	I	Photo detector current input (A3)
6	LPD	I	Non-inverting laser power input
7	LD	O	Laser power auto control output
8	FBL1	I	PD balance adjustment
9	FBL2	I	
10	TBL1	I	Tracking balance adjustment
11	TBL2	I	
12	FOOFS	I	Focus offset adjustment
13	IVA	O	Current/voltage conversion output (A)
14	IVB	O	Current/voltage conversion output (B)
15	FE	O	Focus gain adjustment output
16	FPI	I	Focus error signal input
17	TPI	I	Tracking error signal input
18	C. TPL	I	Tracking error filter capacitor input
19	C. TPH		
20	C. FPL	I	Focus error filter capacitor input
21	C. FPH		

Pin No.	Mark	I/O Division	Function
22	TPO	O	Tracking error signal output
23	FPO	O	Focus error signal output
24	FGC	I	Focus gain up signal input
25	TGC	I	Tracking gain up signal input
26	GD	I	Focus/tracking down signal input
27	PTO	O	Position detecting amp. output
28	PTI	I	Position detecting amp. input
29	PBO	O	Position detecting buffer output
30	POT	I	Position detecting buffer input
31	BDO	O	Dropout detection output
32	RFDET	O	RF detection signal output
33	SDO	O	Dropout detection pulse output
34	C. SBDO	I	Dropout detecting capacitor input
35	ARF	O	RF signal output
36	C. AGC	I	AGC detecting capacitor input
37	VCC	I	Power supply (+5 V input)
38	LDON	I	Laser power control input
39	RF IN	I	RF signal input
40	AMPO	O	RF signal output
41	VREF	O	Reference voltage output
42	GND	I	Ground terminal

17. Pin Description of AN8374S (IC601)

Pin No.	Mark	I/O Division	Function
1	LSA	I	Phase difference input (A)
2	LSB	I	Phase difference input (B)
3	TEOFS	I	Tracking offset adjustment
4	TE	O	Tracking gain adjustment
5	TEG	I	
6	TE OUT	O	Tracking error signal output
7	TE BPF	I	Tracking error gain detecting filter (Not used, open)
8	FEG	I	Focus gain adjustment
9	FE OUT	O	Focus error signal output
10	CLW	O	Triangular wave oscillator capacitor output
11	VREF	I	Reference voltage input
12	ARF	I	RF signal input
13	CDSL	I	Data slice filter capacitor input
14	FPC	I	Frequency difference signal input
15	GND	I	Ground terminal
16	C. PLL	I	PLL loop filter constant
17	VSS	I	Ground terminal
18	CLK	I	Frequency pull-in clock signal (88.2 kHz) input
19	SRF	O	Sliced and digitized RF signal output
20	PCK	O	Clock output extracted from SRF
21	EFM	O	EFM signal output synchronous with PCK

Pin No.	Mark	I/O Division	Function
22	VDD	I	Power supply (+5 V input)
23	SPCNT	O	Track crossing speed control output (Not used, open)
24	SENSE	O	Selector output (track crossing state)
25	TRV	O	Traverse servo control output
26	$\overline{\text{FLOCK}}$	O	Focus lock signal output
27	KICK	O	Track kick signal output
28	LDON	O	Laser power control output
29	VDET	O	Focus/tracking gain up output (Not used, open)
30	CNT1	I	Control input (FOON: Focus servo ON signal)
31	CNT2	I	Control input (TRON: Tracking servo ON signal)
32	CNT3	I	Control input (KICKF: Kick direction (forward) command)
33	CNT4	I	Control input (KICKR: Kick direction (reverse) command)
34	TRVF	I	Traverse forward command signal
35	TRVR	I	Traverse backward command signal
36	$\overline{\text{RFDET}}$	I	RF detection signal input
37	BDO	I	Dropout detection input
38	VCC	I	Power supply (+5 V input)
39	TVPO	O	Traverse position detecting resistor/capacitor inputs
40	TVPI	I	
41	BROUT	O	Tracking drive control output
42	BRIN	I	Tracking error signal input

18. Pin Description of AN8377 (IC602)

Pin No.	Mark	I/O Division	Function
1	PVCC	I	Driver power supply (+5 V input)
2	VCC	I	Power supply (+5 V input)
3	TB	O	External transistor base driving output
4	VMON	O	Voltage (+5 V) output
5	TVDI	I	Traverse error signal input
6	FDI	I	Focus error signal input
7	TDI	I	Tracking error signal input
8	VREF	I	Reference voltage input

Pin No.	Mark	I/O Division	Function
9	TD-	O	Inverting output of tracking driver
10	TD+	O	Non-inverting output of tracking driver
11	FD-	O	Inverting output of focus driver
12	FD+	O	Non-inverting output of focus driver
13	TVD-	O	Inverting output of traverse driver
14	TVD+	O	Non-inverting output of traverse driver
15	$\overline{\text{RESET}}$	O	Reset signal output
16	PC	I	PC input (connect to GND)

1) Principle of the Digital Filter

The way an audio wave is converted to a digital signal, processed digitally, and then returned to a high-purity audio form, is more and more widely understood.

The conversion from analogue into digital signals is often called pulse code modulation, or PCM. The analogue waveform is first sampled (measured at regular intervals). Then it is quantized. This means that it is given the nearest numerical value in the large range of numbers used to transmit the digital sound information. The quantized value is then converted to binary form and encoded into a pulse train representing the series of numbers. This process of analogue to digital conversion is shown in figure 5-1.

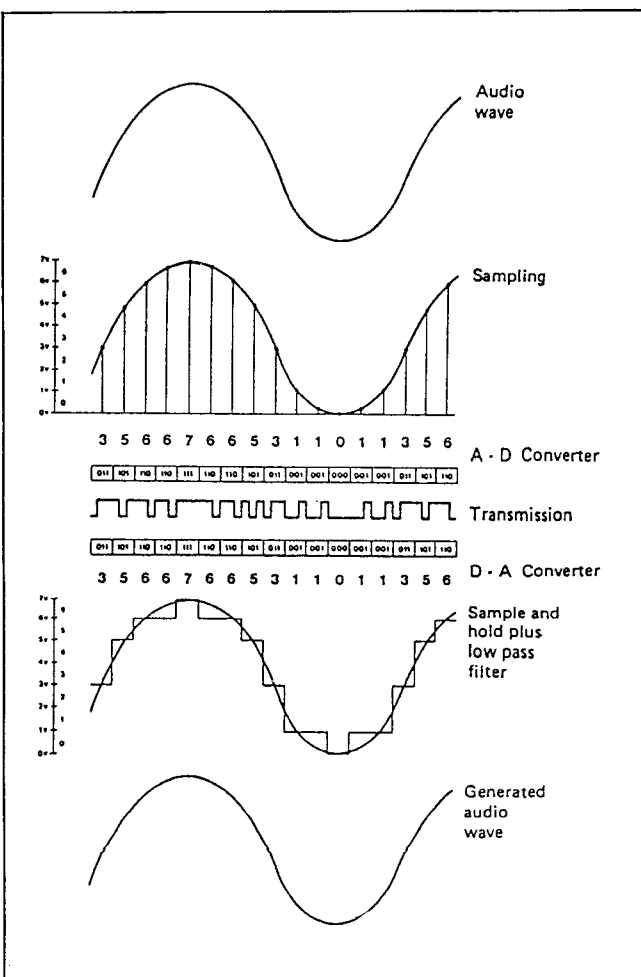


Figure 5-1

Figure 5-1 shows the digital sound reproduction process. The audio signal is sampled, quantized, converted to binary form, and encoded for recording or transmission.

Reversing the process produces a replica of the original signal. It is in this second stage that CD players use digital filtering to achieve exceptionally accurate reproduction.

The pulse train is fed into the transmission system (a cable or radio link, a tape, or a disc etc). At the reception or playback end, the pulse train is detected, decoded and regenerated in analogue form in the reverse process known as digital to analogue conversion.

2) How the Number of Bits Affects S/N

The accuracy of a digital system depends on the number of bits used for the binary number. This is because the number of bits determines the accuracy with which the samples taken from the original waveform can be quantized, i.e. expressed in digital form.

Two bits give four possible values for use in quantization, also three bits give eight possible values.

Every bit added doubles the number of possible values. In audio terms, this adds 6dB to the signal to noise ratio, because it reduces the quantization noise by a factor of 2.

The sixteen bits as used in the CD system give 65536 (2^{16}) possible values for quantization, and a signal to noise ratio of 96dB. The audio signals are encoded in the actual CD with this ratio.

96dB is approximation. An additional factor, which depends on the type of signal, is always present, regardless of the number of bits; for sine wave signals this makes the theoretical figure 97.8dB.

3) The Effect of the Sampling Frequency

Another important factor is sampling frequency, for efficiency in digital operations, this is kept low (44.1KHz)

at a little over twice the maximum audio frequency (20.0KHz). During the sampling process this frequency and its harmonics or multiples become modulated with the original waveform in a series of frequency bands as shows figure 19-2.

The bottom end of the lowest band is rather close to the top of the audio frequency range.

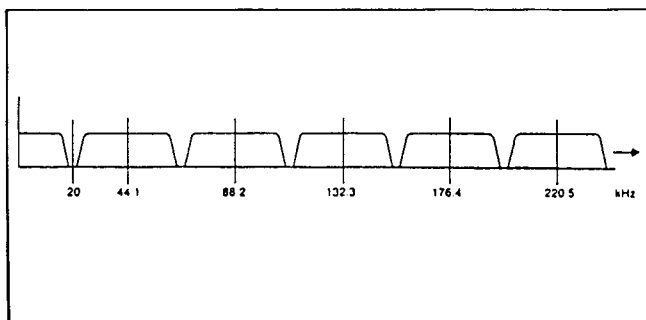


Figure 5-2

Figure 5-2 shows the frequency spectrum produced by sampling showing the periodic repetitions of the audio band. The bottom end of the lowest band is close to the audio frequency range.

4) Quantization Noise

In a digital audio system the audible noise has a different character to the familiar noise in an analogue audio system. The digital audio system background noise is called quantization noise.

The quantization noise of a system can be defined as the error signal between the original waveform before the analogue to digital conversion process, and the restored waveform after the digital to analogue conversion process. This error signal can be considered as both noise and distortion at the same time.

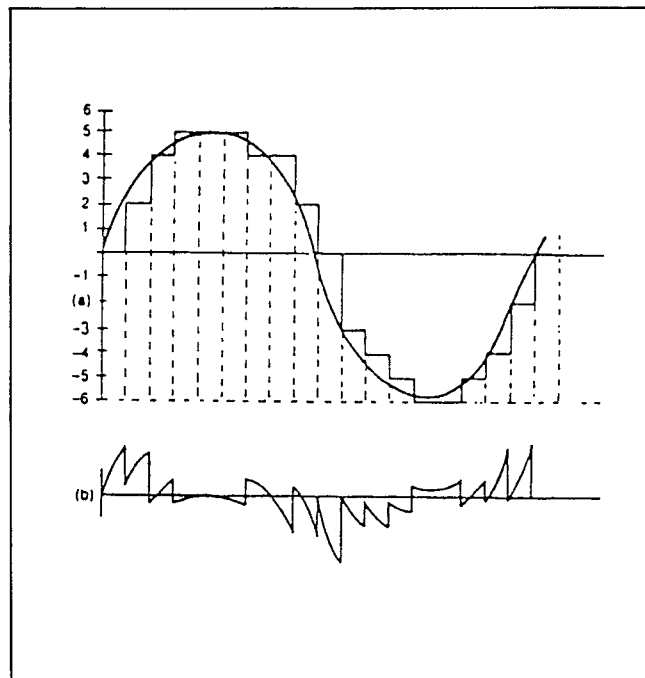


Figure 5-3

Figure 5-3 shows the quantization noise, The difference between the original audio waveform and the regenerated wave after D/A conversion (a) causes quantization noise (b).

The 16 bits analogue to digital conversion at the recording stage delivers a 16 bits digital signal with an intrinsic accuracy or signal to noise ration of maximum 96dB.

An ideal digital to analogue converter (D/A) would deliver an analogue output signal without adding noise to it. However, ideal D/A do not exist a small but finite electronic inaccuracy adds noise to the quantization noise of the digital signal.

The inaccuracy of a 16 bits D/A (the resolution limit of half the least significant bit) can be interpreted as a noise source that reduces the signal to noise ratio of the D/A to 96dB.

The effect of 16 bits analogue to digital conversion followed by 16 bits digital to analogue conversion with 1/2 LSB accuracy is to double the noise. In consequence, the maximum signal to noise ratio achievable in practice with a 16 bit system such as CD digital audio is 93dB.

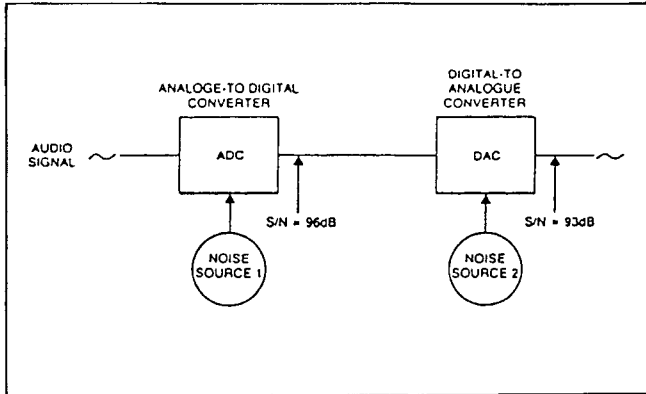


Figure 5-4

Figure 5-4 shows the signal to noise ratio deterioration in a digital chain, consisting of an A/D and a practical D/A.

5) Need for Filtering

With 16 bits (65536 possible numbers) and a sampling frequency of 44.1KHz, the CD system is theoretically capable of far higher reproduction quality than any conventional record player. But to achieve this quality, the output signal that the player derives from the disc must contain only the original audio frequencies, and not the periodic repetitions of the audio frequencies produced by the sampling (see figure 19-2).

These frequency bands can overload the power amplifier, and mix with audio frequencies or other signals to cause audible interference. They must be filtered out.

6) How D/A Conversion Operate

Figure 5-5 shows the generation process. Sampling produces only a series of sample (a) and hold (b) produces a step waveform closer to the original.

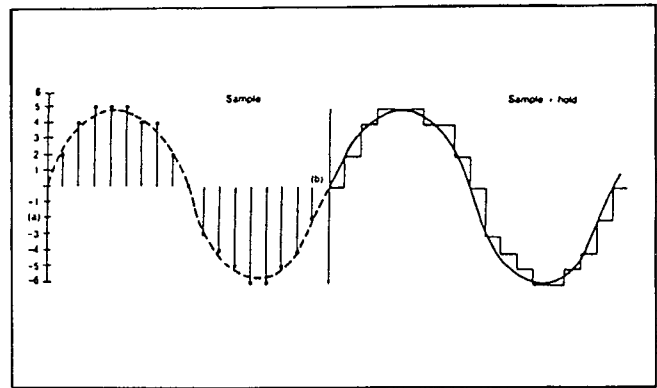


Figure 5-5

In the pulse train a series of binary number represents quantized samples of the original audio frequency. If these are simply converted to analogue values, a series of sample is produced (see figure 19-5).

The outline of this series resembles the audio wave, but the constant on and off switching produces the infinite series of frequency bands which was shown in figure 19-2. In practice, each sample is held in the converter until the next sample arrives, to produce a step waveform which is much nearer the shape of the audio waveform than the series of sample.

This sample and hold process is illustrated in figure 5-5 (b). It substantially reduces the frequency band components, as can be seen from figure 19-6. Significant components still remain, however particularly in the region near the audio band.

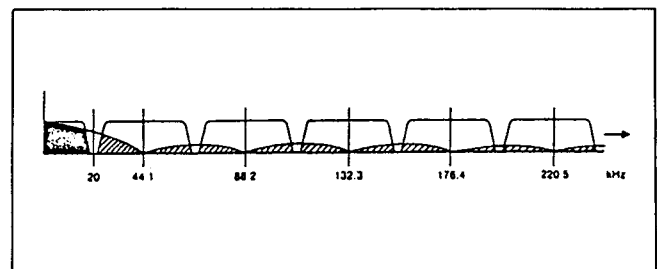


Figure 5-6

Figure 5-6 shows the effect of sample and hold. The remaining signal components are shown shaded.

7) Why Uses Digital Filtering

Most CD players suppress the frequency bands with conventional analogue filtering, after the signal has been converted back to analogue form. Such a filter is always complex, and can not avoid degrading the signal accuracy and introducing phase distortion.

Inevitable factors such as temperature changes and component aging affect performance. Most CD players on the

other hand, carry out this essential filtering process digitally, with no loss of accuracy, and phase distortion.

Digital filtering is a pure numerical calculation process with fixed coefficients. The fixing of the coefficients determines the filter characteristics which cannot afterwards vary.

8) Digital Filtering Process

A schematic diagram of the complete our D/A conversion system including digital filtering, is shows in figure 5-7.

It can be seen that our digital filtering uses the techniques of oversampling and noise shaping. Each plays a vital role in the digital filtering and conversion process.

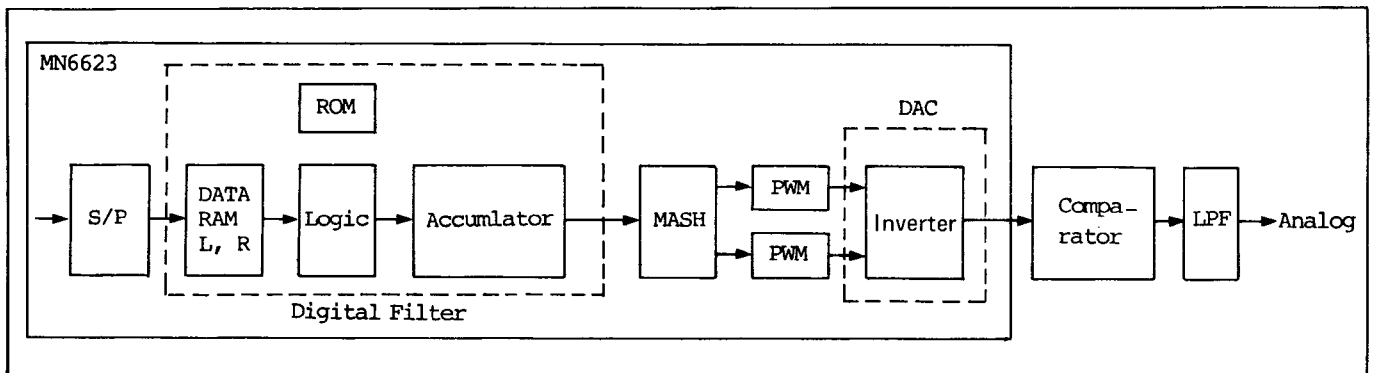


Figure 5-7

Oversampling:

Oversampling simply means multiplying the sampling frequency, in this case by four times. (44.1KHz becomes 176.4KHz.) The values of the intermediate samples are obtained by the calculation process

which determines the filter characteristic. An immediate effect of the oversampling as shows in figure 19-8 is to suppress all the lower frequency bands, centered on f_s , $2f_s$, and $3f_s$.

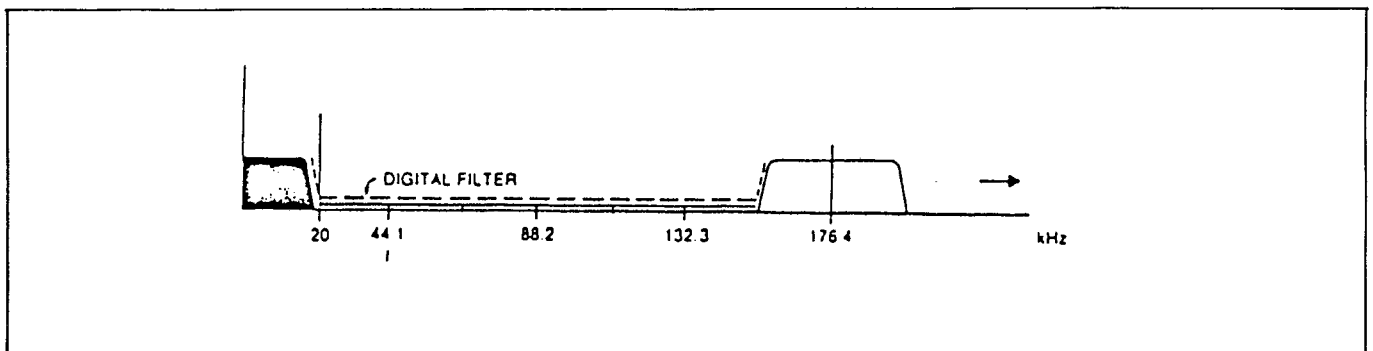


Figure 5-8

9) MN6623 Actual Operation

The MN6623 D/A converter system is composed of a digital filter, MASH, and PWM blocks as shown in figure 5-9.

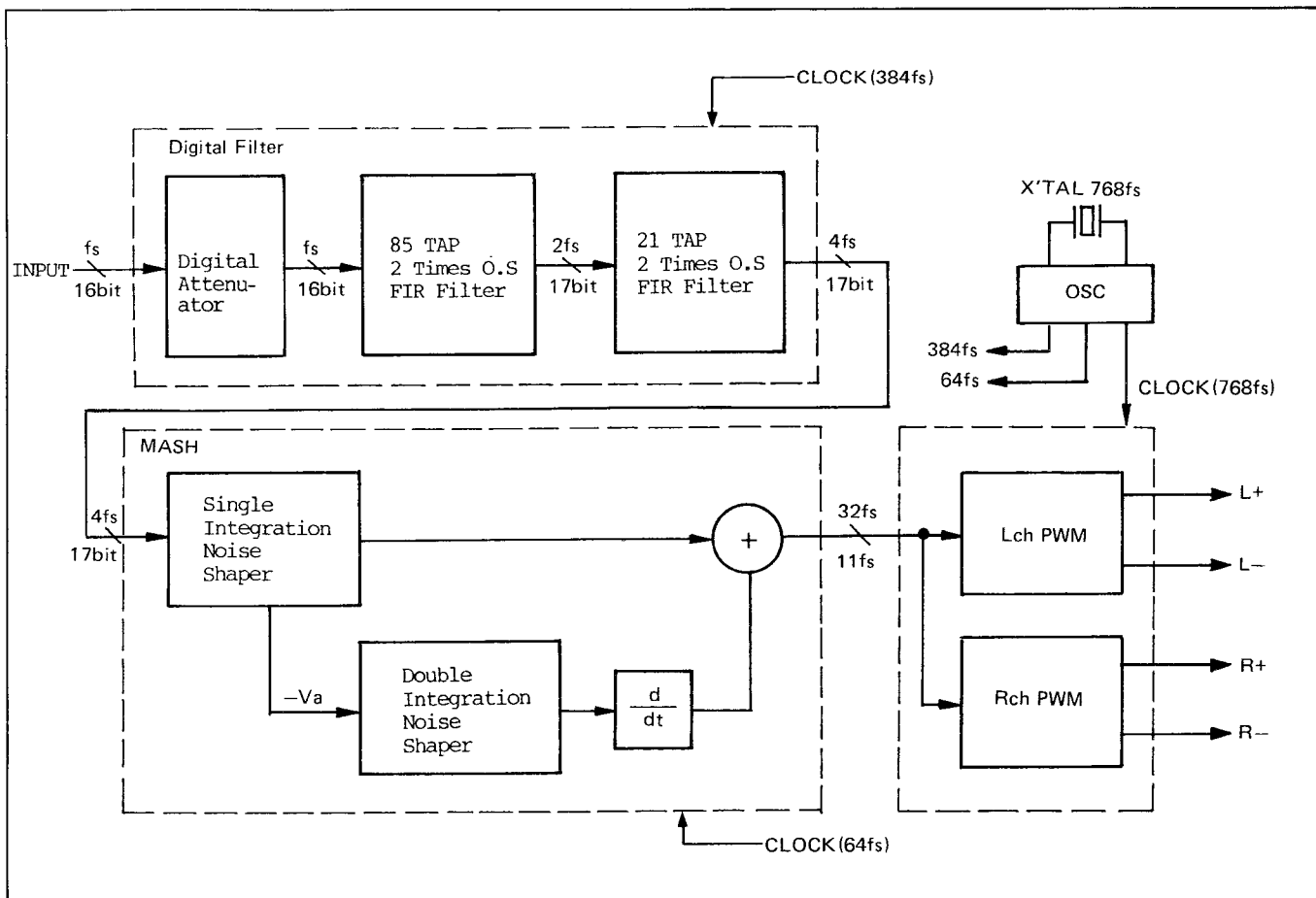


Figure 5-9 MN6623 Block Diagram

The MASH block operates at $32f_s$, where f_s is the sampled input signal frequency. This means that a 32 fold oversampling filter is required. On the MN6623, however, the first digital filter oversamples the f_s signal into a $4f_s$ signal. The following MASH block then performs zero-order holding on the $4f_s$ signal to convert it into a $32f_s$ signal.

The digital filter and MASH blocks use $384f_s$ and $64f_s$ clocks, respectively, to perform time-division multiplexing on the left and right channel signals. The PWM block uses a $768f_s$ clock to process the left and right signals independently. The MASH block converts the 16 bits, $32f_s$ signal into 11 values, which after Pulse-Width Modulation in the PWM block become analog signals. The following section describe the function of each MN6623 block.

Digital Filter Block

The digital filter block consists of three sub-blocks: ATT, FIR1, and FIR2 as shows in figure 5-10. It performs digital attenuation and oversampling to convert the input 16 bits, f_s data into a 16 bits, $4f_s$ signal.

The ATT attenuates the input data in a $1/64$ increment by multiplying the data by any 6 bits data. FIR1 and FIR2 are 2-fold oversampling digital filters with 85 and 21 taps each. Four-fold oversampling is accomplished by cascading the two -fold oversampling filters.

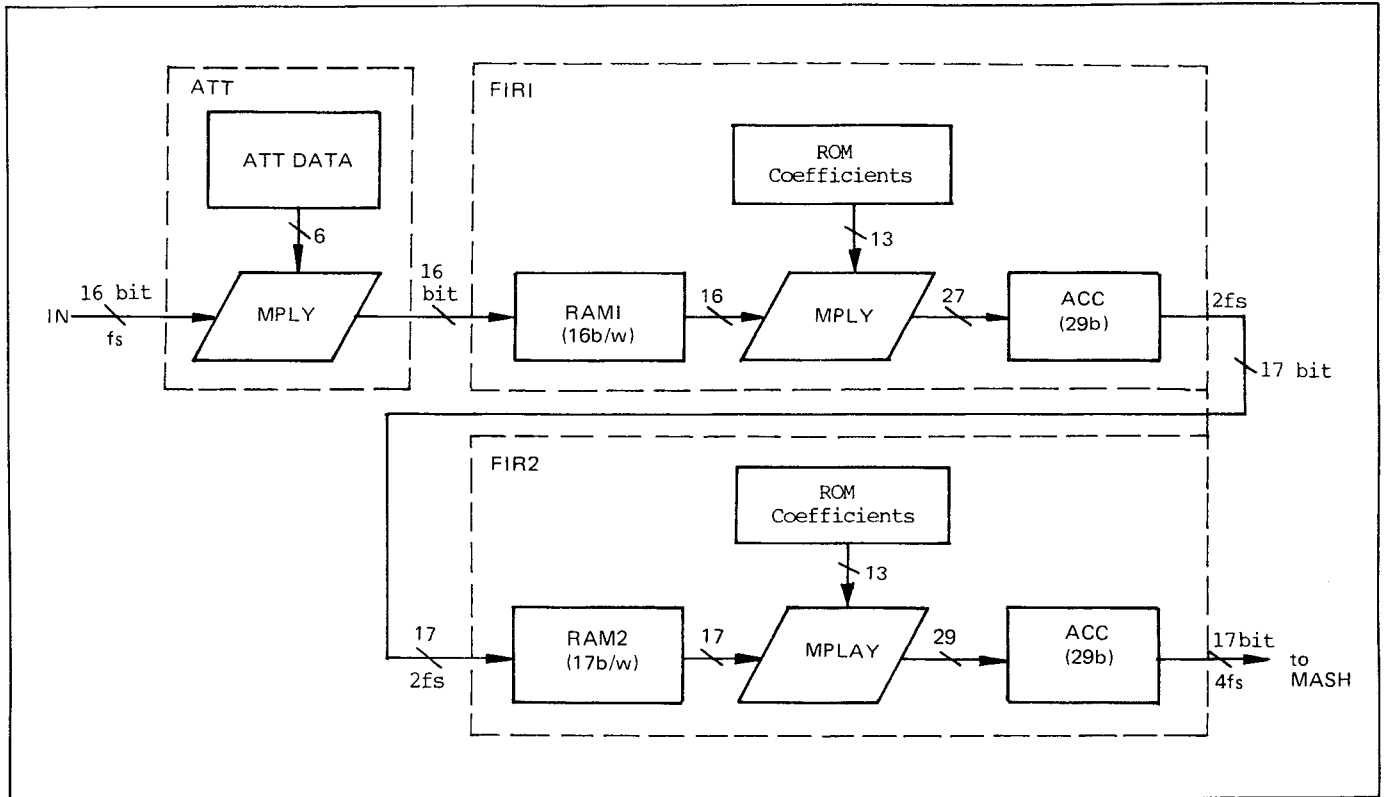
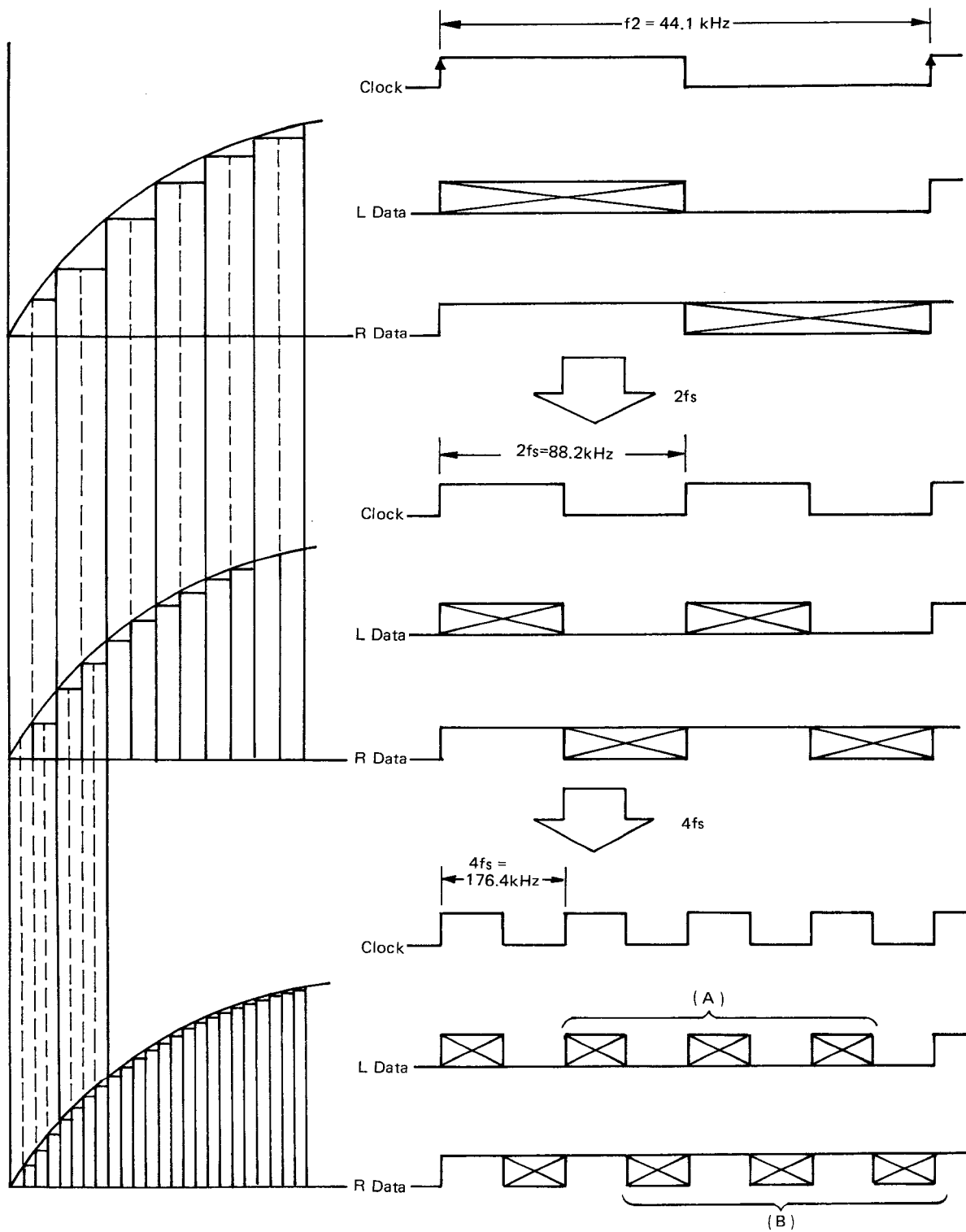


Figure 5-10 Digital Filter Block Diagram

The digital filter block contains many memories (delay logics), as shows in the block diagram above, with which it computes ROM coefficients at high speed while summing the coefficients to interpolate them. For more detail, shows in figure 5-11.



Note: (A) and (B) are the data created in the digital filter.

Figure 5-11

MASH Block

The MASH block consists of many noise shapers connected in series. The noise shapers include the signal and double integration types.

* Single integration noise shaper

The single integration noise shaper uses only a single integrator for noise shaping as shows in figure 5-12. The block marked "Q" in this figure is a local quantizer which adds quantization error to the input data. The relationship between output (Y) and input (X) for this type of noise shaper is:

$$Y = X + (1 - Z^{-1}) \cdot V_q \dots\dots\dots (1)$$

The quantization error, V_q is a random value and the factor $(1 - Z^{-1})$ in this equation represents a differentiation response. Eq. (1) tells us, therefore, that the spectrum of the quantization error, V_q in the single integration noise shaper has an attenuation slope of -6 dB/oct toward lower frequencies as shows in figure 5-14.

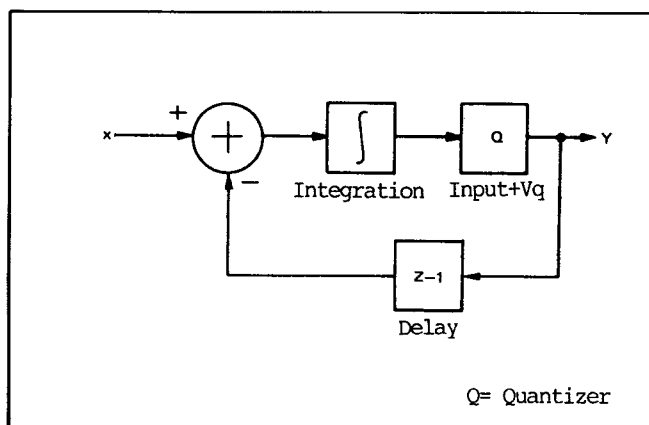


Figure 5-12 Single Integration Noise Shaper

* Double integration noise shaper

The double integration noise shaper uses two integrations in tandem as shows in figure 5-13. When the circuit is viewed at point W toward the output Y, it is

identical to the single integration noise shaper. Therefore, the relationship between Y and W is:

$$W = X + (1 - Z^{-1}) \cdot V_q \dots\dots\dots (2)$$

W expressed as a function of X is:

$$W = \frac{1}{1 - Z^{-1}} (X - Z^{-1} \cdot Y) \dots\dots\dots (3)$$

From Eqs. (2) and (3) we obtain following overall output-input relationship:

$$Y = X + (1 - Z^{-1}) \cdot V_q$$

If we compare this equation with Eq. (1), we find that the element $(1 - Z^{-1})$ is squared in the former equation.

This means that with the double integration noise shaper, the spectrum of the quantization error, V_q , has an attenuation slope of 12 dB/oct toward lower frequencies.

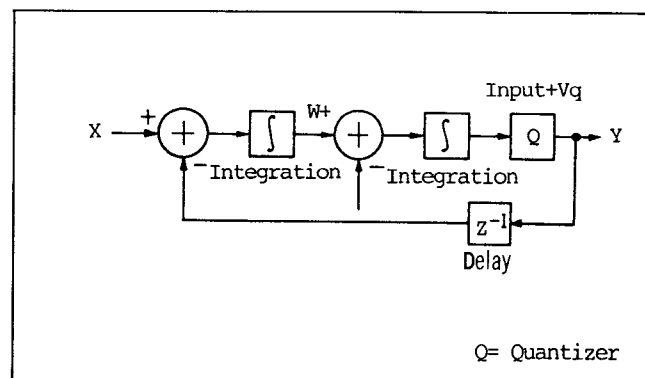


Figure 5-13 Double Integration Noise Shaper

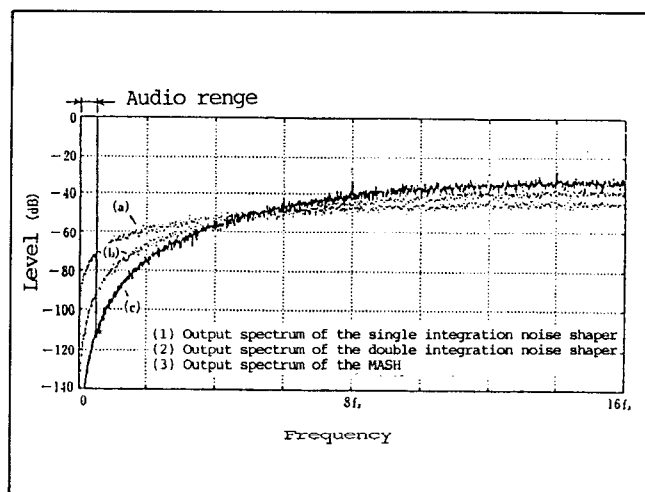


Figure 5-14 Noise Shaper Output Spectrum

The double integration noise shaper thus provides a greater attenuation of quantization noise than that obtained with the single integration type, as shows in figure 5-14.

The MASH block combines the single and double integration noise shapers in tandem so that the quantization error contained in the output of the first stage can be quantized again in the second stage further reduced.

Figure 5-15 is a block diagram of the MASH logic used on the MN6623 D/A converter. The MASH logic uses a single integration noise shaper in the first stage and a double integration noise shaper in the second stage. The outputs of the first and second stages of noise shaper, Y1 and Y2 are given by the following equations:

$$Y1 = X + (1 - Z) \cdot Vq1 \dots\dots\dots(5)$$

$$Y2 = -Vq1 + (1 - Z) \cdot Vq \dots\dots\dots(6)$$

Where

X = Input data

Vq1= Quantization error of local quantizer Q1

Vq2= Quantization error of local quantizer Q2

Y1 = Output of single integration noise shaper

Y2 = Output of single

Y2 = Output of double intergration noise shaper

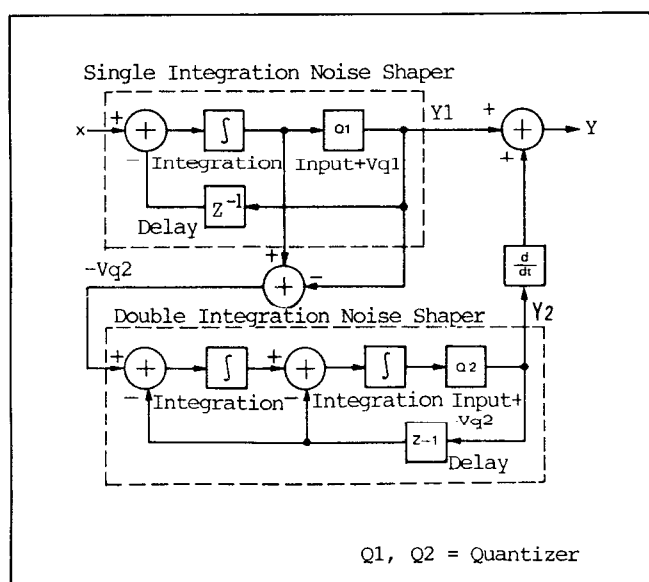


Figure 5-15 MASH Block Diagram

If we multiply both members of Eq.(5) by $(1 - Z)$ and add them to those of Eq. (6), we find that the quantization error is cancelled out.

Applying the output (Y2) of the second noise shaper to the differentiator and then adding the differentiator's output to the first noise shaper's output gives an overall MASH output, Y as represented by:

$$Y = X + (1 - Z) \cdot Vq2 \dots\dots\dots(7)$$

Eq.(7) indicates that the quantization error, Vq, is given in a third-order differentiation response (-18 dB/oct) , which gives further reduction in quantization noise

Table 5-1 lists the quantization levels at the first and second local quantizers, Q1 and Q2.

Q1 OUTPUT	Q2 OUTPUT
33,792	11,264
22,528	
11,264	
0	0
-11,264	-11,264
-22,528	
-33,792	

Table 5-1 Quantization Levels at Local Quantizers

The first quantizer (Q1) quantizes the signal into 7 levels. and the second quantizer (Q2) quantizes it into 3 levels. Actually, the values which are obtained by normalizing by 11264. The output values listed in Table 19-1 appear at Y1 and Y2. i.e. values +-3, +-2, +-1, and +-0 appear at Y1 and values +- and +-0 appear ay Y2. Since the output of the second local quantizer (Q2) is differentiated and added to the output of the first local quantizer (Q1), output Y yields 11 different values.

A high output level and wide dynamic range are thus achieved by quantizing the 16 bit, 4fs data from the digital filter at a sampling frequency 8 times as high (32fs).

The following chart contains the timings of the 11 types of digital data yielded by the MASH block.

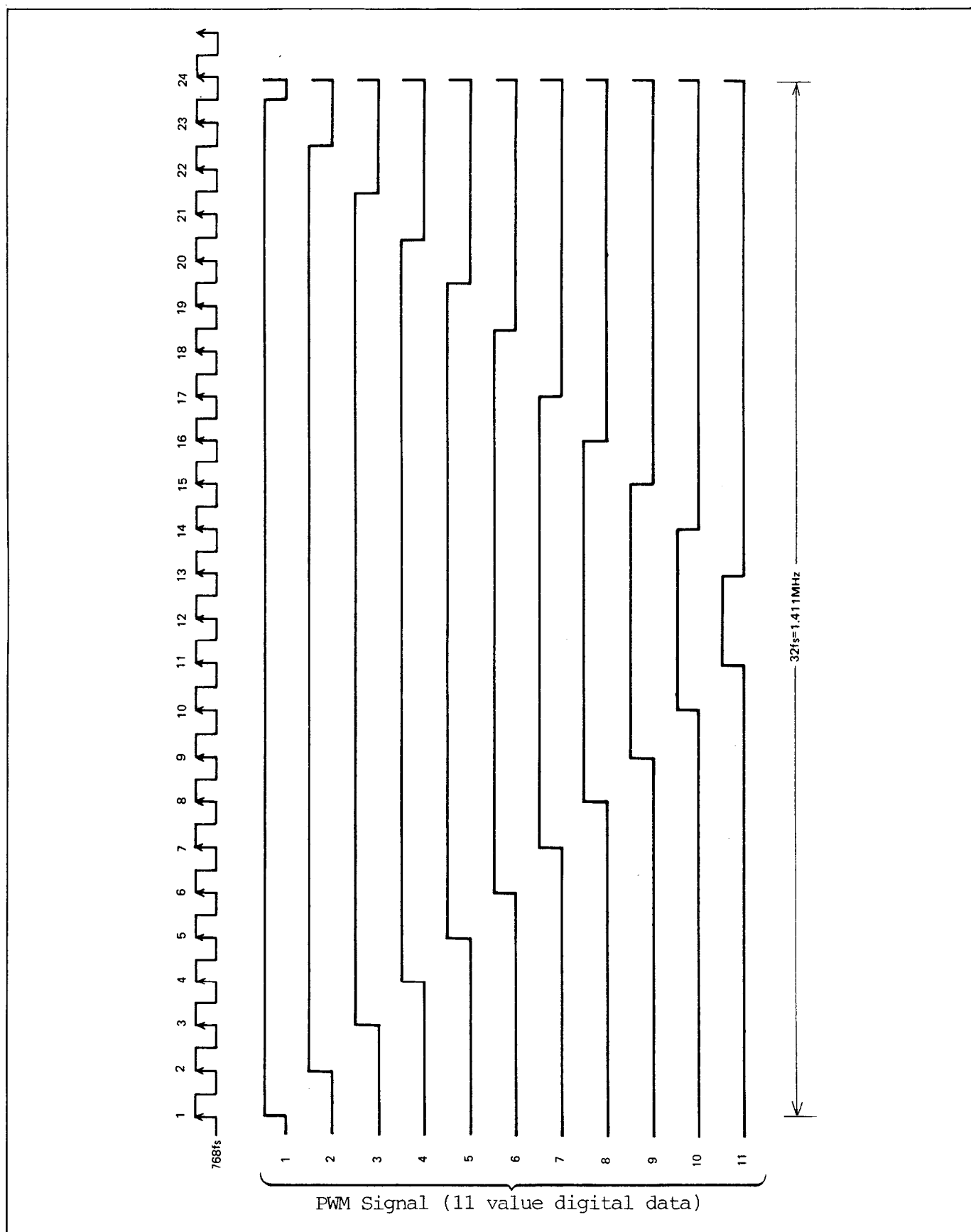


Figure 5-16 Output Data from MASH Part

PWM Block

The PWM block performs Pulse-Width Modulation on the 11 value digital data from the MASH block to convert into analog signals with 11 different pulse widths which are precision controlled by a crystal controlled oscillator.

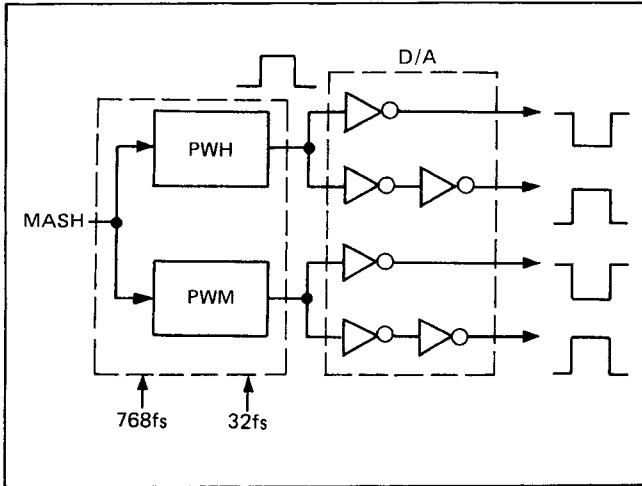


Figure 5-17 PWM Block Diagram

With the PAM, PWM, and PPM techniques, the signal is sampled along the time axis but is not quantized along the amplitude axis. All of these techniques preserve the amplitude information exactly as it was the original analog signal. (see the following figure 5-18)

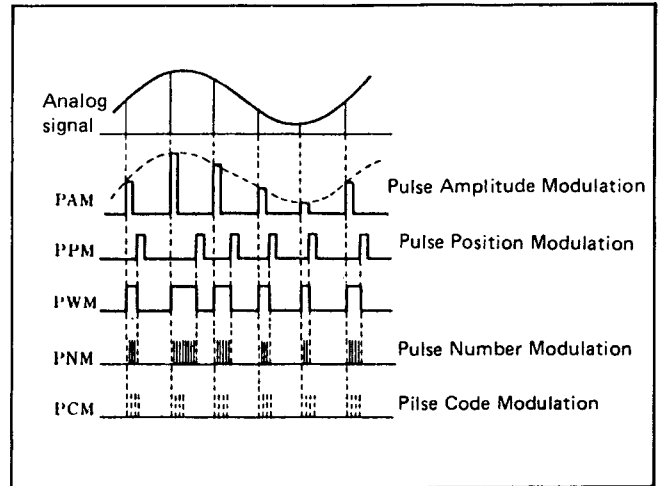
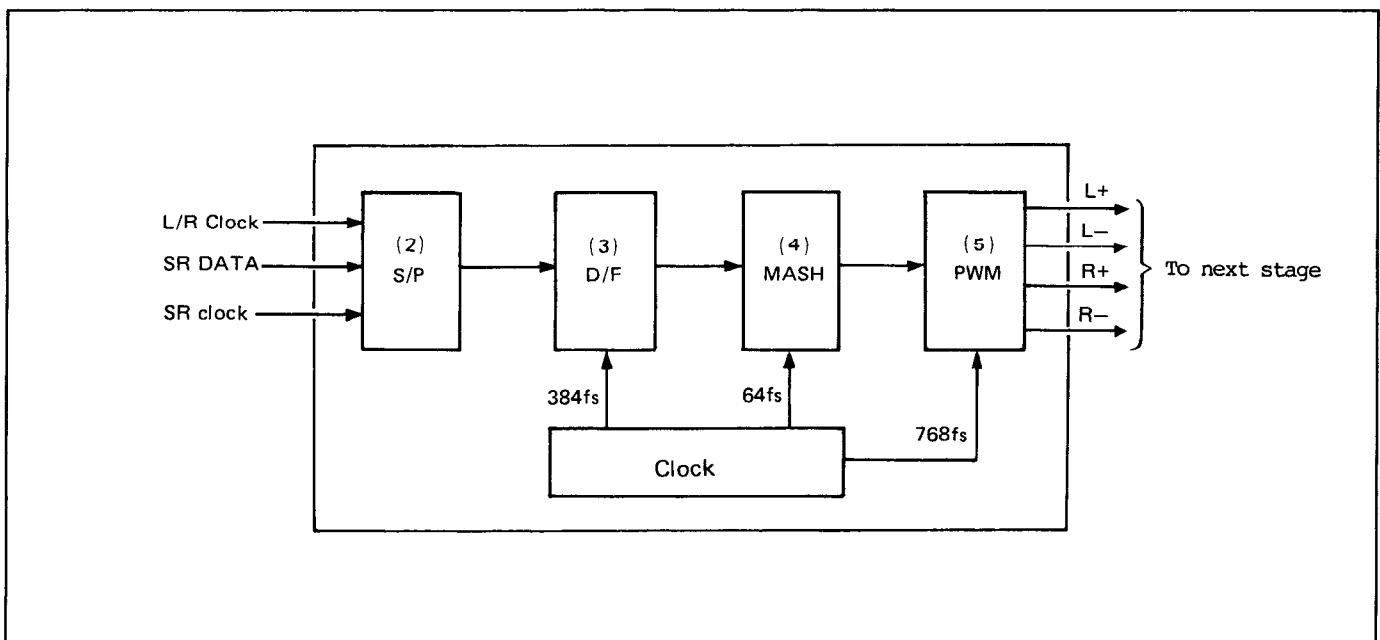


Figure 5-18

There are several different pulse modulation techniques available to convert analog signals into pulse signals.

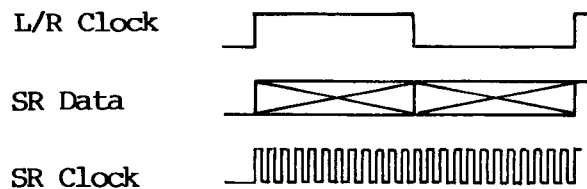
The pulse width modulation (PWM) samples the input signal along the time axis and yields pulses with pulse widths that correspond to sampled signal amplitudes. The resulting pulse signal contains frequency components similar to those obtained with the PAM technique.

Overall MN6623 Operations



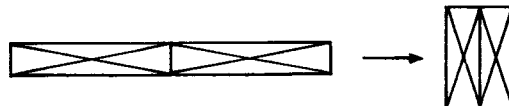
MN6623 Block Diagram

1. Input Signal



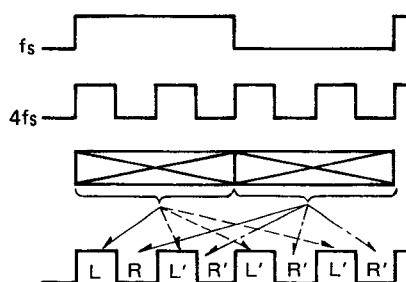
2. S/P Conversion

Converts serial L/R data into parallel L/R data



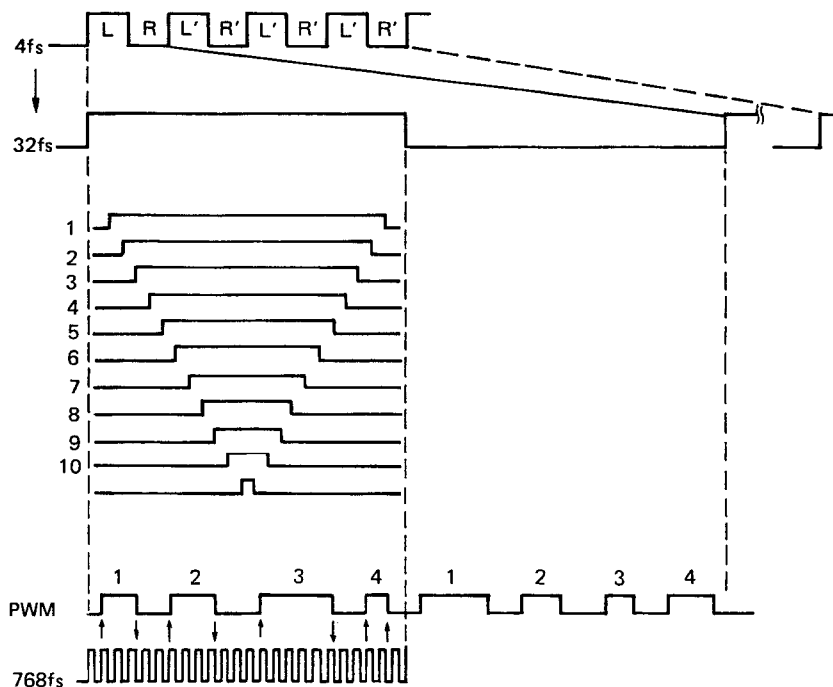
3. D/F Processing

4-hold oversampling



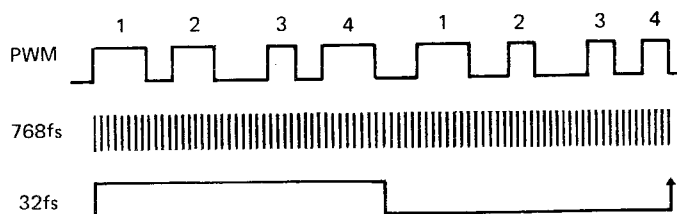
4. MASH Processing

Performs MASH processing on the L channel data from the first stage to determine 4 levels out of 11 levels and transfers them to the PWM block. Does the same processing for the R channel data.



5. PWM Processing

Pulse Width Modulates the 4 level data from the MASH block at a rate of 768fs, then separates the L and R channel data with 32fs clock. The L and R channel outputs of the PWM have the same phase, and are applied to the next stage at the rising edge of the 32fs clock.



10) Pin Description of MN6623A

● IC508: MN6623A (Digital Filter and D/A Converter)

Pin No.	Mark	I/O Division	Function
1	MLD	I	Command load input (load: L)
2	RSTB	I	Reset signal input (reset at "L")
3	IE	I	I ² S select terminal ("H"=I ² S)
4	TP1	—	TEST terminal (Not connected)
5	TP2	—	
6	TEST1	I	TEST terminal 1 (connected to GND)
7	TEST2	I	TEST terminal 2 (connected to GND)
8	NC	—	Not connected
9	NC	—	Not connected
10	AVDD4	I	Power supply (connected to +5V)
11	OUTR(-)	O	Rch data output, (-) terminal
12	AVSS4	I	GND terminal
13	AVSS3	I	GND terminal
14	OUTR(+)	O	Rch data output, (+) terminal
15	AVDD3	I	Power supply (connected to +5V)
16	NC	—	Not connected
17	AVDD2	I	Power supply (connected to +5V)
18	OUTL(-)	O	Lch data output, (-) terminal
19	AVSS2	I	GND terminal (analog system)
20	AVSS1	I	GND terminal (analog system)
21	OUTL(+)	O	Lch data output, (+) terminal

Pin No.	Mark	I/O Division	Function
22	AVDD1	I	Power supply (connected to +5V)
23	AVDD1	I	Power supply (connected to +5V)
24	DVSS1	I	GND terminal (digital system)
25	X2	O	Clock output
26	X1	I	Clock input
27	NC	—	Not connected
28	DVDD2	I	Power supply (connected to +5V)
29	DVSS2	I	GND terminal (digital system)
30	NSUB	I	Not used (connected to GND)
31	768fs	O	768 fs (Not used)
32	192fs	O	192 fs (Not used)
33	LRPOL	I	LR clock selector (Not used)
34	LR	I	L/R clock
35	SRCK	I	Serial data best clock
36	SRDATA	I	Serial data input (MSB first)
37	DVSS 3	I	GND terminal (digital system)
38	NC	—	Not connected
39	384 fs	O	384 fs (16,0344 MHz) output
40	PD	I	Power down terminal
41	MDATA	I	Command data input
42	MCLK	I	Command clock input

4. CIRCUIT OPERATIONS (TYPE C)

C4-1 AN8370S CIRCUIT OPERATION

The optical servo reads the signals recorded on a disk using the laser beam.

AN8370S is the main IC of the optical servo system consisting of laser APC, RF signal amplifier, focus servo (including auto focus), tracking servo (including kick), dropout detector, and the control logic that controls the optical servo in this IC.

A block diagram of AN8370S is shown in Fig. 4-1.

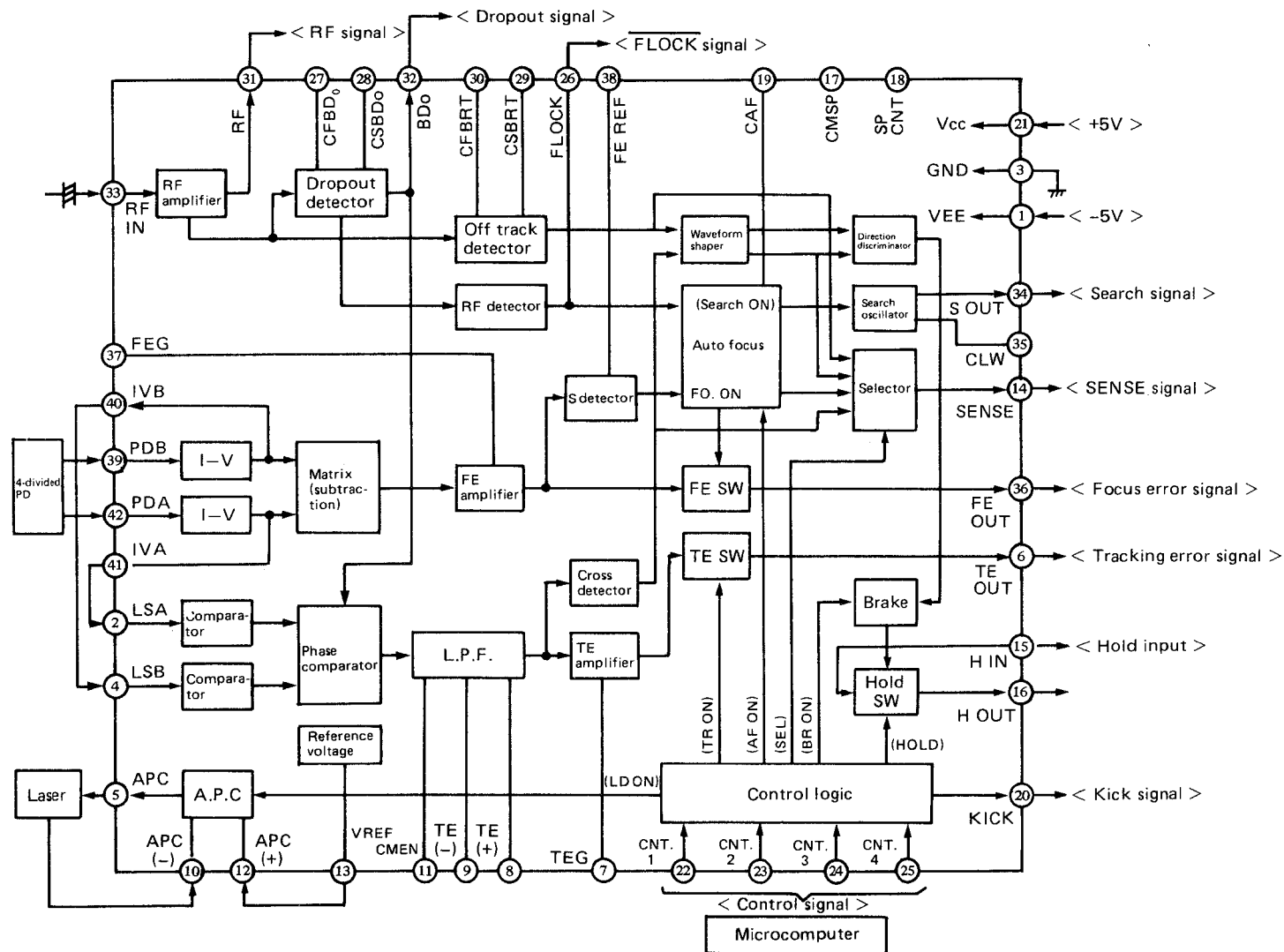


Fig. 4-1 AN8370S Block Diagram

C4-1-1 Control Logic Circuit

Control signals CNT1 to CNT4 are 4-bit parallel output signals from the microcomputer. The logic circuit of AN8370S decodes these control signals and outputs the required control signals to the optical servo system.

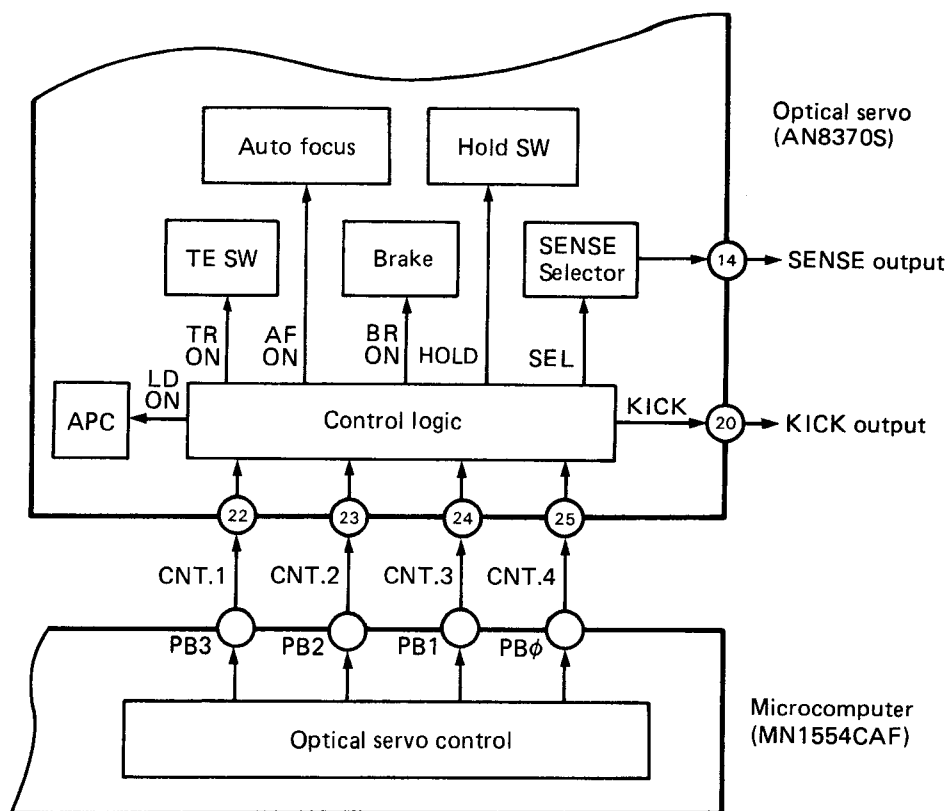


Fig. 4-2 Control Logic Circuit

(1) Control signals and their functions (See Fig. 4-2)

- 1) LD ON Laser ON command
 - Controls laser APC ON/OFF.
- 2) AF ON Auto focus command
 - Controls the focus servo loop to open or close, and the search OSC switch ON/OFF.
- 3) TR ON Tracking servo command
 - Controls the tracking servo loop switch.
- 4) HOLD Tracking hold command
 - Holds tracking error at preceding value during track kick.
- 5) BR ON Tracking brake command
 - After high-speed feed or kick, brakes the tracking actuator to absorb unnecessary vibration in a short time to facilitate tracking.

- 6) SEL SENSE signal select command
- Selects SENSE signal (that indicates optical servo status) to be sent from AN8370S to the microcomputer.
- SENSE signal changes depending on the system mode.
- 7) KICK Track kick command
- Controls track kick output signals.
- There are KICK FWD (outward) and KICK REV (inward), and three modes of 1 kick, 10 kicks, and 100 kicks according to kick distance.
- The logic circuit has $16 (2^4 = 16)$ decoding modes, of which eight are used as shown in Table 4-1.

(1 "H", 0 "L")

HEX No.	Mode	Logic input				Logic control signal output							Remarks
		CNT1	CNT2	CNT3	CNT4	LD ON	AF ON	TR ON	HOLD	BR ON	KICK	SENSE	
0	PLAY	0	0	0	0	ON	ON	ON	OFF	OFF	OFF	Off track	Play mode
3	BRAKE	0	0	1	1	ON	ON	ON	OFF	ON	OFF	Off track	Brake mode
4	KICK CROSS	0	1	0	0	ON	ON	OFF	ON	OFF	OFF	CROSS	Kick mode
5	KICK REV	0	1	0	1	ON	ON	OFF	ON	OFF	ON (REV)	Shaping CROSS	
6	KICK FWD	0	1	1	0	ON	ON	OFF	ON	OFF	ON (FWD)	Shaping CROSS	
7	ACCESS	0	1	1	1	ON	ON	OFF	OFF	OFF	OFF	Shaping CROSS	Coarse feed
8	DISK DETECT	1	0	0	0	ON	ON	OFF	OFF	OFF	OFF	S latch	Disk detection
F	STANDBY	1	1	1	1	OFF	OFF	OFF	OFF	OFF	OFF	Shaping CROSS	Forced return

Table 4-1 Control Logic Table

LD = LASER DEVICE
 AF = AUTO FOCUS
 TR = TRACKING SERVO
 HOLD = TRACKING HOLD
 BR ~~OK~~ = TRACKING BRAKE
 SEL = SENSE
 KICK = TRACK KICK

C4-1-2 Laser APC Circuit *Auto Laser power control output.*

Fig. 4-3 shows the laser APC circuit.

When a laser ON mode command (H, L, L, L) is received, pin 5 goes high to turn the laser on.

A loop circuit is formed by Q and D (the laser paired with a PIN diode) and the APC in the IC, and laser beam intensity is detected by the PIN diode in APC operation.

If laser beam intensity increases, the PIN diode's photo-electric current increases so that the pin to voltage rises (Pin 12 voltage is the reference voltage) to decrease laser beam intensity.

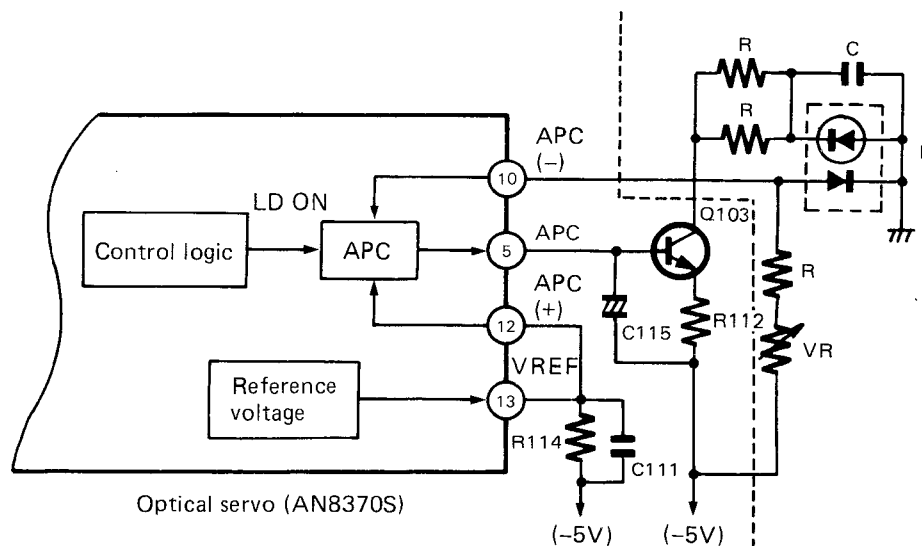


Fig. 4-3 Laser APC Circuit

C4-1-3 RF Amplifier and Dropout Detector Circuits

(1) RF amplifier circuit

RF signal is the total sum of all the 6-segment photo-detector ($A1 + A2 + A3 + A4 + B1 + B2$) of the optical pickup. The resulting RF signal is applied to pin 33, via Q, (Q is located in the optical pickup.) where it is amplified by an RF amplifier having gain of approx 34 dB.

The output of the RF amplifier appears at pin 31, and couples to pin 15 of IC AN8371S.

(2) Dropout detector circuit

The dropout detector detects signal dropouts in the RF signal, stabilizes the tracking servo against dropouts, and provides the defection signal to pin 32.

The detection dropout process is shown in Fig. 6-5. The level goes high if a dropout is detected.

C4-1-4 Auto Focus Circuit and Focus Error Circuit

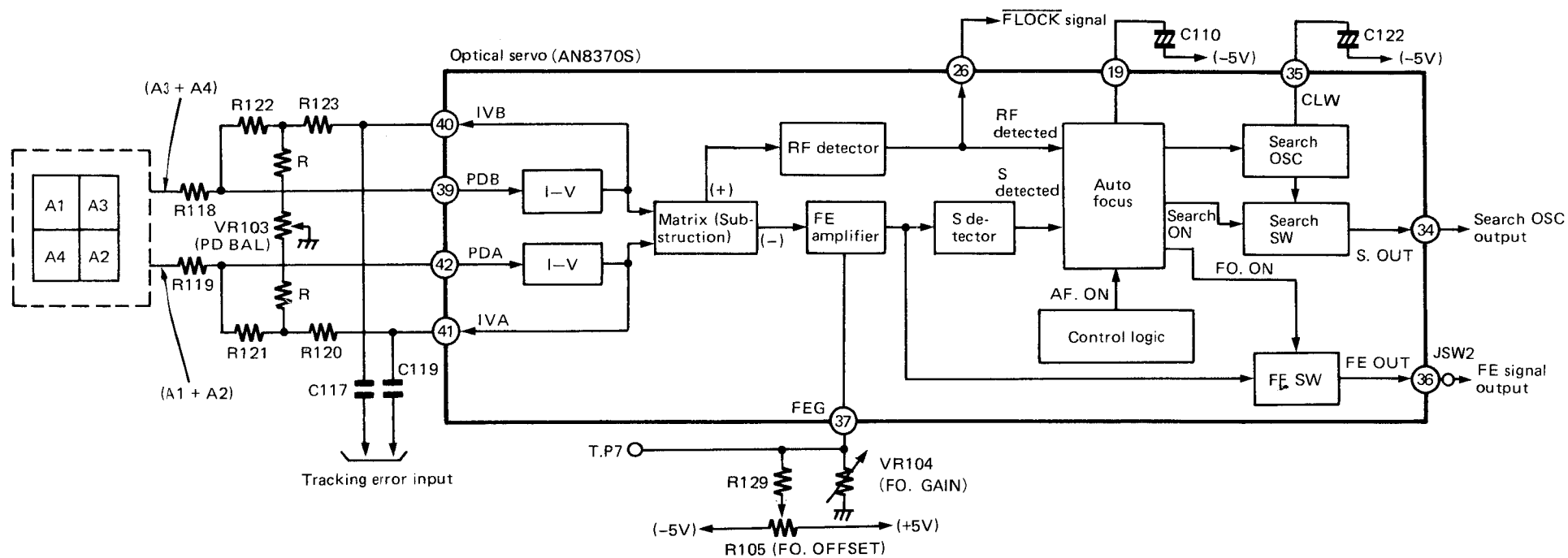


Fig. 4-7 Focus Error Circuit and Auto Focus Circuit

(1) Auto focus circuit

The auto focus is operated by the AF ON (auto focus command) signal output from the control logic circuit. It has two inputs, RF detection and S-curve detection, and regulates search SW and FE SW.

FE = FOCUS SIGNAL ERROR

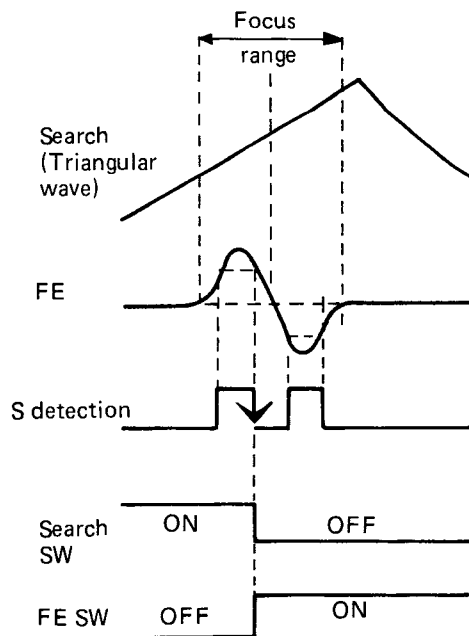


Fig. 4-8 Fo ON Timing

1) Disk detection

AF ON turns search SW on and FE SW off.

In this state, the search signal (triangular wave) drives the focus coil (raising or lowering the lens).

When the distance between the disk and the lens nears the focus length by focus search, the FE signal develops an S-curve, which is detected to turn search SW off and FE SW on to close the focus servo loop as shown in Fig. 4-8.

SW = SWITCH

Identifica la funzione
di "Search SW" = Ricerca
focus on e di loop
FE SW (FOCUS ERROR
CONTROL)

2) Re-focusing

The auto focus circuit checks the servo status by RF detection when the focus servo loop is closed. When RF detection is no longer possible due to defocusing, search SW is turned ON to start the focus search. The search signal is input to the microcomputer.

Whether the auto focus is working can be checked the voltage at pin 26 (F LOCK). (Low when the focus is correct.)

FIG. 4-7

(2) Focus error circuit

The diagonal sums $(A1 + A2)$ and $(A3 + A4)$ of the 4-segment photo-detector are input to pins 39 and 42 of IC101. The I/V convertor converts the photo-detector output currents to voltages. An external resistor is used for gain setting. VR101 is a photo-detector balance (best eye) adjuster, for setting off the sensitivity difference of the photo-detector.

The difference between the two converted signals $(A1 + A2) - (A3 + A4)$ is a focus error signal (FE). The FE signal applied to pin 36 (FE out) of IC101 via the FE amplifier and FE SW.

VR104 is the focus gain (FO GAIN) control, and VR105 the focus offset (FO OFFSET) control.

C4-1-5 Tracking Error Circuit and KICK, HOLD Circuits

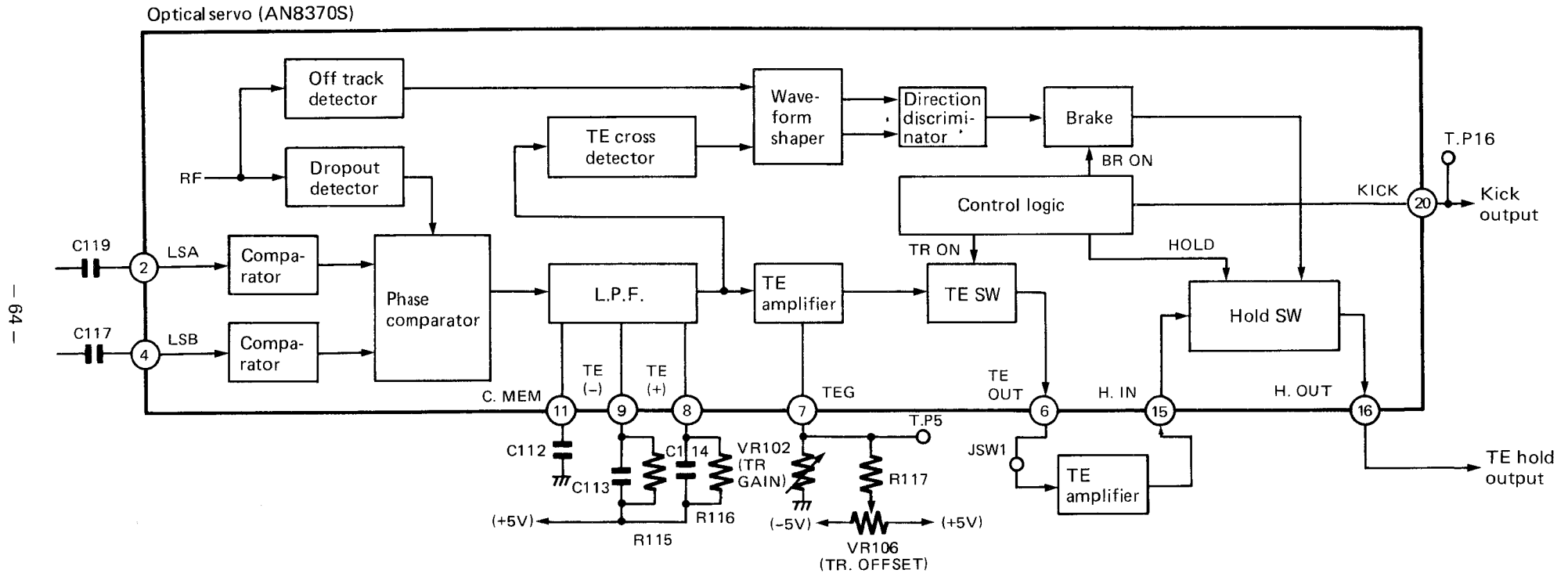


Fig. 4-9 Tracking Error Circuit and Kick, Hold Circuits

(1) Tracking error circuit

Tracking error circuit inputs the I-V converted signals to pin 2 and 4 via C119 and C117. The comparators shape the waveforms.

In the tracking error process of the single beam type, the phase difference between the two signals is detected and a pulse corresponding to the phase difference is generated as described in section 1-4. Deviating direction is judged from the polarity of the pulse. The generated pulse is smoothed by the LPF into an analog tracking error (TE) signal. This is how the phase comparator and LPF operate.

Then, the TE signal is output from pin 6 (TE OUT) via the TE amplifier and TE SW.

VR102 is the tracking gain (TR GAIN) control, and VR106 the tracking offset (TR OFFSET) control.

(2) Kick and hold circuits

Kick means jumping the beam from one track to another during a pause or manual search. The kick timing is set by a control signal from the microcomputer, and a kick signal is appeared pin 20. The waveform of each kick signal is shown in Fig. 4-10.

The hold circuit holds the preceding TE signal during kick made. TE SW is turned off during kick mode. Signal wave forms in pause kick are shown in Fig. 4-11.

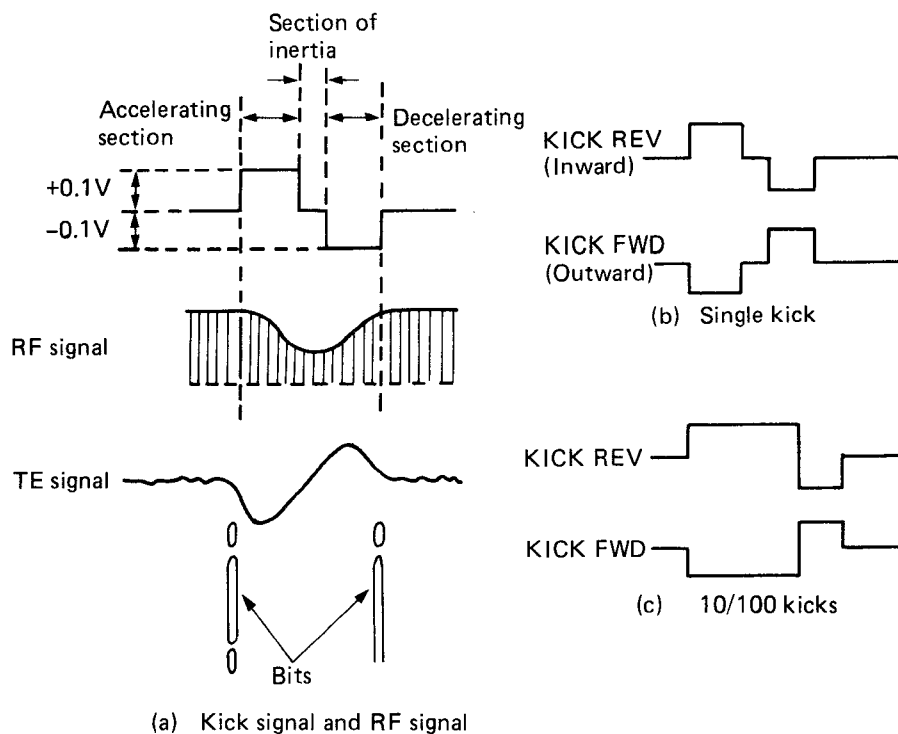


Fig. 4-10 Kick Signal Waveforms

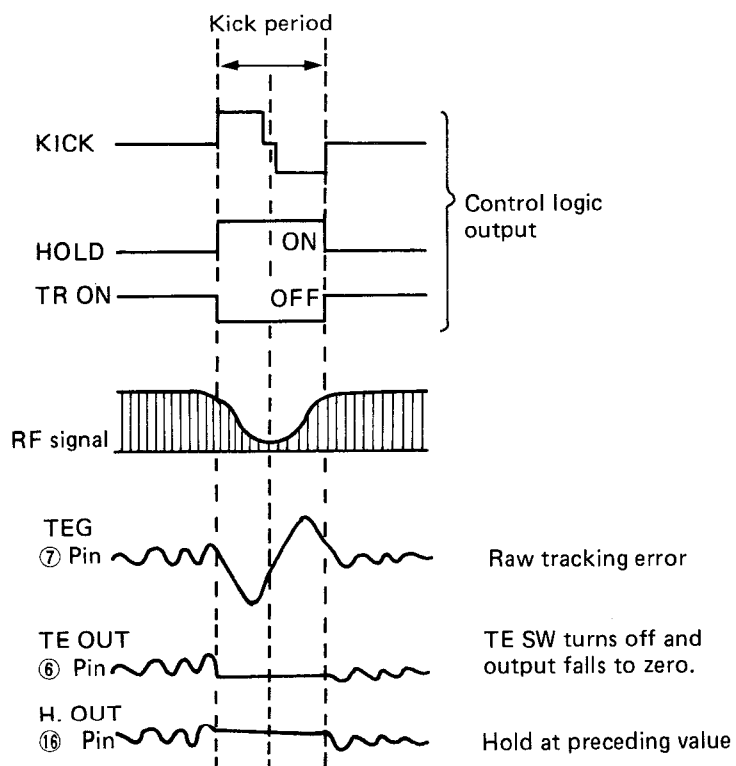


Fig. 4-11 Pause Kick

(3) Brake circuit

The brake circuit brakes the tracking actuator to quickly retract the tracking servo after rapid feed or kick (10/100). The process is shown in Fig. 4-12.

The brake circuit applies a single limiter (indicated by the dotted line in the figure) to the TE signal to absorb the energy after kick.

The figure shows the brake in the KICK REV mode. In the KICK FWD mode, the brake is applied with the single limiter inverted in polarity. Therefore, the brake circuit requires a kick direction discriminating signal, which is generated from a tracking cross signal and off track signal as shown in Fig. 4-13.

The cross signal phase leads in the KICK FWD mode and lags in the KICK REV mode. The brake direction is determined by detecting the above phase difference.

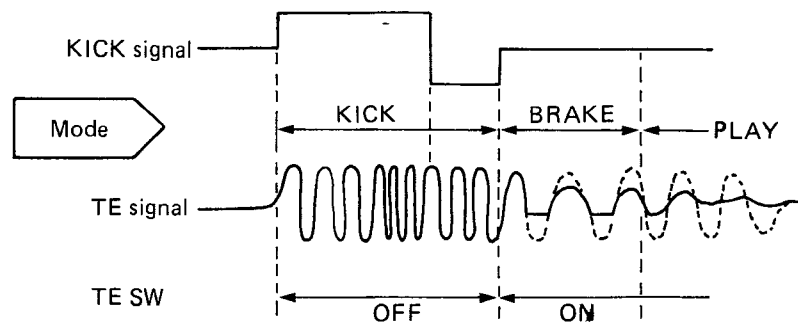


Fig. 4-12 Brake Timing

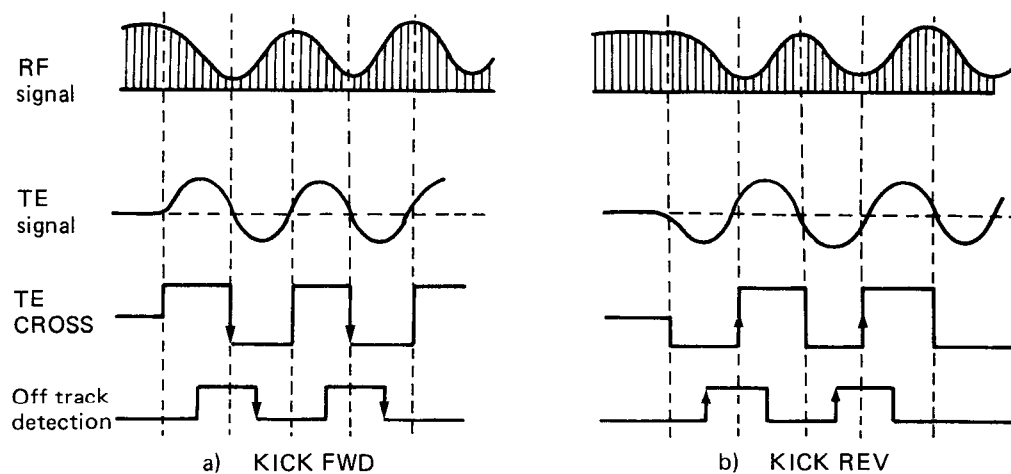


Fig. 4-13 Kick Direction Discrimination

C4-1-6 Focus Tracking Drive Circuit

Both servos have the same drive circuit except that the focus servo has a search circuit and that the tracking servo has a kick hold circuit.

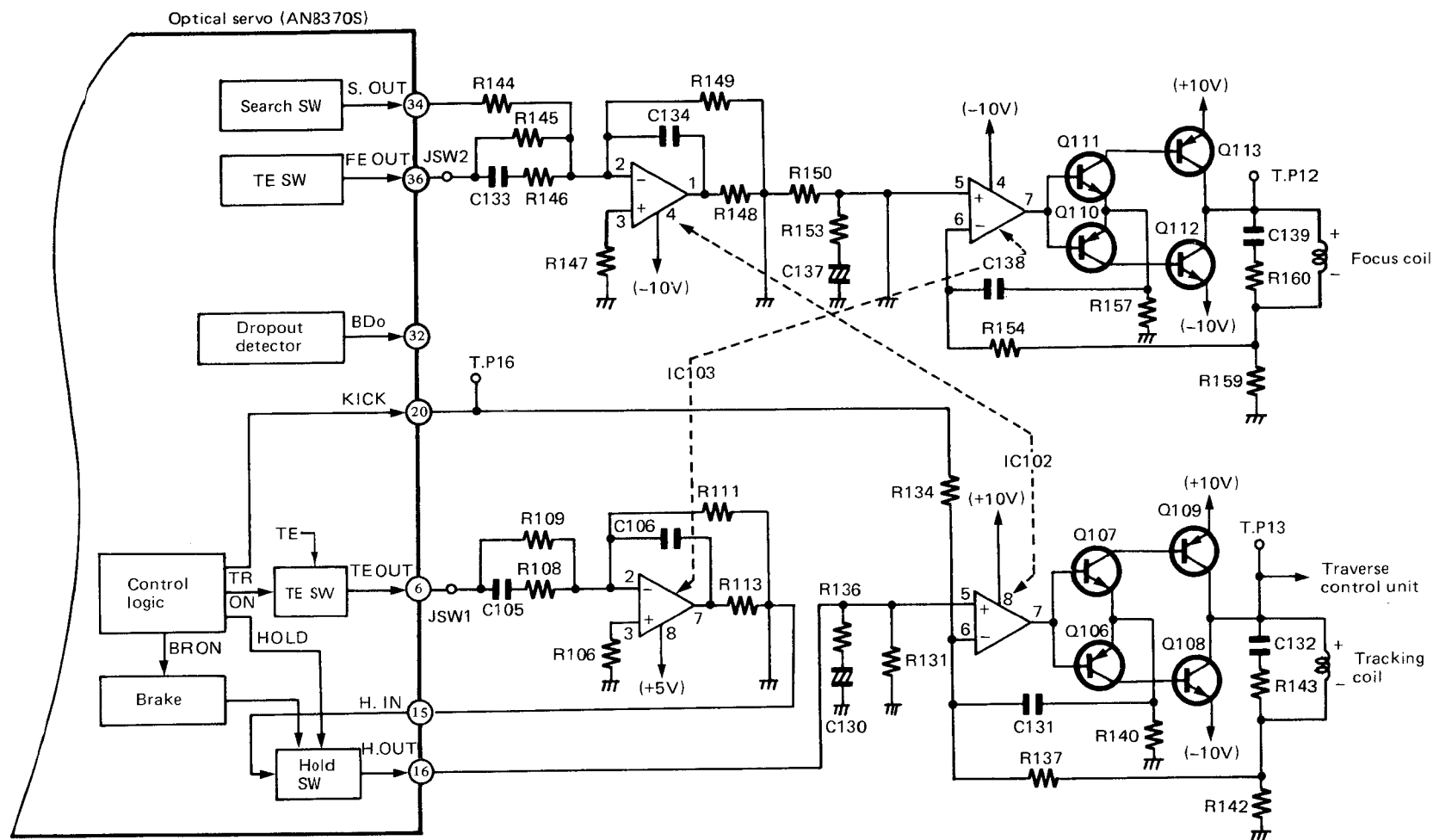
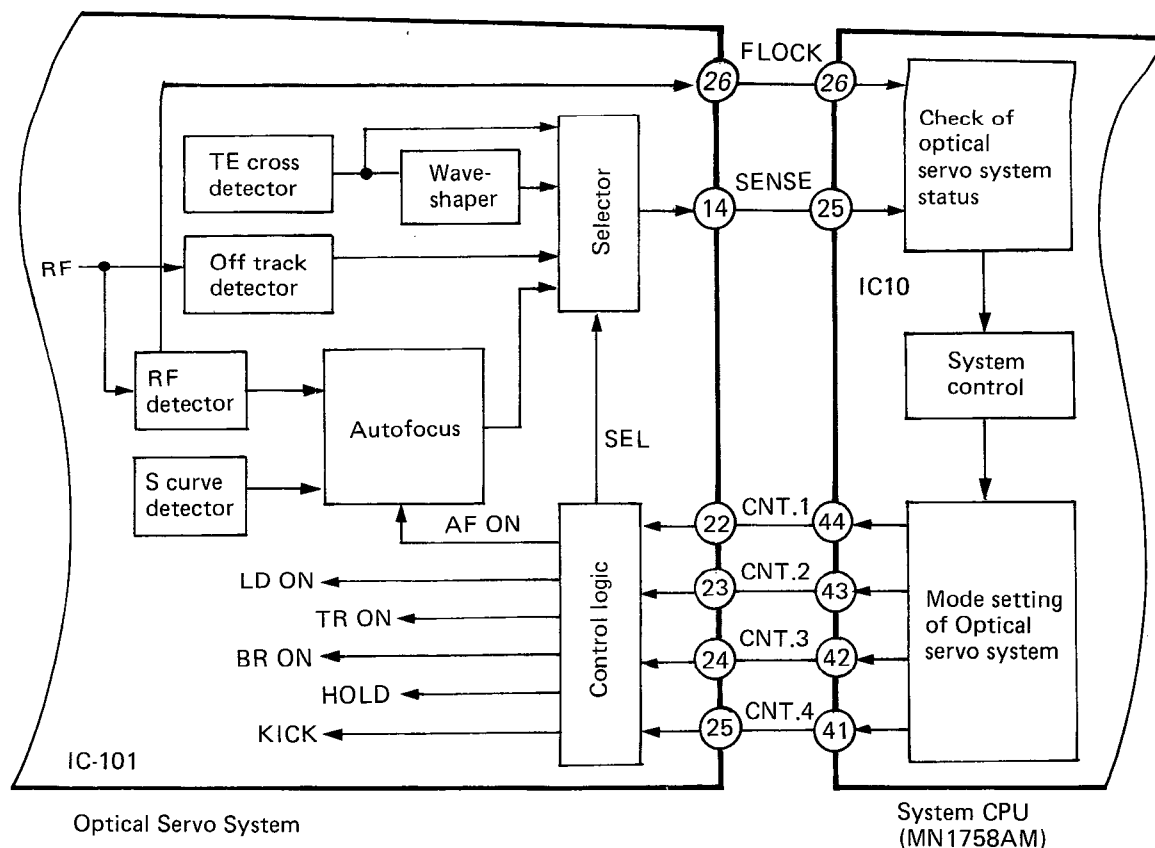


Fig. 4-14 Focus Tracking Drive Circuit

Relation with Servo IC and System CPU

The figure 4-15 shows the interconnection diagram of servo IC-101 (AN8370S) and system IC10. There are for buses for carrying signals from the system IC to AN8370S and two buses for carrying signal from AN8370S to the system IC.



P.49A Fig. 4-15 Connection Diagram of System CPU and Optical Servo System

- (a) Output signal from system: — Control logic signals (CNT1 — CNT4)
 (b) Input signal from system: — RF signal (FLOCK) Active "L"

Servo system status signal (SENSE) . . .

Type of SENSE signal:

- (1) Off-track signal
- (2) Tracking cross signal
- (3) Waveshaped tracking cross signal
- (4) Focus S-curve detection signal.

C4-2 TRAVERSE SERVO CIRCUIT OPERATION

The traverse servo detects the inclination of the objective lens and drives the entire optical pickup so that keeps the objective lens within the range of tracking servo.

The traverse servo circuit is shown in Fig. 4-1. It consists of a traverse detector circuit, mode select circuit, and traverse drive circuit.

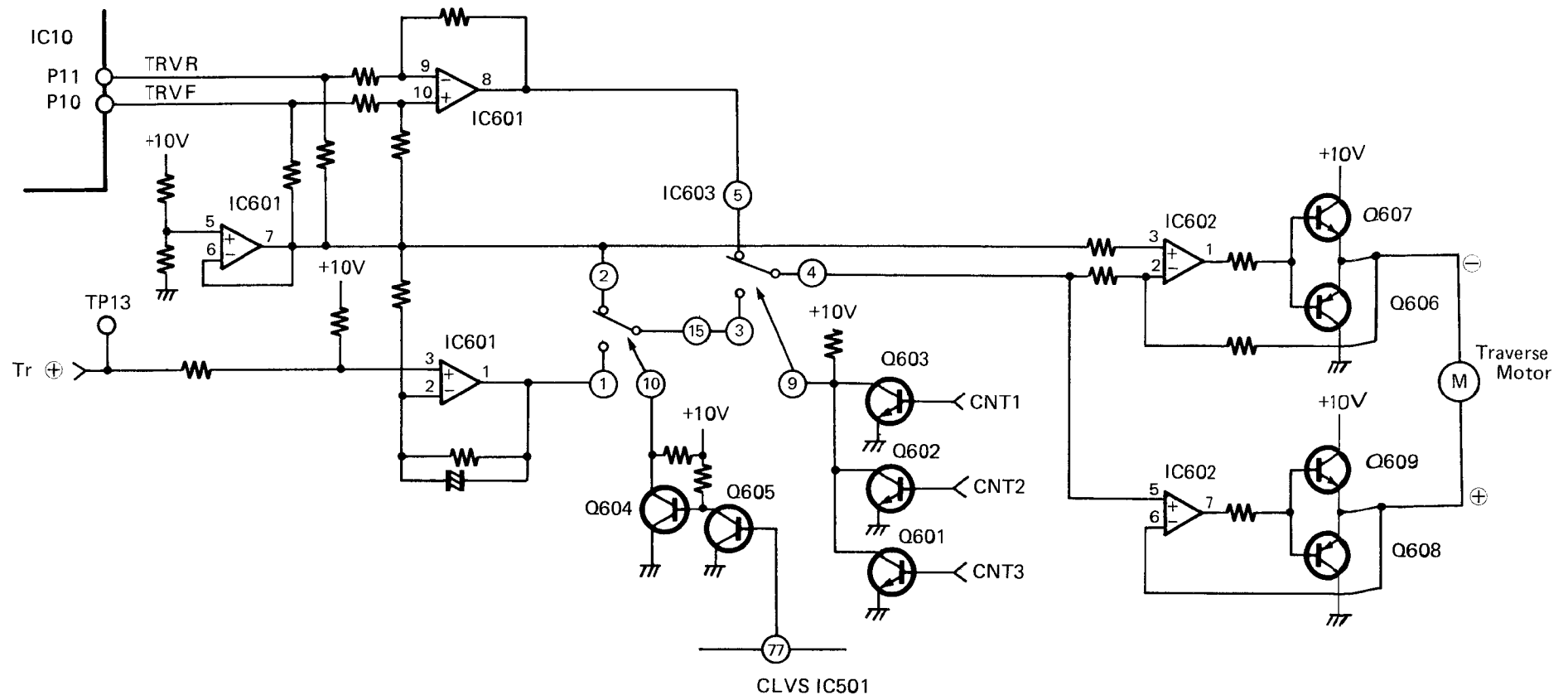


Fig. 4-1 Traverse Servo Circuit

C4-2-1 Traverse Detector Circuit

The tracking error signal is a signal of the positional deviation component of the disk as far as one turn of the disk is concerned. In a long period, however, DC voltage (servo error voltage) is generated as shown in Fig. 4-2. This DC voltage drives the traverse motor.

The traverse detector circuit in here range from the tracking coil output to the lowpass filter (pin 1 of IC601) output. Pin 8 of IC601 outputs forced drive signals, that is, forced forward (H) and reverse (L), by microcomputer commands.

Pin 7 of IC601 is set to a reference voltage of +5V.

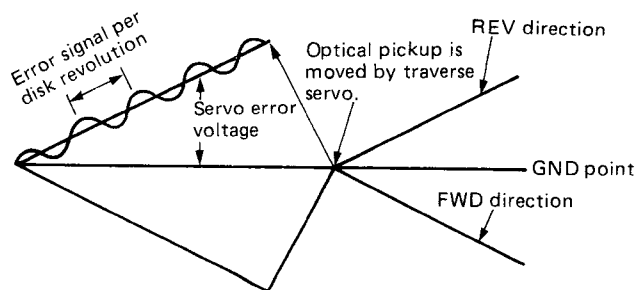


Fig. 4-2 Traverse Servo Voltage

C4-2-2 Mode Select Circuit

The multiplexer (IC603) is used for mode selection.

- 1) Servo error voltage selection (Play mode)

The output from pin 1, IC601 is applied to the drive circuit when pin 10 and pin 9 of IC603 are high.

- 2) Brake selection by reference voltage (KICK mode)

The output from pin 7, IC601 is applied to the drive circuit when pin 10 of IC603 is low and pin 9 of IC603 is high.

- 3) Forced forward/reverse selection (SKIP mode)

The output from pin 8, IC601 is applied to the drive circuit when pin 9 of IC603 is high.

The input to pin 9 of IC603 is controlled by CNT 1 to CNT3 of the microcomputer via the NOR circuit. Thus, pin 9 goes high when all of CNT1 to CNT3 are low (during disk play), or goes low when one of CNT1 to CNT3 is high (other than during disk play).

The input to pin 10 of IC603 is controlled by CLVS signal from pin 77 of LSI (IC501). Thus, pin 10 goes high when T11 is OK.

C4-2-3 Traverse Drive Circuit

The traverse drive circuit consists IC602, Q606 to Q609 to drive the traverse motor using the reference voltage applied to pin 3 of IC602 and the polarity corresponding to the voltage difference applied to pin 2. Q606 to Q609 comprise a dual direction drive amp. Q607 and Q609 are NPN and Q607 and Q608 are PNP. By controlling the polarity of the voltage applied to the Bases of ALL TRANSISTORS, it will determine whether the motor is supplied with a (-) or (+) voltage thus determining the direction of motor movement.

C4-3 AN8371S CIRCUIT OPERATION

AN8371S receives the RF signal from the optical servo IC (AN8370S), and generates the EFM signal and PCK signal for inputting to the digital signal processing LSI (MN6617).

An internal block diagram is shown in Fig. 4-1. AN8371S consists of an AGC circuit, data slice circuit, PLL circuit, and dropout detector circuit.

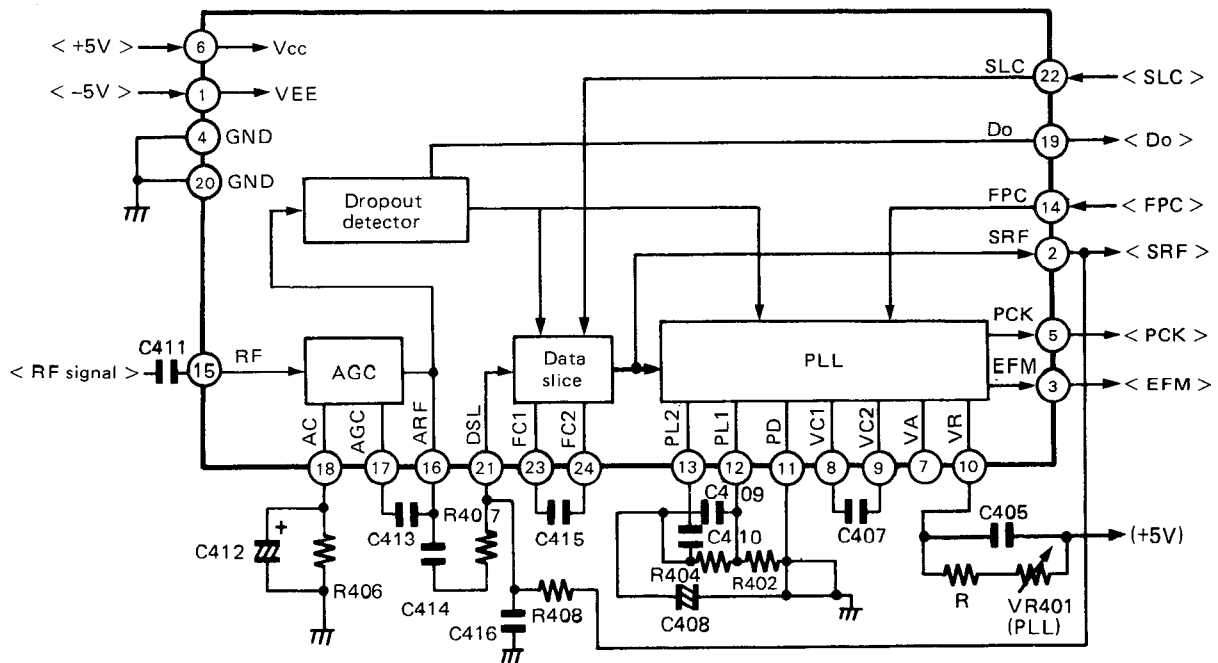


Fig. 4-1 AN8371S Block Diagram

C4-3-1 EFM Signal

The RF signal is an analog continuous signal as shown in Fig. 4-2.

Therefore, it cannot be processed as a digital signal unless it is shaped into a digital signal. This waveform shaping is performed by the data slice circuit, whose output signal is EFM signal.

C4-3-2 PCK Signal

The PCK signal is a clock signal synchronized with EFM signal.

The EFM signal includes a jitter component until it is written into the memory. Therefore, a clock signal synchronized with EFM signal is necessary in order for the digital signal processing circuit to accurately receive data.

The PCK signal is generated from the PLL circuit with an EFM signal input. Fig. 4-3 shows a timing chart of EFM signal and PCK signal.

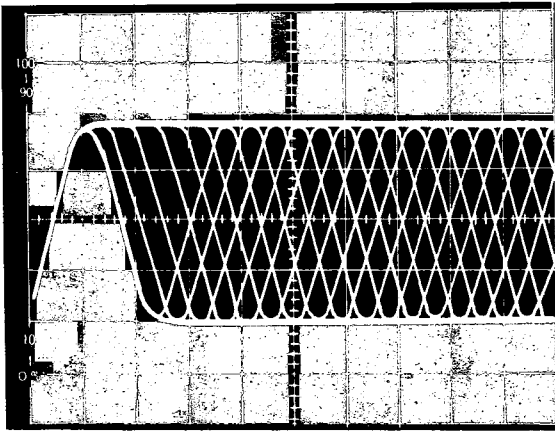


Fig. 4-2 RF Signal

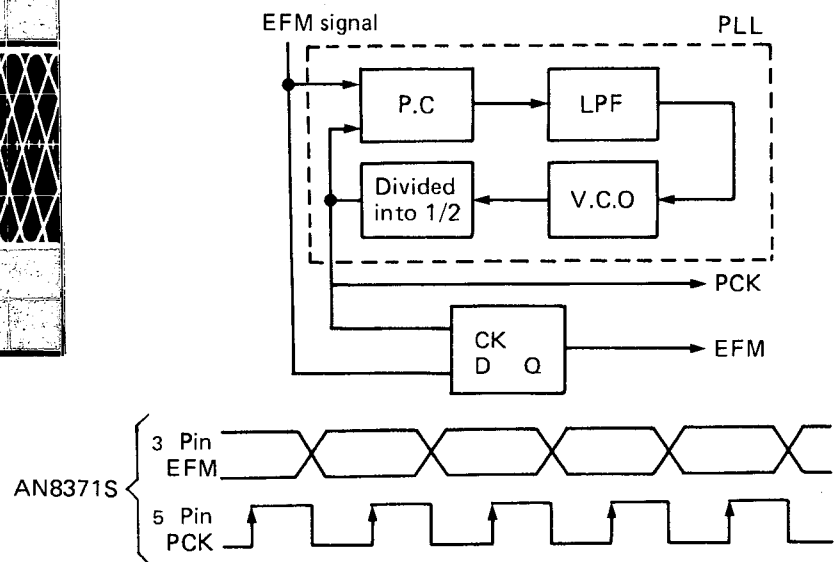


Fig. 4-3 EFM Signal and PCK Signal

C4-3-3 Other Signals

IC AN8371S outputs SRF signal and D0 signal besides the EFM and PCK signals. FPC and SLC signals are input signals to IC AN8371S.

The SRF signal is generated by data-slicing RF signal. The D0 signal is the dropout detecting signal in the RF signal (high at dropout). Both are input to the CLV control LSI (MN6617).

The FPC signal is the VCO control signal of the PLL circuit that is fed back from the LSI. The SLC signal is the slice level control signal from the microcomputer (MN1554CAC).

Their functions are described in Section 9-3.

C4-4 MN6617 CIRCUIT OPERATION

Prior to the circuit operation description, the signal processing of the recording system will be outlined.

C4-4-1 Signal Processing in Recording System

(1) Encoder

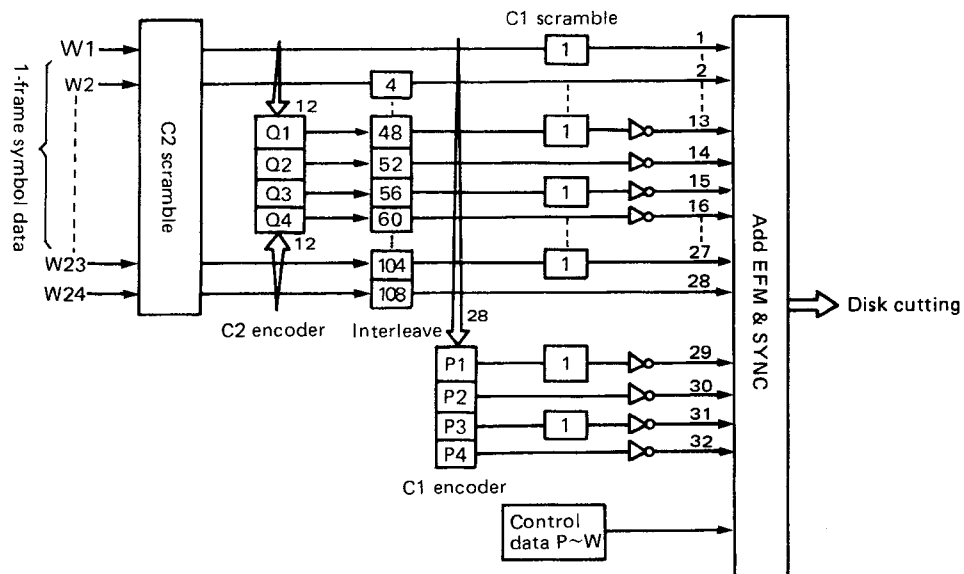


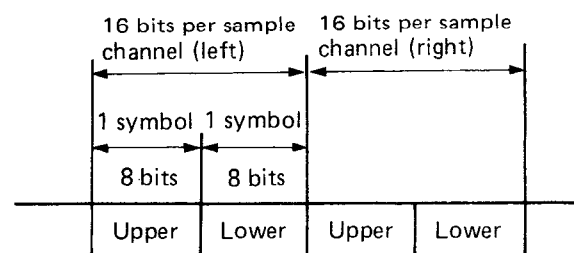
Fig. 4-1 Signal Processing in Recording System

Fig. 4-1 shows a block diagram of CD recording system signal processing.

A brief description of CD recording system signal processing is given in the following.

- 1) The CD sampling frequency is 44.1 kHz, and quantizing involves A—D conversion to 16 bits per sample.

The 16 bits are divided into upper 8 bits and lower 8 bits.



* 16 bits per sample are called a data word, and 8 bits per symbol are called symbol data. In other words, two symbols constitutes one data word.

Fig. 4-2 Symbol Data and Data Word

- 2) 12 samples of data (six each for the right and left) are taken as a frame for signal processing. In other words, 24-symbol data constitutes a frame. W1 to W24 shown in Fig. 4-3 correspond to it.
- 3) Data W1 to W24 are first scrambled by C2. Scramble may be considered to be a small interleave. In C2 scramble, the data are delayed over two frames alternately in terms of sample data.
- 4) After C2 scramble, the data enter the C2 encoder, which generates 4-symbol parity signals Q1 to Q4 from the 24-symbol data using reed solomon code, and arranges them in the frame center.

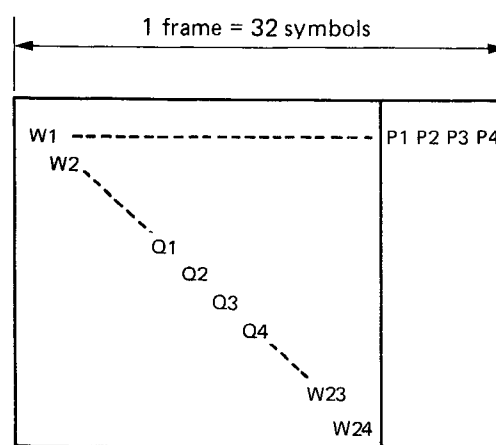
The symbol data count in one frame is 28 in total.

- 5) Next is the interleave.

As shown in the figure, the interleave delays the 28-symbol data, including the Q parity signals, over the range of 4 frames minimum to 108 frames maximum.

- 6) The C1 encoder generates 4-symbol parity signals P1 to P4 from the 28-symbol data, which have been rearranged by interleaving, using reed solomon code, and arranges them at the frame end.

Here, the symbol data count in a frame is 32 in total. (See Fig. 4-3.)



* Symbol parties P1 to P4 are added to the 28 symbols arranged horizontal in a frame, and four symbol parities are added to the 24 symbols arranged diagonally.

Fig. 4-3 CIRC Code Composition

- 7) The C1 scramble delays only the even-numbered symbols of the 32-symbol data in a frame to the next frame.
- 8) Control data P to W are otherwise called subcodes or users bits, containing time information for random access or display and consisting of a single symbol (8 bits).
These signals are arranged at the head of a frame.

Entry of these control signals, increases the symbol data count from 1 frame to 33 in total.

- 9) Finally, the 1-frame data are subjected to EFM-modulated and are recorded on the disk with a frame synchronizing signal is added.

EFM modulation is described in detail in the next paragraph.

All these are intended for improving the correction ability.

This system is called CIRK (Cross Interleave Reed Solomon Code).

(2) EFM modulation

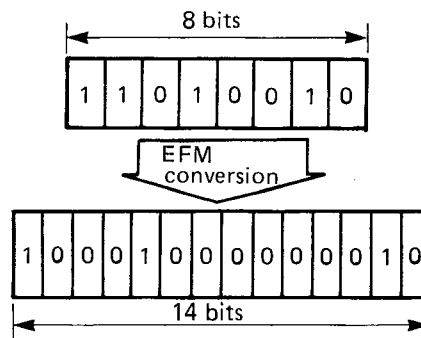


Fig. 4-4 EFM Modulation

EFM is a new modulation system for CD, and stands for Eight to Fourteenth Modulation meaning modulating 8 symbol data bits to 14 channel bits.

EFM is necessary because the data bit arrangement of 1's and 0's cannot be defined if symbol data remains as such.

If 1's or 0's continue so long, data will be a direct current so that self-clocks cannot be generated by the tracking servo (using pits) and reproducing system.

Conversely, if 1's and 0's are frequently reversed, it will be necessary to reduce the laser beam to a very small diameter.

Therefore, signals to be recorded must be such that 1's and 0's are reversed at appropriate intervals and that the DC component is as small as possible.

8 bits has 256 combinations (2^8), while 14 bits has 16,384 combinations (2^{14}).

EFM selects bit patterns, in which 2 to 10 0's are present between two 1's, from among 2^{14} combinations for correspondence to 2^8 data patterns.

Therefore, signals to be recorded have definitions with 3T to 11T (T meaning 1, 1 clock after EFM = 4.32 MHz).

Fig. 9-5 shows the process of up to the recording signal format.

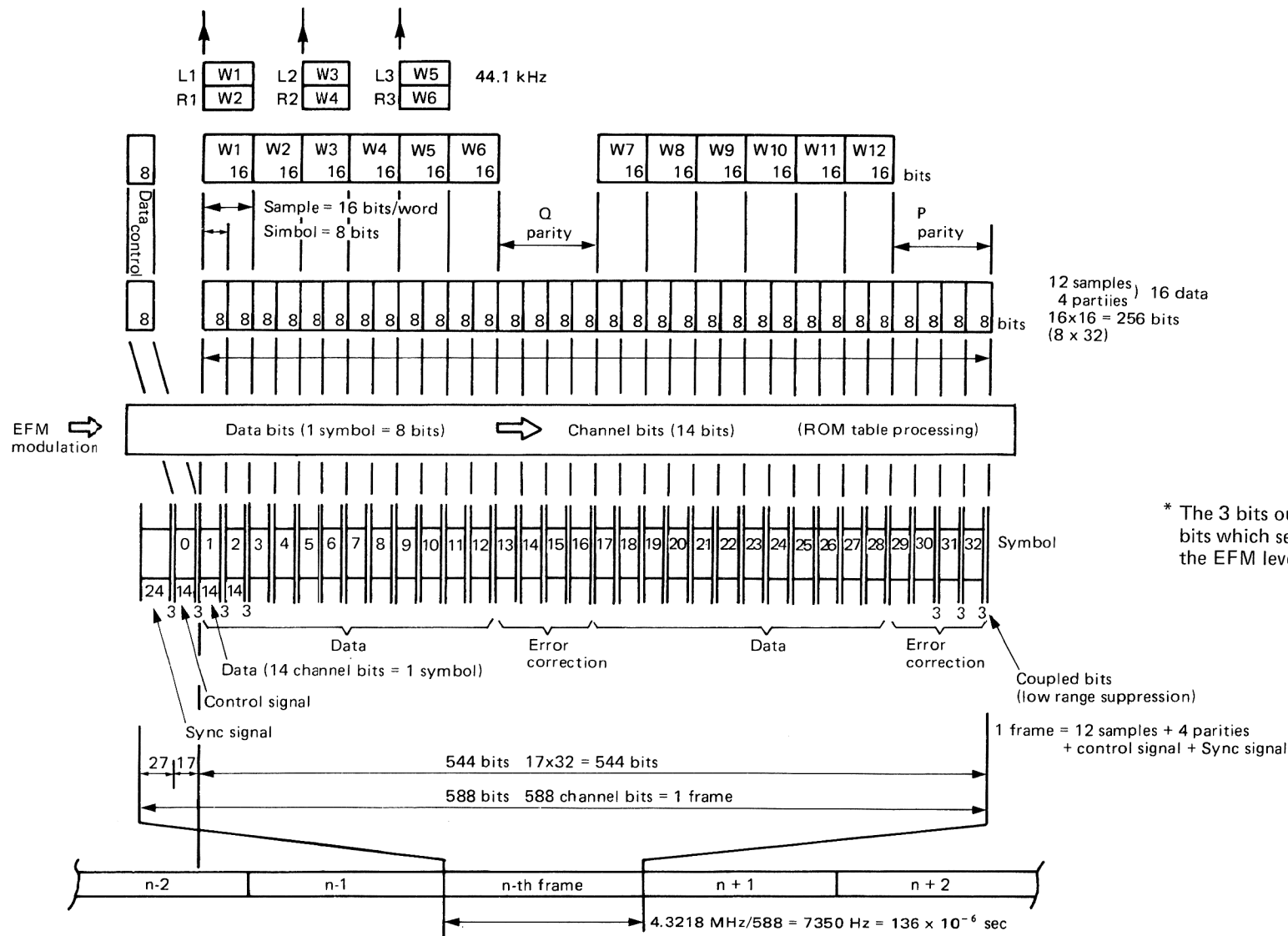


Fig. 4-5 CD Signal Format

(3) Control data

Control data is recorded in one symbol (8 bits) in one frame.

The 8 bits are named P, Q, R, S, T, U, V, and W in this order. Actually, however, only bits P and Q of the 8 bits are used.

Subcode channel No.	P	Q	R	S	T	U	V	W
ϕ	Sync signal							
1	Sync signal							
2								
			Unused					
96								
97								

1/75 second
(13.3 ms)

Fig. 4-6 Control Data Contents

Control P signal

This signal discriminates program signals (music signals) or intervals between tunes on the lead-in track, and indicates the program end on the lead-out track using a 2 Hz repetitive pulse.

Control Q signal

The contents of tunes recorded on the lead-in track, program track, and lead-out track of the disk are recorded for use in various kinds of display and control.

Lead-out track

Music No.	Point (Tune No.)	Tune time (Minutes, seconds, frame)	Zero	Overall time (Minutes, seconds, frame)
-----------	------------------	-------------------------------------	------	--

Program track and Lead-in track

Control emphasis	ID No.	Music No.	Music No. index	Tune time (Minutes, seconds, frame)	Zero	Overall time (Minutes, seconds, frame)
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Fig. 4-7 Control Q Details

C4-4-1 Reproducing System Signal Processing

(1) Decoder

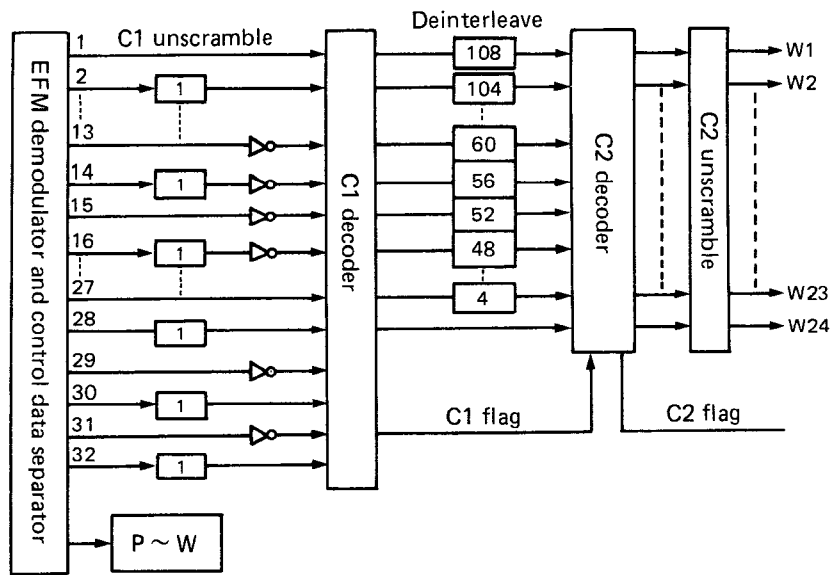


Fig. 4-8 Playback Signal Processing

The flow of playback signal processing is just opposite to the flow of recording system signal processing. Here, decoders for error detection and correction are used in place of the C1 and C2 encoders that are used for parity generation.

Both the C1 and C2 decoders are capable of error detection and correction. If data that cannot be corrected by C1 decoder is marked with C1 flag, and after deinterleaving, the data is sent to the C2 decoder for accurate correction.

If data that cannot be corrected by C2 decoder, the average value obtained from the adjacent data is switched to interpolation after the process of C2 unscramble.

(2) LSI signal processing

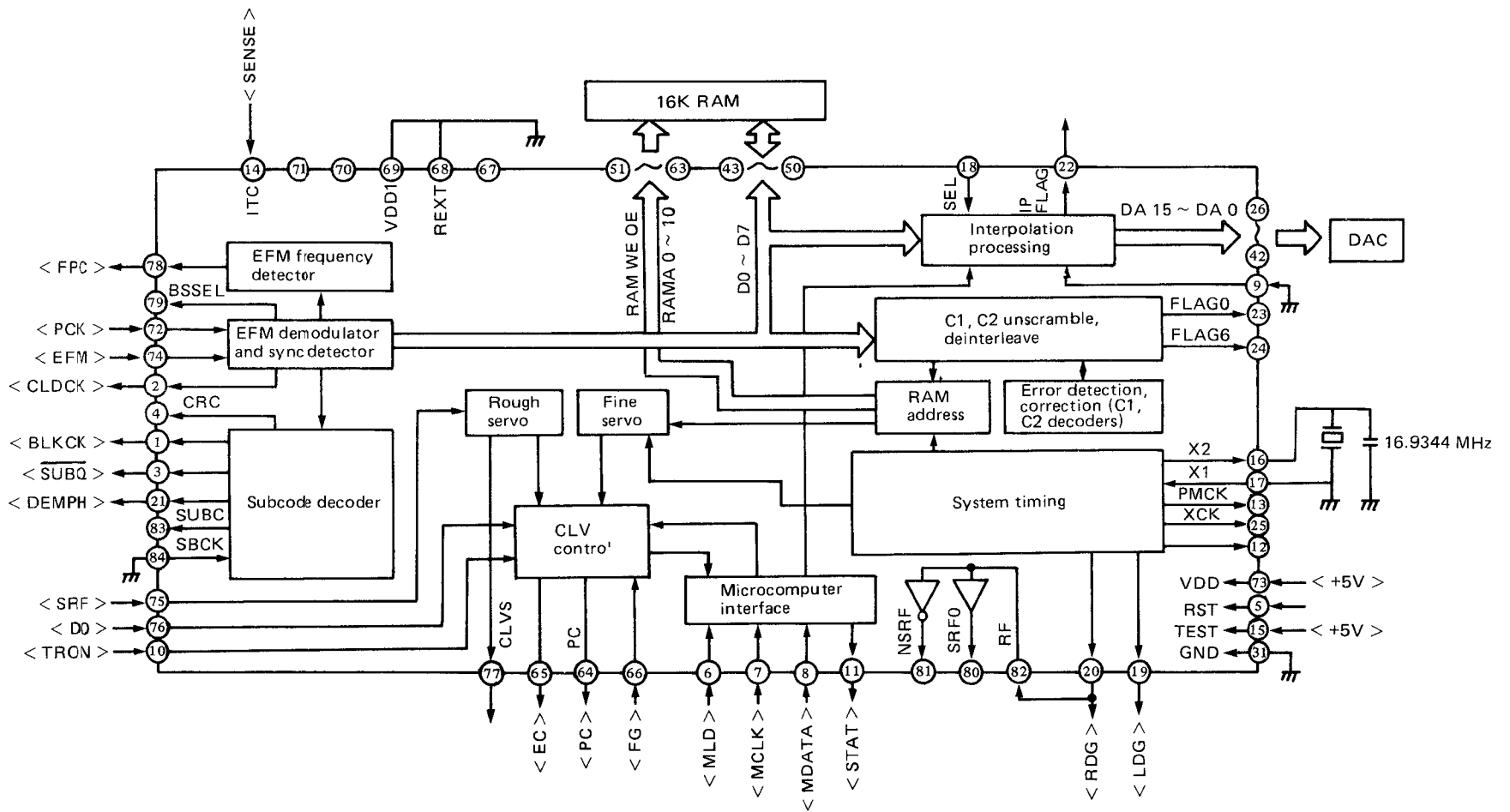


Fig. 4-4 LSI Block Diagram

C4-4-3 LSI and Associated Operations of Peripheral Circuits

(1) Operations associated with AN8371S

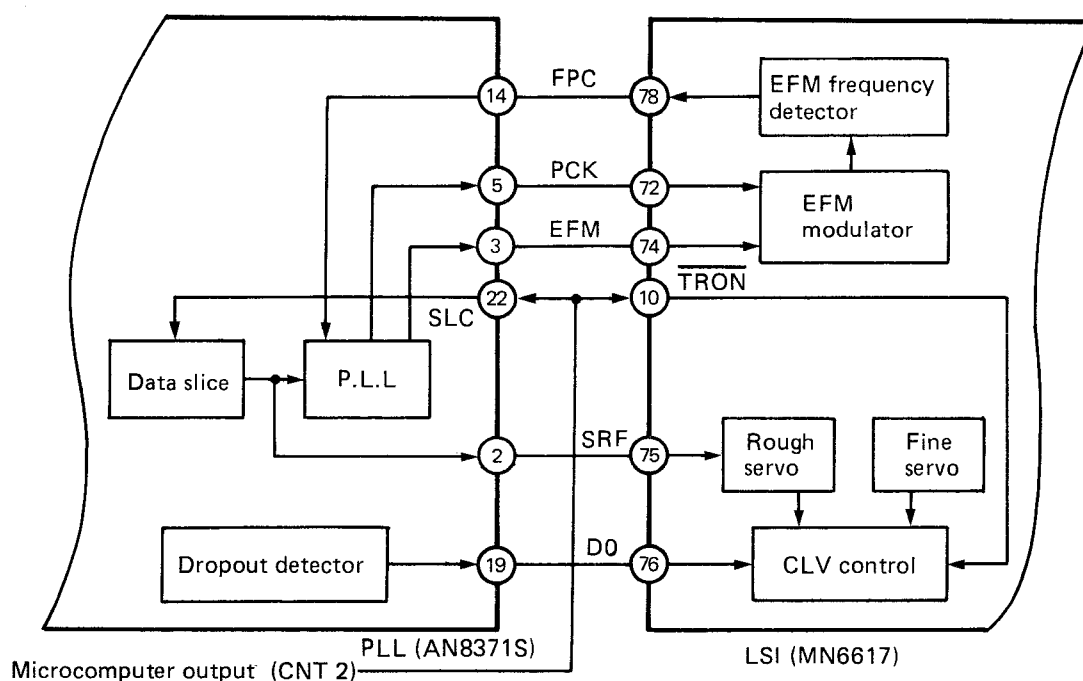


Fig. 4-10 AN8371S and LSI Connection Diagram

Fig. 4-10 shows an LSI (MN6617) and PLL IC (AN8371S) connection diagram.

There are four inputs and one output as viewed from the LSI, and the output (CNT2) from the microcomputer is input to both ICs.

1) PCK, EFM, FPC signals

PCK and EFM signals have been described in Section C4-3.

FPC signal results from measuring the pulse width of EFM signal (3T to 11T) with PCK signal (1T).

If EFM signal is a signal of 2T or less, a feedback operation takes place to raise the VCO frequency of the PLL. If EFM signal is 12T or over, the VCO frequency is lowered.

During play (PLL has completed operating), pin 78 of LSI is constant at 2.5V. If the PLL is being pulled in, a control pulse of $2.5V \pm 2.5V$ is output from pin 28.

2) SRF, D0 signals

SRF signal is generated by data-slicing RF signal (data slice output of AN8371S), and is input to the CLV rough servo (11T) servo) of LSI. The CLV servo is described later.

The D0 signal is a dropout detection signal (dropout detection output of AN8371S) of AN8371S, and is used as CLV control signal of LSI. If dropout is detected, the 11T servo is turned off to allow free rotation. Fig. 4-12 shows SRF signal and D0 signal.

- 3) The CNT2 (SLC or TR ON) signal is one of the 4-bit control logic signals of the optical servo IC (AN8370S). This signal remains high until the tracking servo is in the OFF mode (See Table 6-1).

This is used to control both ICs as follows.

- (a) The CLV control of LSI

The mode is switched over to the rough servo mode to control the rotation speed even in a high speed driving so that the rotation speed can be locked in by the CLV servo system immediately after the tracking servo system is turned on.

- (b) The data slice circuit of AN8371S

The slice level is forcibly raised so that the SRF signal input to the rough servo system can be extracted exactly.

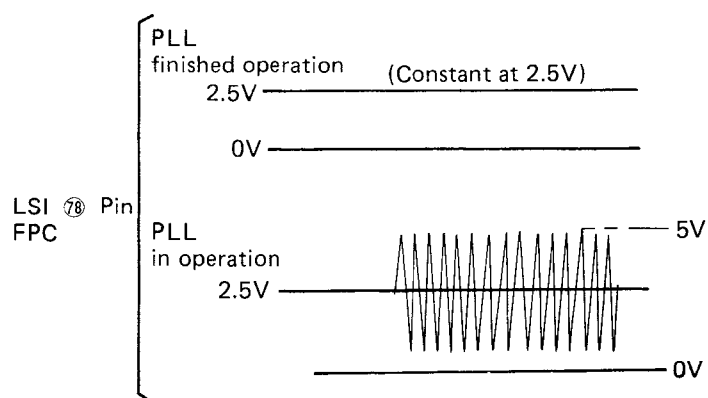
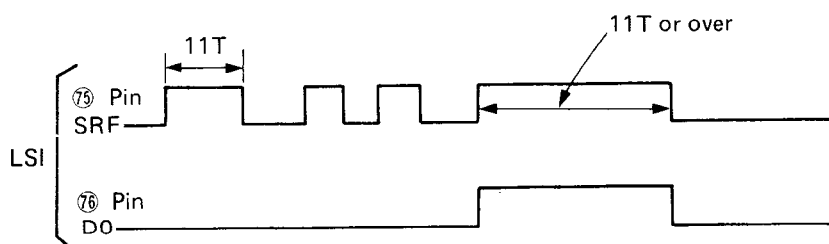


Fig. 4-11 FPC Signal



- SRF signal has a period of 11T or longer when a dropout occurs, which is interpreted as slow rotation, causing the spindle motor to run away. Thus, D0 signal makes rotation free.

Fig. 4-12 SRF Signal and C0 Signal

(2) Operations associated with microcomputer

Fig. 4-13 shows an LSI (MN6617) and microcomputer (MN1554CAF) connection diagram. There are four inputs and five outputs as viewed from the LSI.

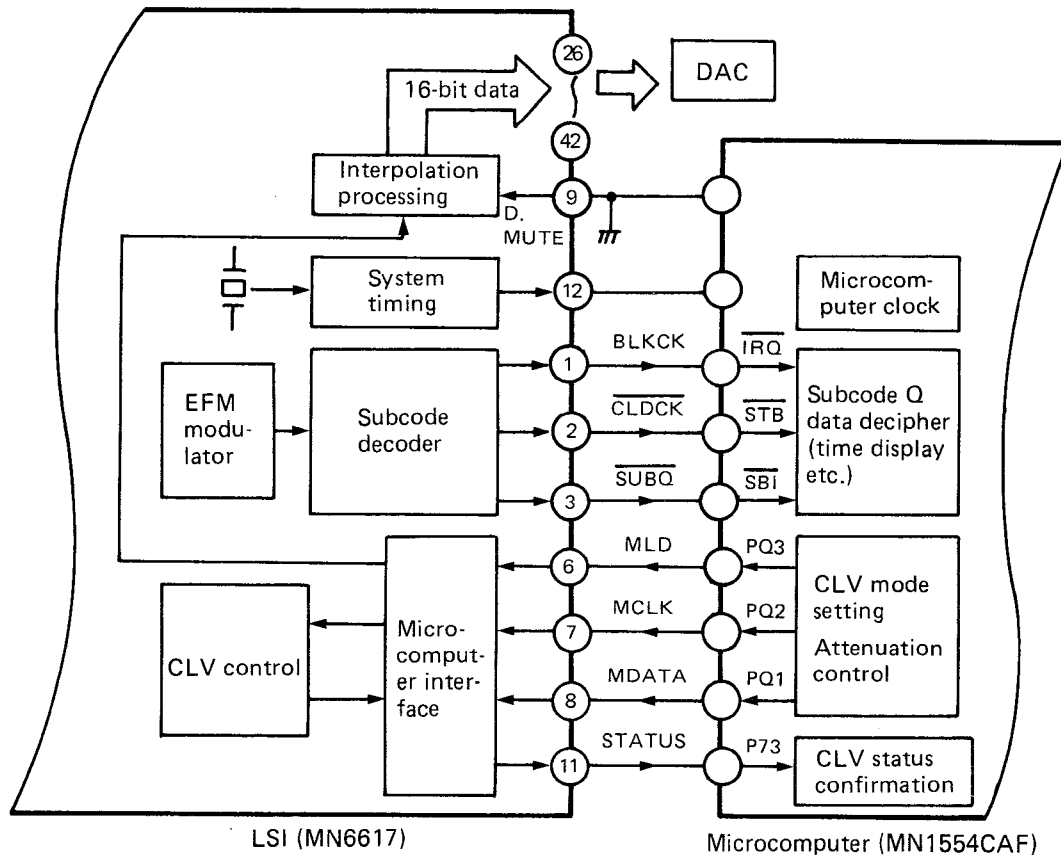


Fig. 4-13 LSI and Microcomputer Connection Diagram

1) BLKCK, CLDCK, SUB Q signal

These three signals transfer the subcode Q data demodulated from the EFM data by LSI to the microcomputer. The microcomputer decodes the subcode Q data and carries out time base control and display.

The BLKCK is a clock signal of one block (98 frames) extracted from EFM data; the CLDCK is a clock signal of one frame similarly extracted, and the SUB Q becomes subcode Q data.

Data are transferred serially.

Fig. 4-14 shows a timing chart of the three signals.

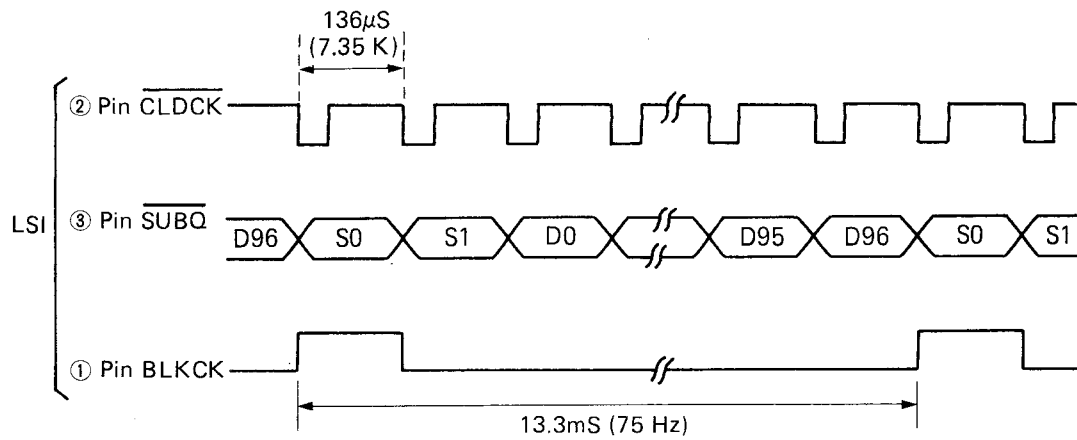


Fig. 4-14 Subcode Q data Transfer (LSI to Microcomputer)

2) MLD, MCLK, MDATA signals

These three signals transfer data from microcomputer to LSI.

- (a) MLD : Data loading signal
- (b) MCLK : Data transfer clock signal
- (c) MDATA : Transfer data

Data is transferred to set the modes of the CLV control and interpolation circuit in LSI.

Fig. 4-15 shows MDATA contents.

No.	8 bits								Remarks
	A4	A3	A2	A1	D4	D3	D2	D1	
①	0	0	0	1	1 (with FG)	1 (TT ON)	0 (STOP)	0	CLV mode
					0 (without FG)	0 (TT OFF)	0 (ACC)	1	
							1 (BRAKE)	0	
							1 (PLAY)	1	
②	0	0	1	0	—	—	1 (ATT-12dB)	1 (MUTE ON)	Interpolation circuit
							0 (ATT OFF)	0 (MUTE OFF)	
③	0	1	1	1	—	—	0 (CRC)	0	Status output switching
							1 (CLVOK)	0	
							1 (TT STOP)	1	

Fig. 4-15 MDATA Contents

The MDATA signal comprises 8 bits, and is serially transferred.

Of the 8 bits, the upper 4 bits express the type of data being transfer data and the lower 4 bits carry data.

Fig. 4-16 shows a timing chart of the three signals.

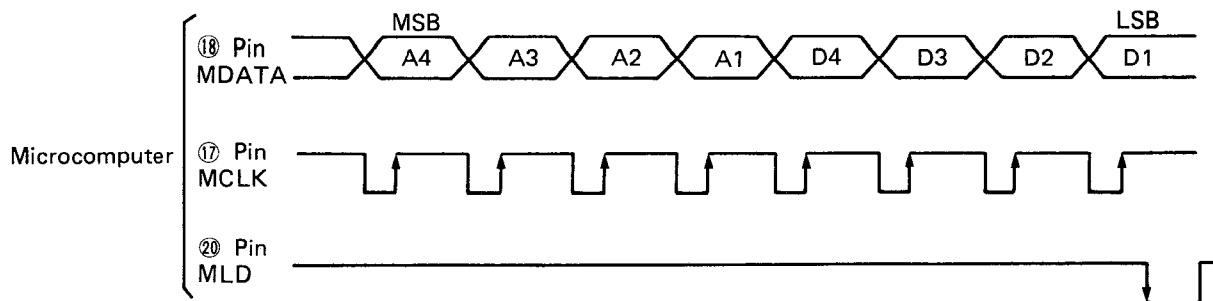


Fig. 4-16 MDATA Transfer (Microcomputer to LSI)

STATUS signal is data to be transferred from LSI to microcomputer, and its contents are checked by microcomputer by switching over the output signal as in Item 3), Fig. 4-15.

(3) CLV servo

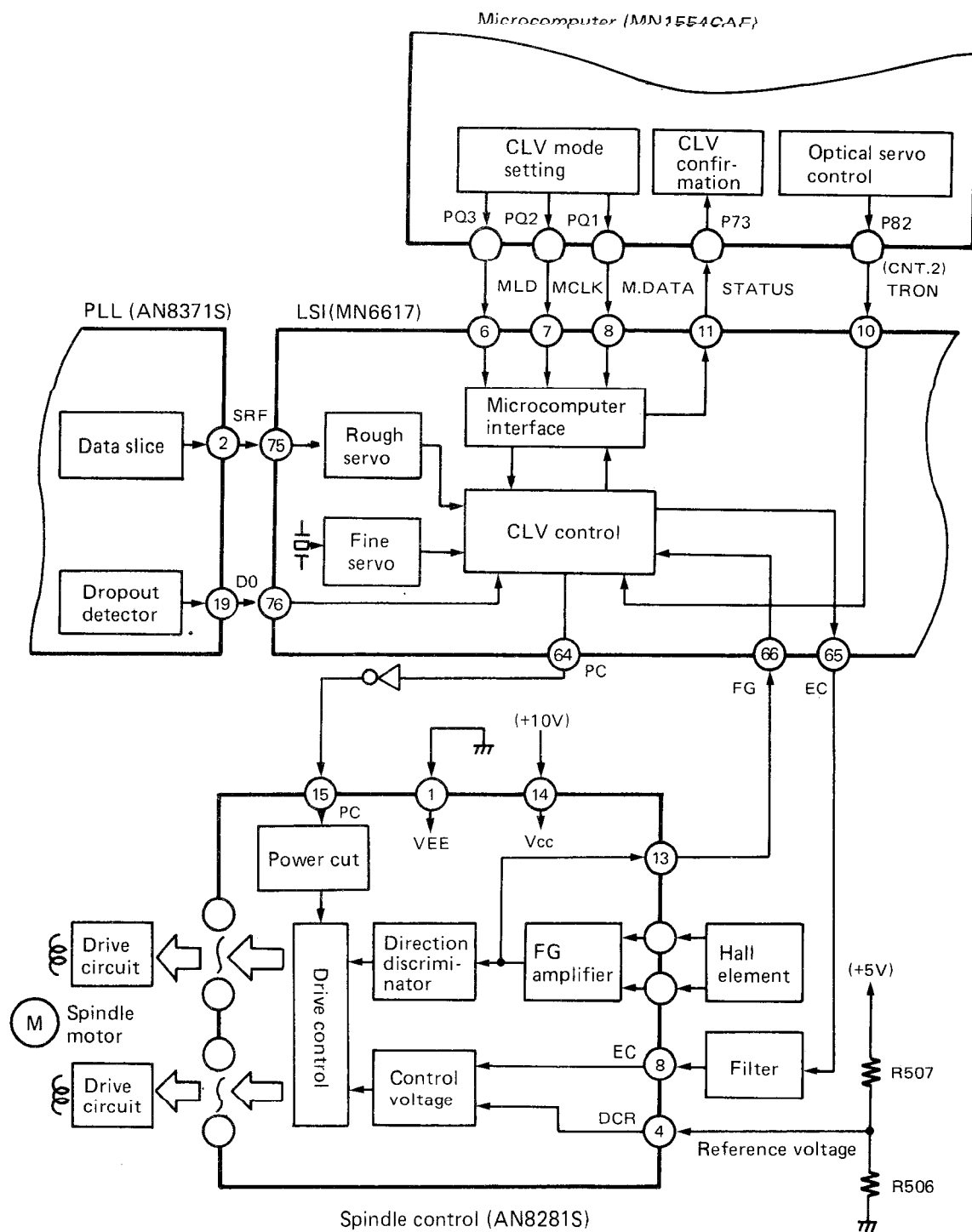


Fig. 4-17 CLV Servo Circuit

The CLV servo is a rotation speed control servo for maintaining linear speed of the disk constant. The servo is divided into three parts, that is 11T servo, CLV servo, and traverse tracking servo. The traverse tracking servo and 11T servo are rough servos, and the CLV servo is a fine servo. The CLV control which is the primary part of the circuit in the LSI. The CLV servo circuits consists of micro-computers AN8371S and AN8281S.

Figure 9-17 shows the CLV servo circuit.

1) AN8281S circuit operation

AN8281S is the spindle motor drive IC. The control voltage EC is applied from the CLV control circuit is applied to pin 8, and the reference voltage divided by R506 and R507 (DCR) is applied to pin 4.

FG and rotating direction are detected by the Hall element, and the FG signal is fed back to the CLV control circuit from pin 13.

The ON/OFF function of this IC (pin 15) controls start and stop of the motor.

Fig. 4-18 shows EC signal versus the motor.

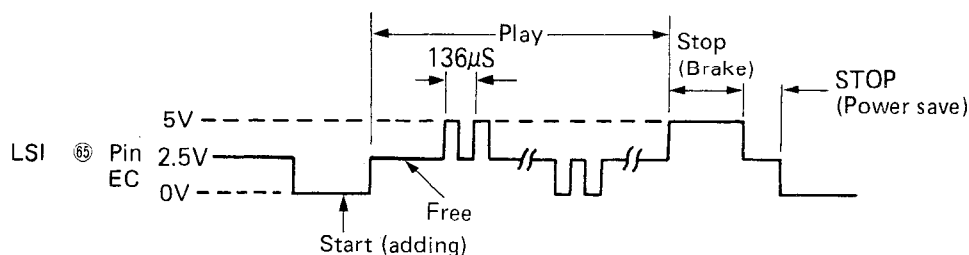


Fig. 4-18 CLV Control Output

2) 11T servo

The 11t servo changes the constant speed to constant linear speed.

This requires some signal extracted from the disk, that is, SRF signal. (SRF signal is generated by data-slicing RF signal.)

The 11T servo controls the SRF signals that the longest period of the signal will be equivalent 11 times the period of the reference clock 1T (4.3 MHz).

When the 11T servo operates, the spindle motor comes close to the control speed of the fine servo.

3) CLV servo

The CLV servo is started by the 11T servo when digital signals are ready for processing in the LSI.

For digital signal processing, data is once written in an external RAM and read out again. A RAM address generator circuit is provided in the LSI for this purpose. Because a crystal clock is used for reading data and a clock synchronized with reproducing signals (corresponding to PCK) is used for writing data, the address locations fluctuates in RAM. The fine servo detects the address location signals and operates to keep the read/wirte pace constant.

4) Traverse tracking servo

The traverse tracking servo follows the pick-up and controls the linear speed of the pick-up constant when the traverse mechanism at high speed.

This servo operates in such a way that two continuous minimum periods of the SRF will be 6 times the period of the reference clock 1T (4.3 MHz).

TRON is used for switching the traverse tracking servo to the 11T servo or vice versa. That is, the traverse tracking servo is selected when TRON is high and the 11T servo is selected when TRON is low.

is low.

AUDIO CIRCUIT

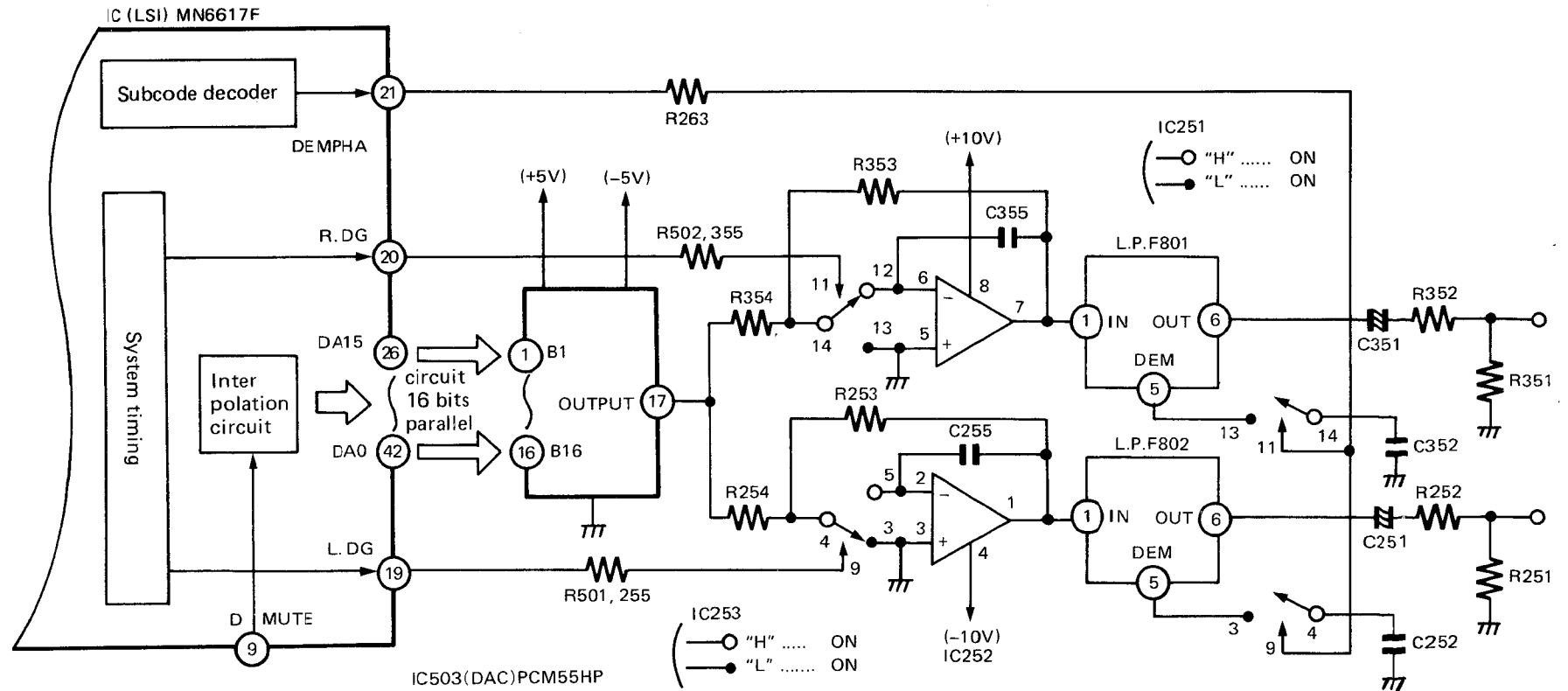
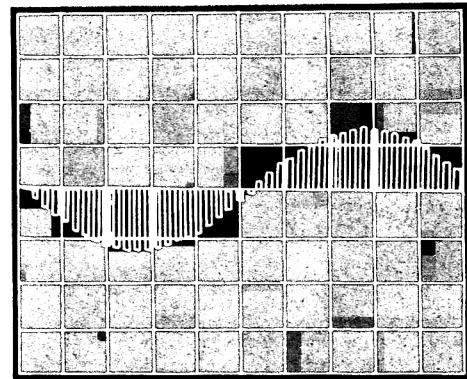


Fig. 4-1 Audio Circuit

C4-5-1 D—A Converter

IC503, is a D—A converter, which converts digital signals to analog signals. Digital signals are input to pins 1 to 16 of IC503, 16 bits parallel. Pin 1 (B1) is MSB, and pin 16 (B16) is LSB. PAM signal is output from pin 17 (OUT).



(Lch = 1 kHz, Rch = 0)

Fig. 4-2 PAM Signal

C4-5-2 Deglitch Circuit

The deglitch circuit consists of IC253 (analog switch), IC252 (operational amplifier), C255 and C355 (for hold).

Pin 19 (L DG) of IC601 (LSI) is the deglitch signal output for the left channel, and pin 20 (R DG) of IC601 is the deglitch signal output for the right channel.

Fig. 4-3 shows a demodulation timing chart of the deglitch circuit.

When L DG signal is low, the left channel circuit takes signal in, while the right channel circuit outputs the signal that have been taken in and held by C225.

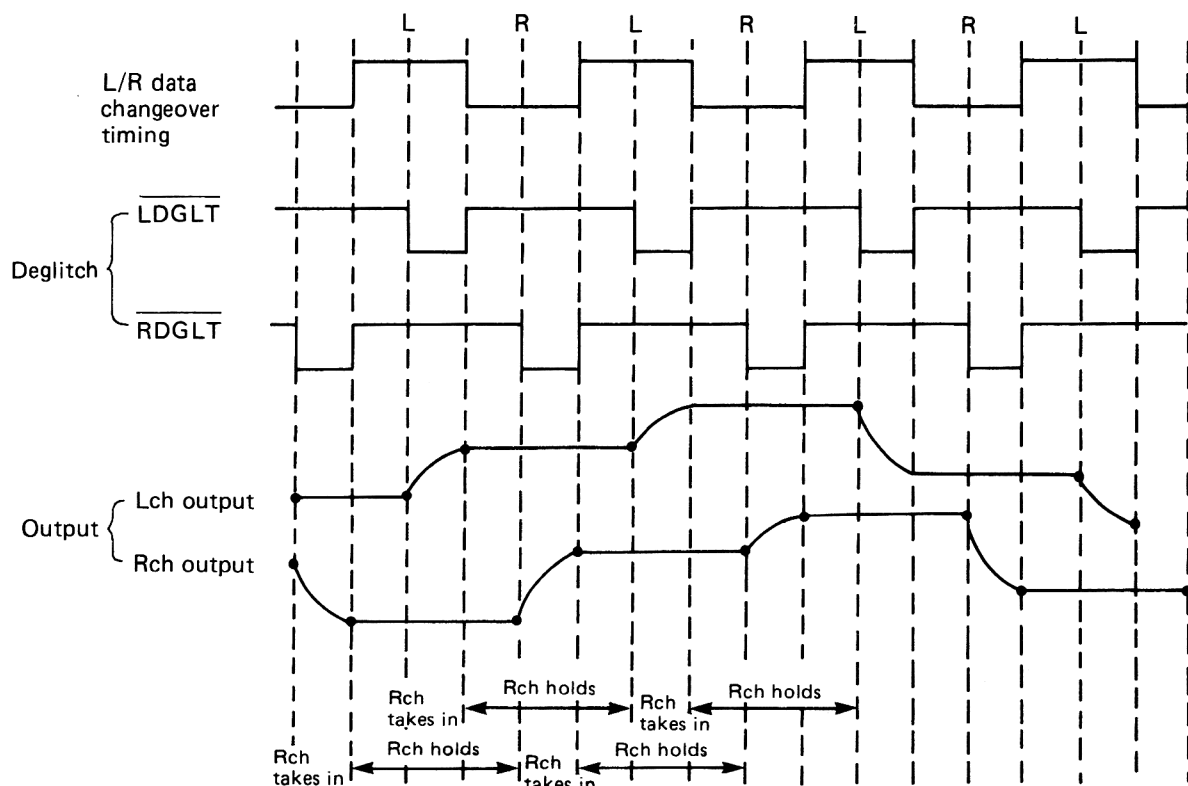


Fig. 4-3 Delitch Timing

C4-5-3 LPF and De-emphasis Circuit

The lowpass filter (LPF) restores the signals containing harmonics (step waves) received from the deglitch circuit to a complete audio signal.

The LPF also has a de-emphasis function. It reads emphasis or no emphasis from subcode (Q channel) recorded on the disk, and the control signal is output from pin 21 (DEMPHA) of IC601 (LSI). De-emphasis is applied when the control signal is high.

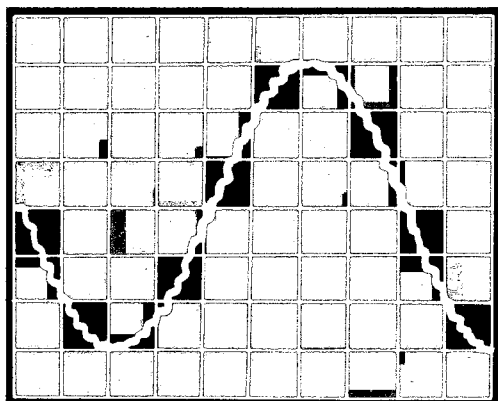


Fig. 4-4 LPF Input

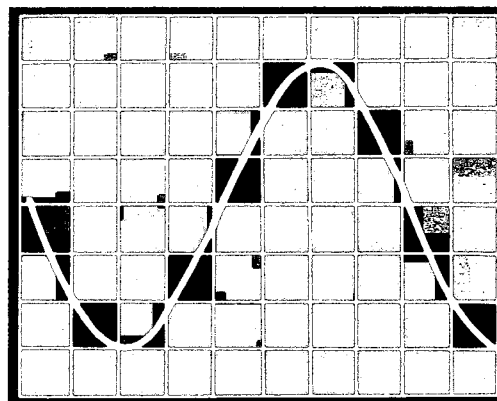


Fig. 4-5 LPF Output

ALIGNMENT INSTRUCTIONS

A5-1 ELECTRICAL ADJUSTMENT

A5-1-1 Measuring Instructions and Tools

1. CD gain alignment jig YESFZ5019
2. Two-channel oscilloscope (30 MHz or over)
3. Low-frequency oscillator
4. Screwdriver for adjusting semi-fixed variable resistors

5. Test Disc SZZP1014F or SZZP1054C and Black Band Disc SZZP1057
6. QRB Power cord ass'y YESFZ5017

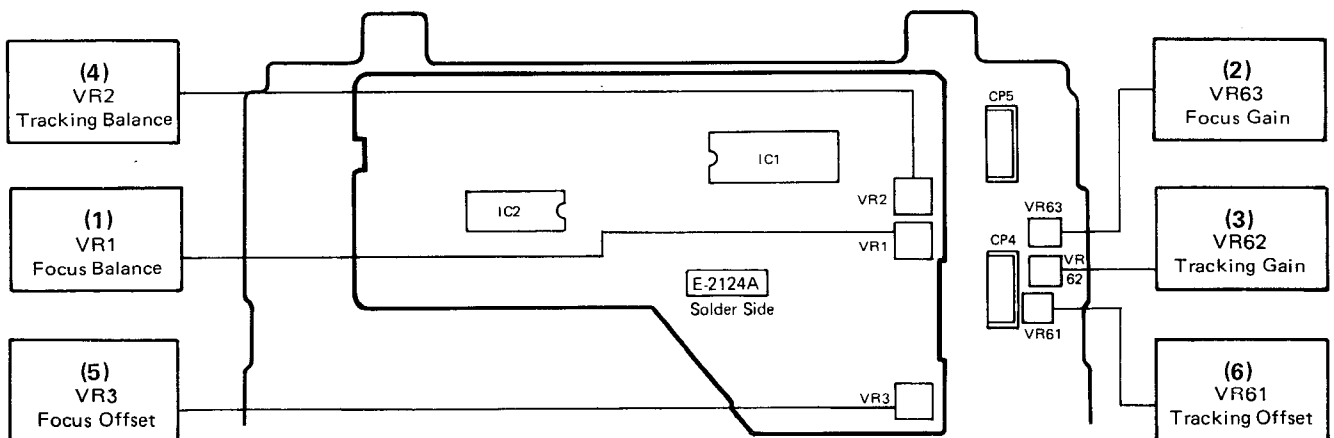
Caution: Use of controls making adjustments, or performing procedures other than specified herein may result in hazardous exposure to radiation.

A5-1-2 Alignment Procedures and Points

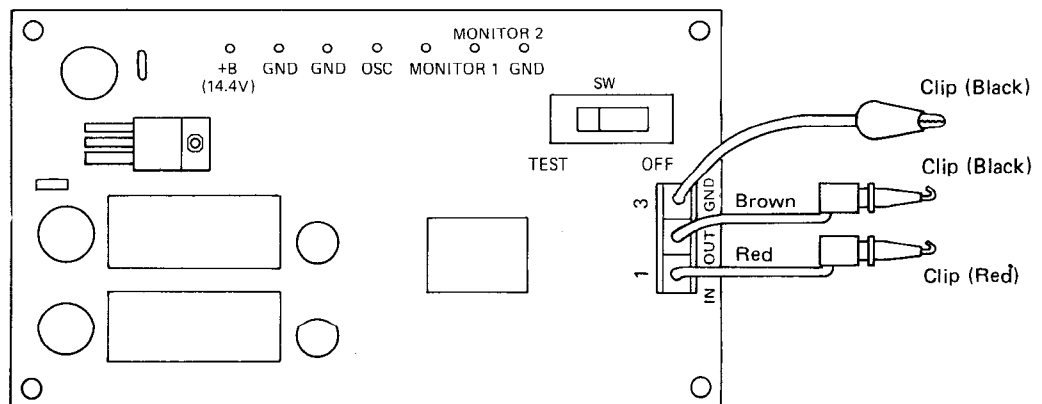
Remove the upper cover and bottom cover, solder the leads to test points, and follow the sequence below.

Note: Do not attempt alignments when the deck is disconnected.

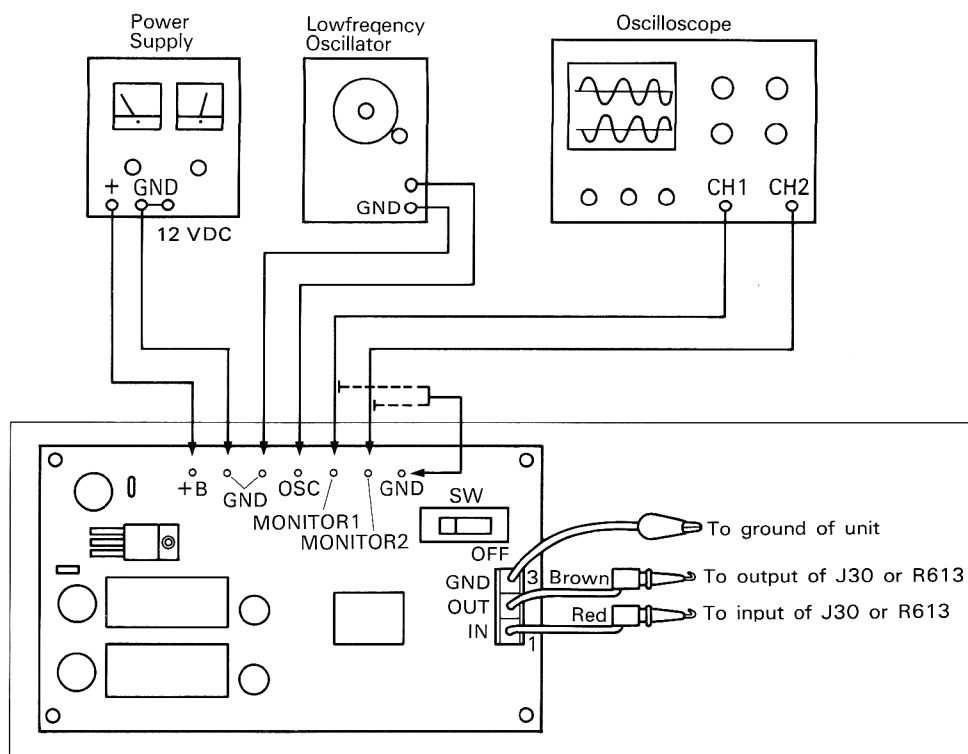
- | | | |
|-------------------------------|------|--|
| 1) Focus balance alignment | VR1 | [TP (IC601-Pin12) (ARF)] |
| 2) Focus gain alignment | VR63 | [TP (IC601-Pin9) (FE OUT)
TP (IC1-Pin16) (FPI)] |
| 3) Tracking gain alignment | VR62 | [TP (IC601-Pin6) (TE OUT)
TP (IC1-Pin17) (TPI)] |
| 4) Tracking balance alignment | VR2 | [TP (IC601-Pin6) (TE OUT)
TP (IC1-Pin17) (TP1)] |
| 5) Focus offset alignment | VR3 | [TP (IC601-Pin12) (ARF)
TP (IC601-Pin8) (FEG)] |
| 6) Tracking offset alignment | VR61 | [TP (IC601-Pin12) (ARF)
TP (IC601-Pin5) (TEG)] |



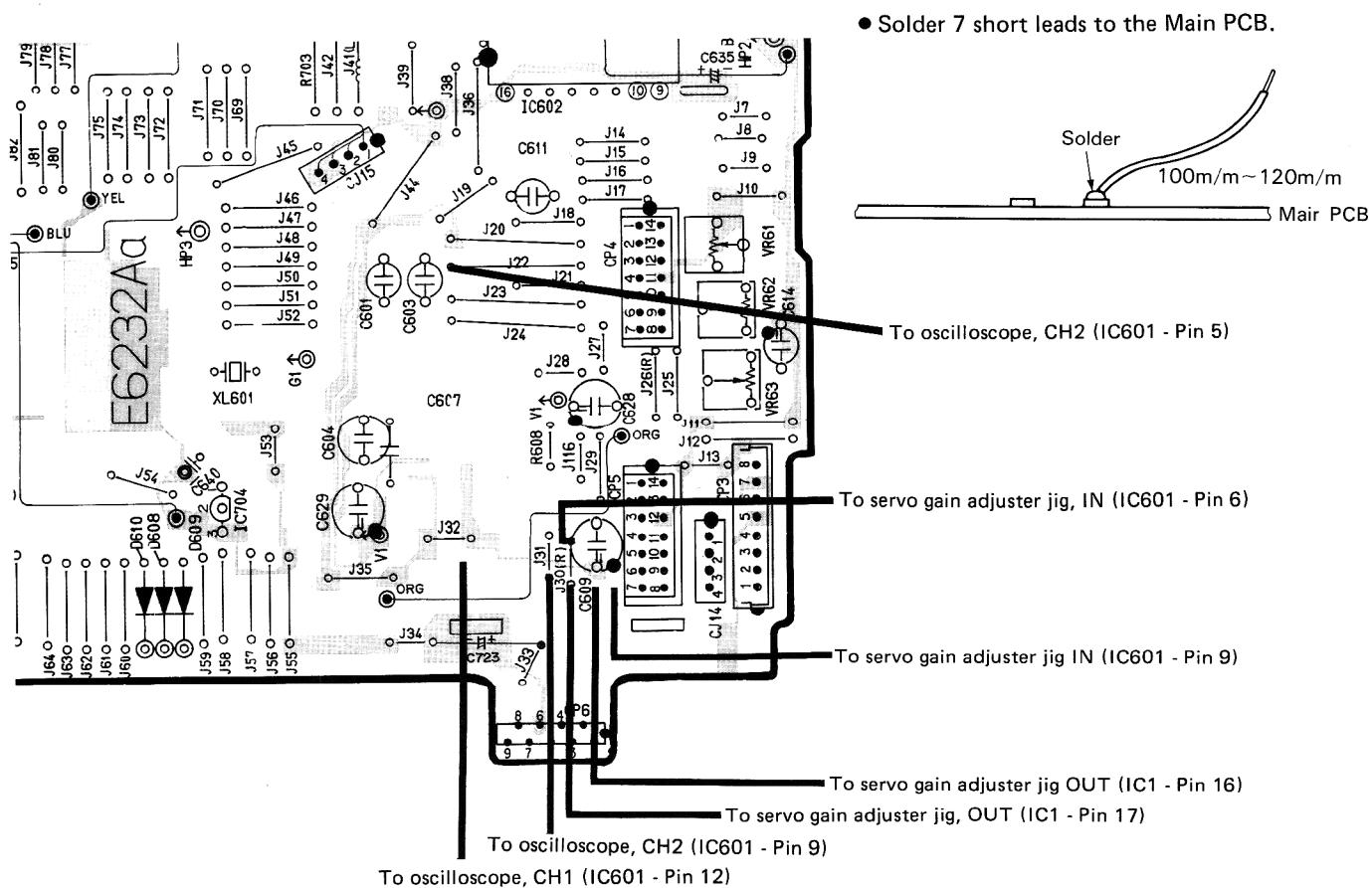
A5-1-3 CD Gain Alignment Jig



1) Wiring of Servo Gain Adjuster



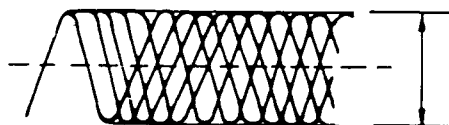
2) Solder the leads of Test Point on the Main PCB, as shown below.



A5-1-4 Adjusting Methods

1) Focus balance alignment (Best eye)

- Test point : IC601-Pin12.
- Oscilloscope setting : Volt 20 mV. Sweep 0.5 μ S Input selector AC
- Operation : Play test disc track # 1.
- Alignment : RF signal appears. Adjust VR1 until the eye pattern extends widest.
- Waveform sample :

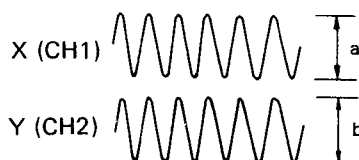


Adjust to fully widen the waveform

2) Focus gain alignment (Use of CD gain alignment jig)

- Test point : Both ends of J30
-
- CD gain alignment jig connection
 - 1) Clip Chassis
 - 2) Lead IN (Red) J30 IN (IC601-Pin9)
 - 3) Lead OUT (Brown) J30 OUT (IC1 Pin-16)
 - 4) OSC terminal Low frequency OSC
 - 5) Monitor Ch1, Ch2 terminals Oscilloscope probe Ch1, Ch2
 - Low frequency oscillator setting
 - 1) Frequency 1 kHz
 - 2) Level 200 mVp-p
 - Oscilloscope setting (Ch1, Ch2) :

Volt	20 mV	Sweep	0.5 ms	Input selector	AC
----------------	-------	-----------------	--------	--------------------------	----
 - Operation : Set the lever switch on the CD gain alignment jig to TEST. Insert a disc and play.
 - Alignment : Adjust VR63 until both channels are equal in amplitude.
 - Waveform sample :

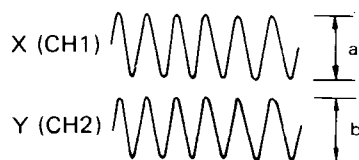


(Adjust to a = b)

3) Tracking gain alignment (Use of CD gain alignment jig)

- Test point : Both ends of R613
-
- CD gain alignment jig connection
 - 1) Clip Chassis
 - 2) Lead IN (Red) R613 IN (IC601-Pin6)
 - 3) Lead OUT (Brown) R613 OUT (IC1-Pin17)
 - 4) OSC terminal Low frequency OSC
 - 5) Monitor 1, Monitor 2 terminals Oscilloscope probe Ch1, Ch2
 - Low frequency oscillator setting
 - 1) Frequency 1 kHz
 - 2) Level 400 mVp-p
 - Oscilloscope setting (Ch1, Ch2) :

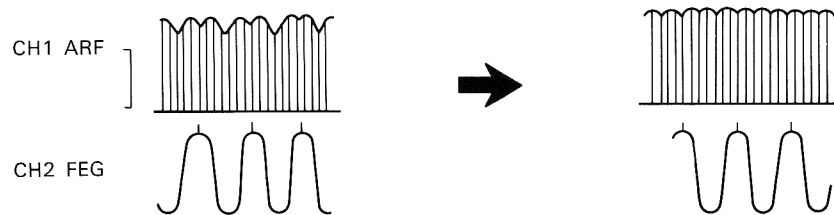
Volt	20 mV	Sweep	0.5 ms	Input selector	AC
----------------	-------	-----------------	--------	--------------------------	----
 - Operation : Set the lever switch on the CD gain alignment jig to TEST. Insert a disc and play.
 - Alignment : Adjust VR62 until both channels are equal in amplitude.



(Adjust to a = b)

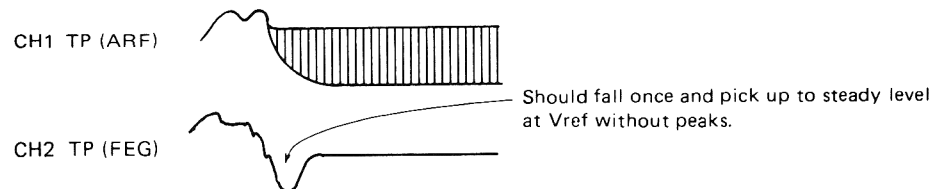
4) Tracking balance alignment (Use of CD gain alignment jig)

- Test point : Both ends of R613
- CD Adjusting jig connection :
 1. Clip Chassis
 2. Lead IN (Red) R613 IN (IC601-Pin6)
 3. Lead OUT (Brown) R613 OUT (IC1-Pin17)
 4. Monitor 2 Oscilloscope CH2
 5. Connect CH1 of oscilloscope to TP (IC601-Pin12)
- Low Frequency Oscillator setting
 - Frequency 1 kHz
 - Level 400mVp-p
- Oscilloscope setting : Volt 20 mV/div. Sweep 1ms/div. (Normal) Input Selector AC
- Operation : Set the level switch on the CD gain alignment jig to TEST. Insert a test disc and play.
- Alignment : Adjust VR2 until the peaks of CH1 (ARF) envelope evenly cover the peaks and bottoms of CH2 (TEG).



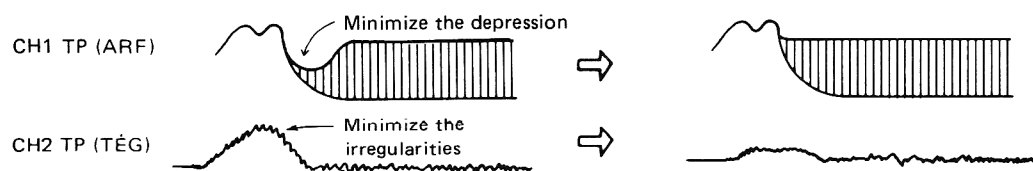
5) Focus offset alignment

- Test point : TP (ARF) (IC601-Pin12), TP (FEG) (IC601-Pin8)
- Oscilloscope setting
 - CH1 : Volt 20 mV/div. Sweep 0.5 ms/div. (Normal)
 - CH2 : Volt 5 mV/div. Sweep 0.5 ms/div. (Normal)
- Operation : Play the seventh track of disc (CDT - 002). Connect CH1 of oscilloscope to TP (IC601-Pin12) and CH2 of the same to TP (IC601-Pin8).
- Alignment : Adjust VR3 until the waveform is as shown below.
- Waveform Sample :



6) Tracking offset alignment

- Test point : TP (ARF) (IC601-Pin12), TP (TEG) (IC601-Pin5)
- Oscilloscope setting (CH1, CH2) :
 - Volt 20 mV/div. Sweep 0.2 ms/div. (Normal) Input Selector AC
- Operation : Play the seventh track of black band disc. Connect CH1 of oscilloscope to TP (IC601-Pin12) and CH2 to TP (IC601-Pin5).
- Alignment : Adjust VR61 to minimize the depression (CH 1) and to make the FE signal as flat as possible (CH 2).



Note: Eliminating the groove of ARF takes precedence to all others.

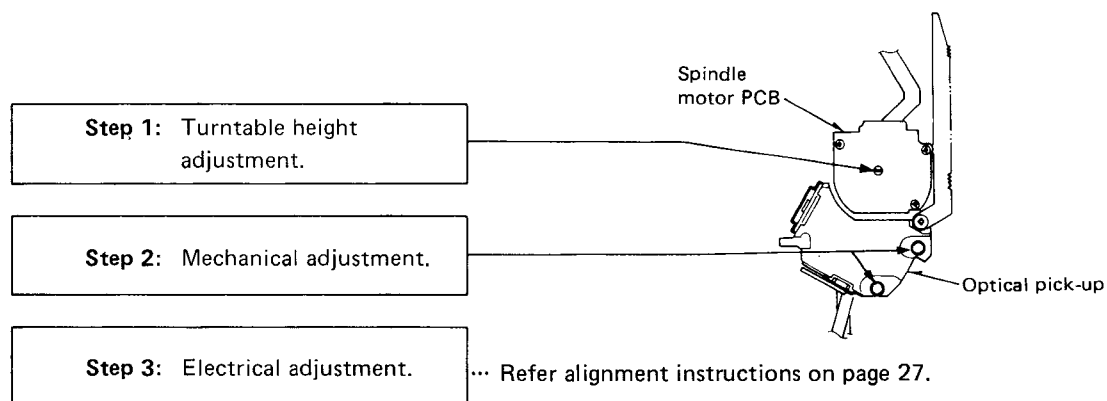
A5-2 OPTICAL PICKUP ADJUSTMENT INSTRUCTION (TYPE A)

A5-2-1 Measuring Instruments and Special Tools

- Two-channel oscilloscope (with trigger) of 30 MHz or over
- Test disc (SZZP1014F) new type CDT-002
Inspection test disc (SZZP1054C) CDT-D16
Uneven test disc (SZZP1056C) CDT-D18
- Hexagonal wrench (2mm)
- Screw lock paint (RZZ0L01)
- Lead wires
- Extension cord (Jig) YESFZ5020
(Supplied from APD) (See page 25.)
- Resistor (4.7 k Ω 1/4W)

A5-2-2 Adjust Procedure and Adjustment points

- If the optical pickup is replaced, adjust it according to the following procedure.

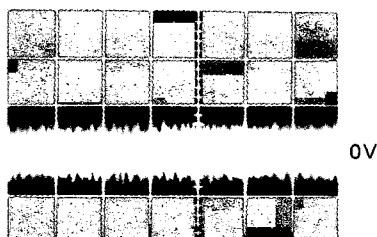
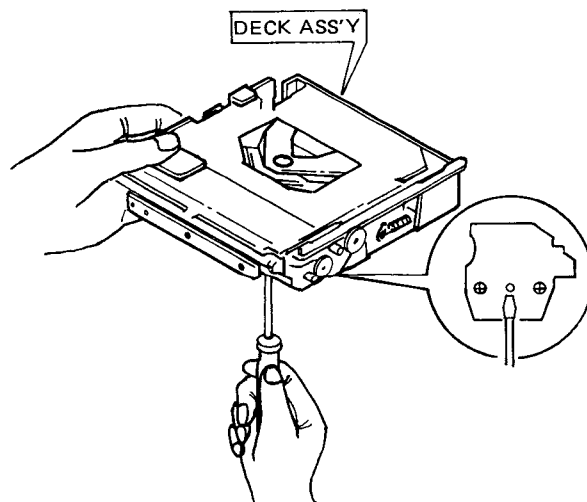
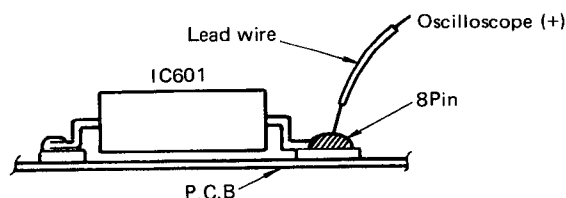


A5-2-3 Turntable Height Adjustment

- 1) Turn off the power of the player.
- 2) Solder a lead wire to IC601 Pin 8 of the SERVO PCB for oscilloscope connection.
- 3) Connect the extension cords between the Deck Ass'y and the unit.
- 4) Connect CH 1 of the oscilloscope to the IC601 Pin 8 (+) and Ground (-).

Oscilloscope setting: VOLT 200 mV
SWEEP 5 msec.
INPUT DC

- 5) Set the oscilloscope to DC zero balance.
- 6) Place the **test disc (SZZP1014F or SZZP1054C)** and turn on the power switch of the player.
- 7) Hold the CD Deck by the hand with the top face up.
- 8) The player to the play mode.
- 9) Turn the **adjustment screw** at the bottom of the spindle motor drive P.C.B. with a flat screwdriver so that the waveform is 0 ± 50 mV.
- 10) Turn off the power switch of the player.
- 11) After the adjustment, apply **screw lock paint (RZZ0L01)** to the adjusting screw.

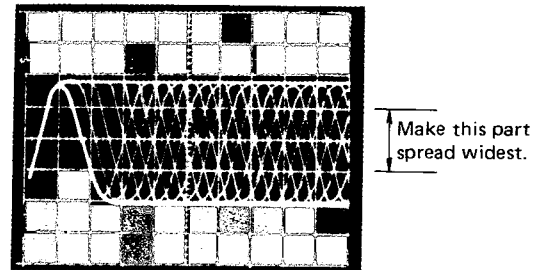
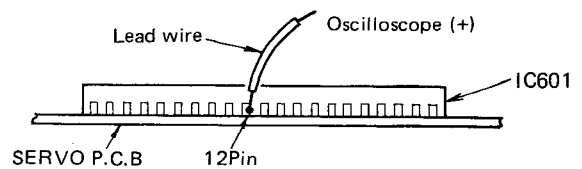


A5-2-4 Mechanical Adjustment

1. Connect CH1 of the oscilloscope to IC601 Pin 12 and Ground.

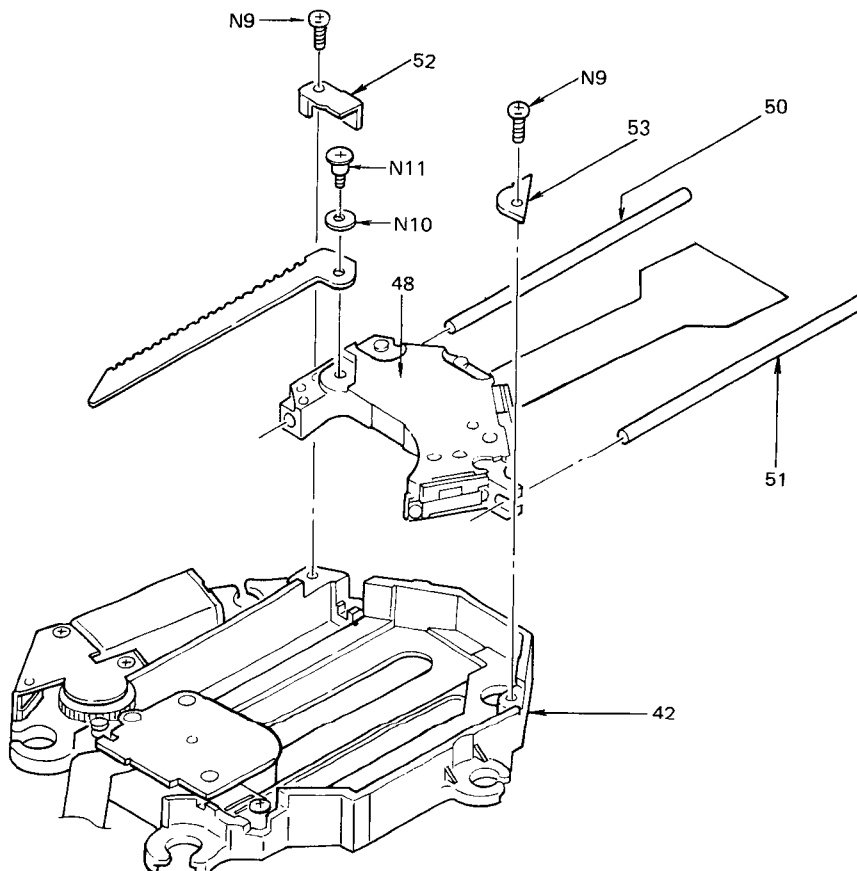
Oscilloscope setting: VOLT 200 mV
SWEEP 0.5 μ sec.
INPUT AC

2. Turn on the power of the player and play the Uneven Test Disc.
3. Monitoring the RF signal on the oscilloscope, adjust the **two adjusting screws** alternately so that the vertical fluctuation of RF signal is minimized and the eye pattern most open.
4. Turn off the power switch of the player.
5. After the adjustment, apply **screw lock paint (RZZ0L01)** to the two adjusting screws.



A5-2-5 Disassembly Instructions

1. Remove the screw N11.
2. Remove two screws N9.
3. Remove brackets 52 and 53
4. Raise Optical pickup assembly 48 and remove it.
5. Pull out the guide shafts 50 and 51 from the optical pickup assembly.



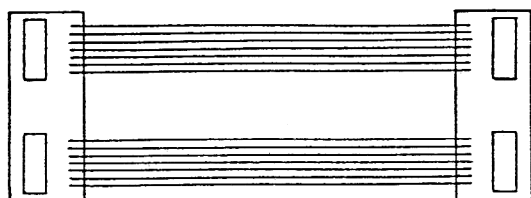
A5-2-6 How to Use the Extension Cord

1. Part Kit Name

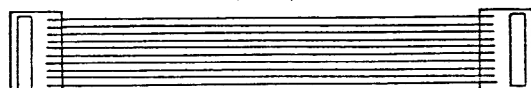
YESFZ5020

2. Contents

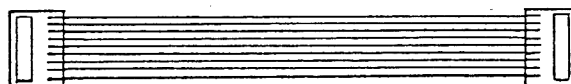
CONNECTOR-1 (14Pin, 2 pcs.)



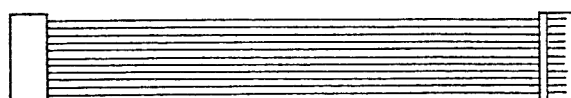
CONNECTOR-2 (9Pin)



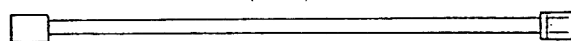
CONNECTOR-3 (8Pin)



CONNECTOR-4 (8Pin)

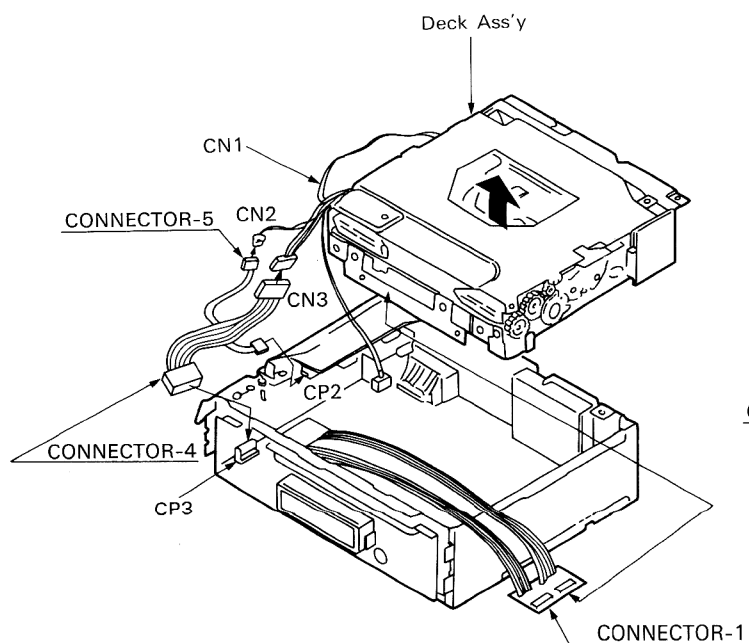


CONNECTOR-5 (2Pin)

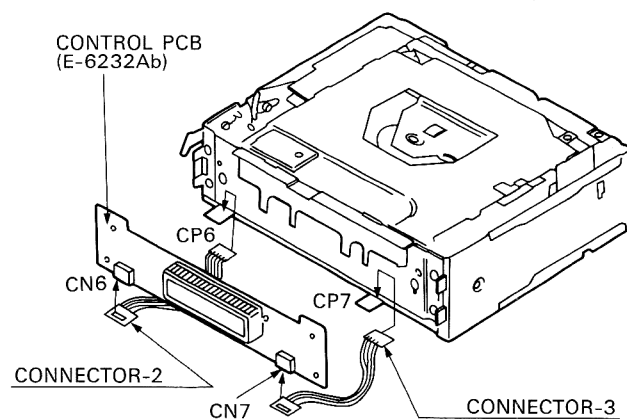


3. Connections

1) In the case of checking CD deck block.



2) In the case of checking control PCB.

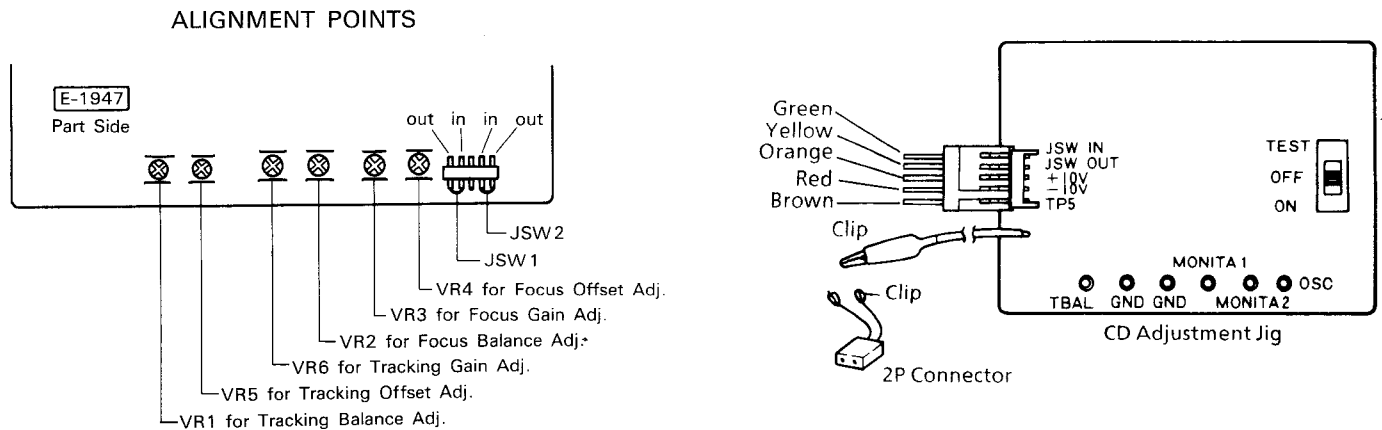


B5-1 ALIGNMENT INSTRUCTIONS (TYPE B)

B5-1-1 Measuring Instruments and Tools

- (1) CD adjusting jig W-2 YESFZ155
- (2) 2-channel oscilloscope
- (3) Low-frequency oscillator
- (4) Screwdriver for adjusting semi-fixed VRs
- (5) Test disc (SZZP1014F or SZZP1054C and SZZP1057C)

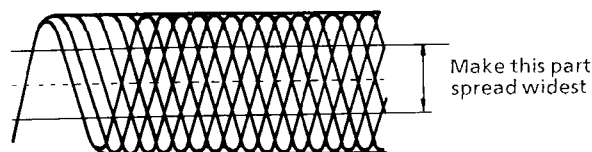
B5-1-2 Adjustment Points



B5-1-3 Electrical Alignment Procedures

1. Focus Balance Adjustment (Best Eye)

- Test Point: Connect scope probe CH1 to CN106 pin 8 or J58 on main PCB.
- Scope settings: Set probe to X10 (for the entire procedure)
VOLT 200—500 mV
SWEEP 0.5 μ s
INPUT AC
- Adjustment: Play test disc CDT-002 or CDT-016 track #1.
Adjust VR2 so that the eye pattern of RF signal is stretched to maximum.
Play track #6 after the alignment for confirmation check.



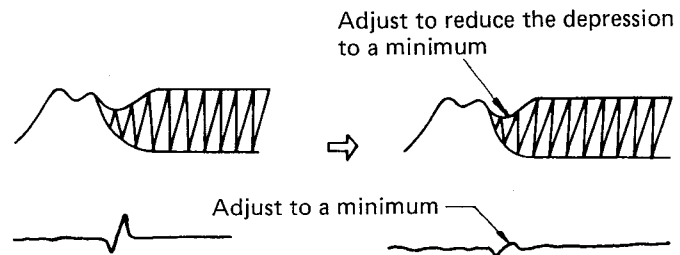
-





4. Focus Offset Adjustment

- Test Point: Connect scope probe CH1 to CN106 pin 8 or J58 on main PCB.
Connect scope probe CH2 to JSW2 (with plug)
- Scope Settings: CH1 and CH2: VOLT 500 mV
SWEEP 0.5 ms (NOR)
INPUT AC
*** (CH1 set at normal trig) ***
- Adjustment: Play the Black Band test disc track #9.
Adjust VR4 to minimize the depression. Also make a fine adjustment of VR4 to make the FE signal of CH2 as flat as possible.

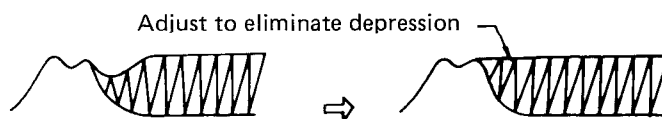


*** Continued in step 5 ***

5. Tracking Offset Adjustment

*** Continued from Step 4 ***

- Adjustment: Adjust VR5 to eliminate the depression. If necessary, adjust VR4 and VR5 alternately.



6. Tracking Offset Balance Adjustment (using test jig)

- Test Point: Connect the brown lead (TP5) from test jig to R3 (jump wire) or J44 behind VR1 on Servo PCB.
- Scope Settings: VOLT 100 mV
SWEEP 0.1 ms
INPUT DC
- Test Jig Connection:

Clip	TO	CD chassis GND
Brown Lead (TP5)		TP above
TBAL Terminal		Scope Probe CH1
Green Lead (JSW in)		JSW1 out (PCB)
Yellow Lead (JSW out)		JSW1 in (PCB)
- Adjustment: Set the test jig level switch to ON.
Play the test disc track #1.
Adjust VR1 within 15 seconds after the test jig level switch is set to OFF so that the center of the waveform will be 0V.



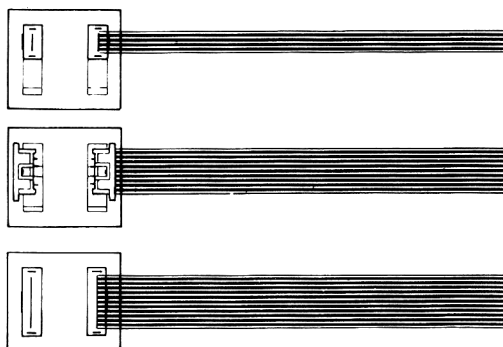
* * * * * NOTE * * * * *

1. If kicking noise is heard when playing the disc after having the alignments done, adjust the VR6 (tracking gain) for fine tuning until the kicking noise disappears.
2. Do not lose the JSW shorting plugs. Replace the shorting plug after each alignment step.
3. Turn off the power before changing any connection.

B5-2 OPTICAL PICKUP ADJUSTMENT

B5-2-1 Measuring Instruments and Special Tools

- Two-channel oscilloscope (with trigger) of 30 MHz or over
- Test disc (Supplied from HAD)
Test disc (SZZP1014F) new type CDT-002
Inspection test disc (SZZP1054C) CDT-016
Uneven test disc (SZZP1056C) CDT-018
- Screw lock paint (RZZ0L01)(Supplied from HAD)
- Hexagonal wrench (2 mm)
- Lead wires
- Extension cords
- Extension cords [YESFZ5003]
(Supplied from APD)

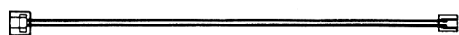


for CN4 (4P) 1PC

for CN1 (9P) 1PC

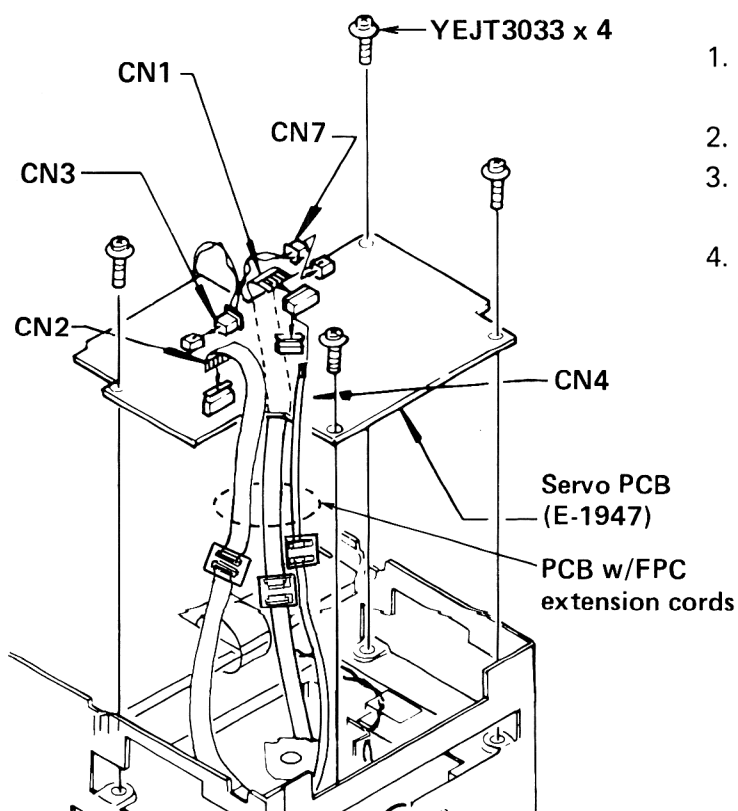
for CN2 (11P) 1PC

(2) Extension connectors



for CN3, CN7 (2P) 2PCS

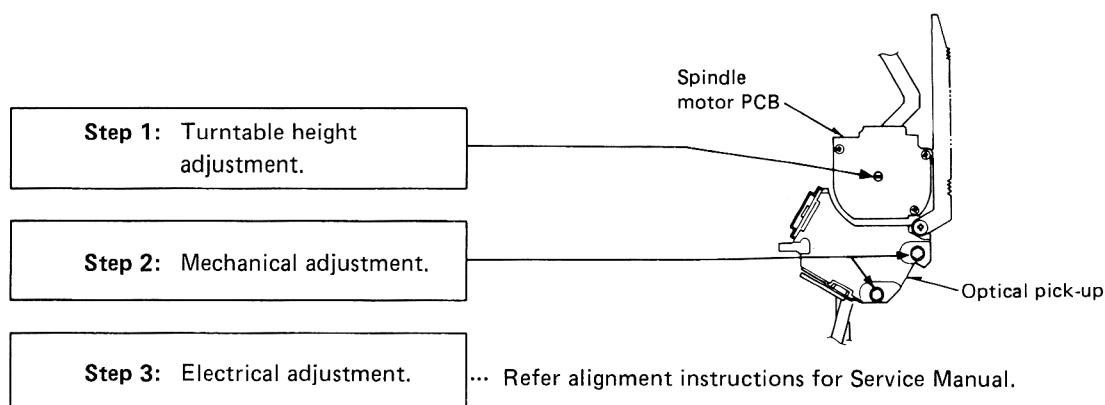
• Instruction for Jig



1. Remove the four ⊕ Screws and disconnect each connectors.
2. Remove the PCB (E-1947)
3. Connect each Jig connectors and FPC extension cords to optical pickup.
4. Connect each Jig connectors and FPC extension cords to Servo PCB (E-1947).

B5-2-2 Adjustment Procedure and Adjustment Points

- If the optical pickup is replaced, adjust it according to the following procedure.

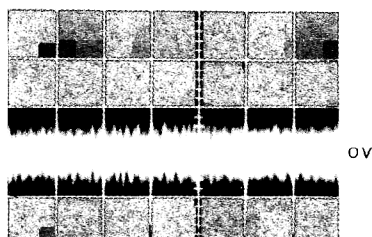
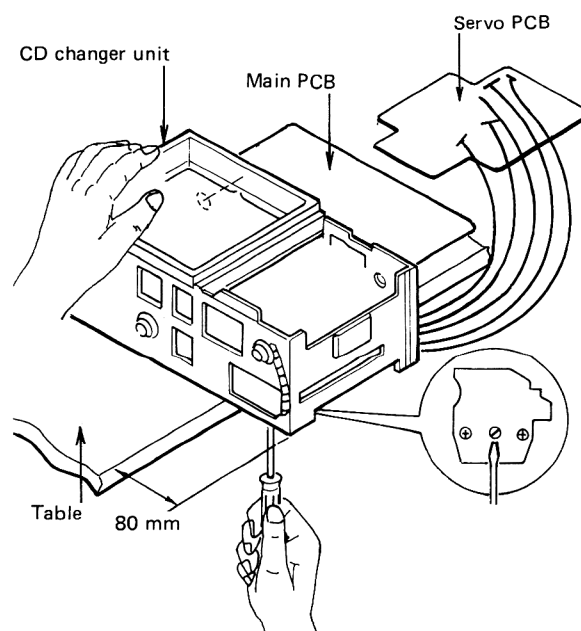
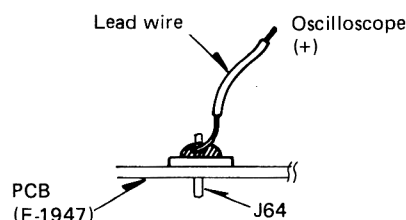


B5-2-3 Turntable Height Adjustment

1. Turn off the power of the player.
2. Solder the lead wires to J64 (IC2, 37 pin) of the PCB (E-1947) and Ground so that the probe of oscilloscope can be connected.
3. Connect the extension cords between the Deck Ass'y (Refer instruction for Jig).
4. Put the CD changer unit in a posture that it is protruded about 80 mm from the edge of the table (repair servicing bench).
5. Connect CH1 of the oscilloscope to the J64 (+) and Ground (-) of the lead wires.

Oscilloscope setting: VOLT 200 mV
 SWEEP 5 m sec.
 INPUT DC

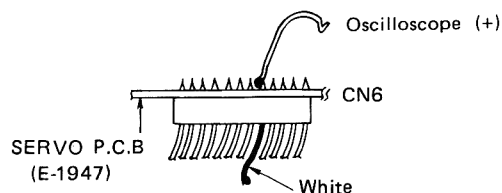
6. Set the oscilloscope to DC zero balance.
7. Place the test disc (SZZP1014F or SZZP1054C) and turn on the power switch of the player.
8. The player to the play mode.
9. Turn the adjusting screw at the bottom of the spindle motor drive PCB with a flat screwdriver so that the waveform is 0 ± 50 mV.
10. Turn off the power switch of the player.
11. After the adjustment, apply screw lock paint (RZZ0L01) to the adjusting screw.



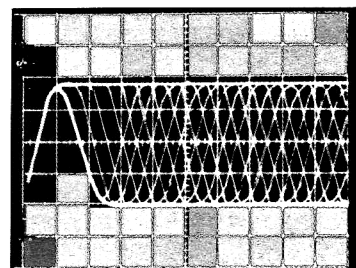
B5-2-4 Mechanical Adjustment

1. Solder a lead wire to CN6 for 8 pin (white) of the SERVO PCB (E-1947).
2. Connect CH1 of the oscilloscope to the lead wire (+) and Ground (-).

Oscilloscope setting: VOLT 200 mV
 SWEEP 0.5 μ sec.
 INPUT AC

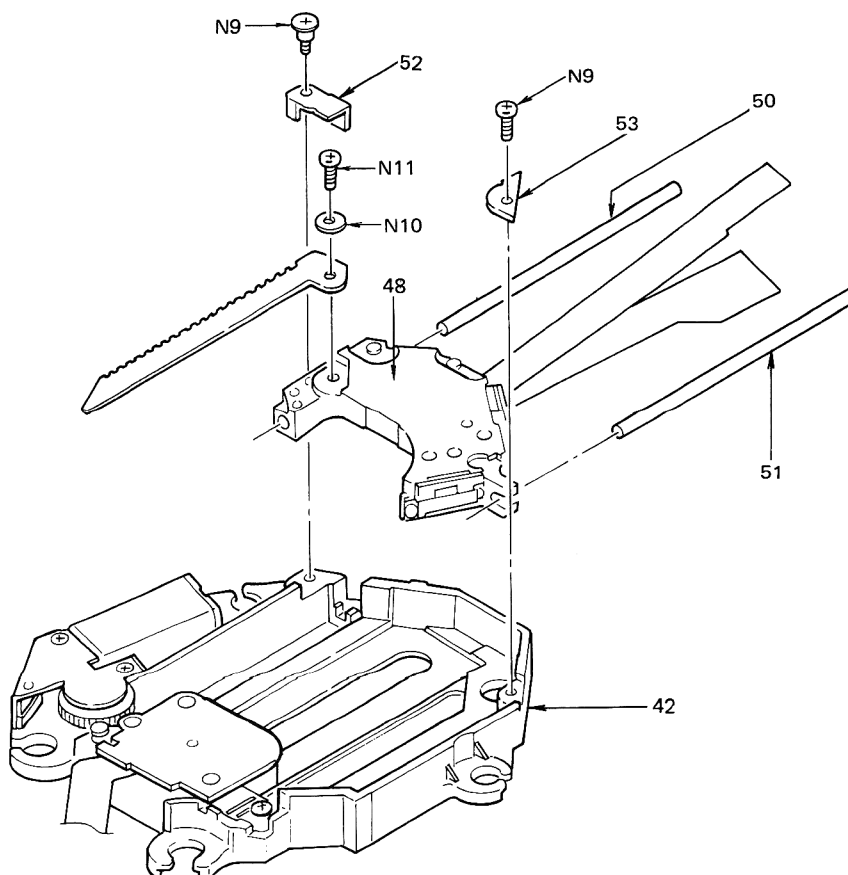


3. Place the **test disc (SZZP1056C uneven disc)** and turn on the power switch of the player.
4. Monitoring the RF signal on the oscilloscope, adjust the **two adjusting screws** alternately so that the vertical fluctuation of RF signal is minimized and the eye pattern most open.
5. Turn off the power switch of the player.
6. After the adjustment, apply **screw lock paint (RZZ0L01)** to the two adjusting screws.



DISASSEMBLY INSTRUCTIONS

1. Remove the screw N11.
2. Remove two screws N9.
3. Remove brackets 52 and 53.
4. Raise Optical pickup assembly 48 and remove it.
5. Pull out the guide shafts 50 and 51 from the optical pickup assembly.



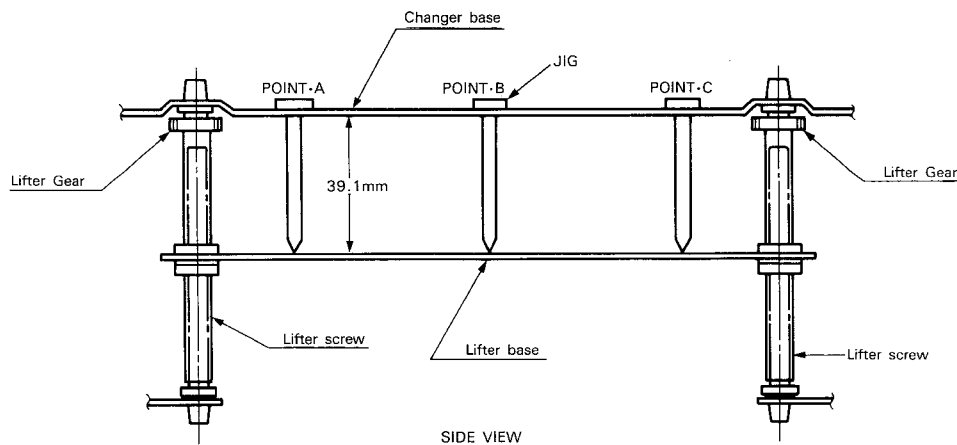
B5-3 MECHANICAL ALIGNMENT PROCEDURES:

Test Fixtures Required:

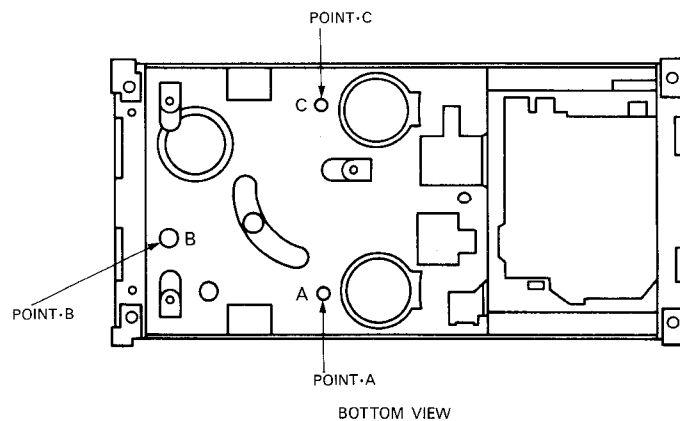
- YESFZ5001 — Lifter Base Height Adjustment Jig (B)
- YESFZ5002 — Lifter Base Level Adjustment Jig (A)

1. Lifter Base^{*} Level Adjustment (Perform this adjustment first)

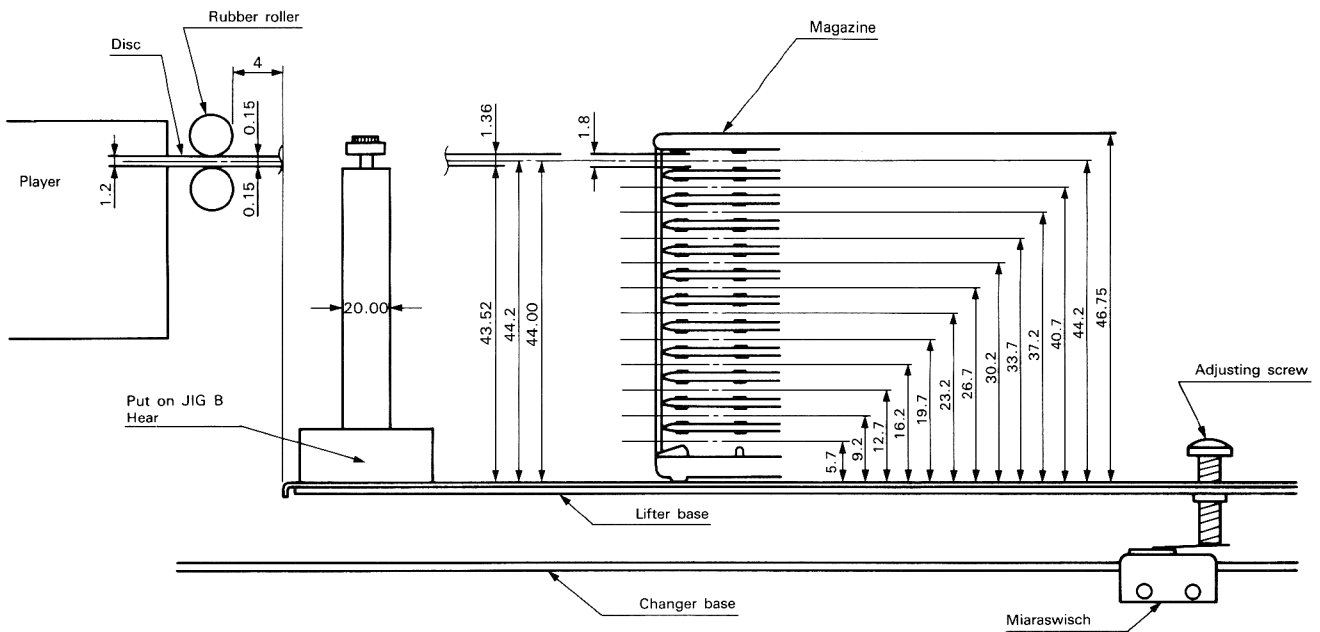
- Turn the power ON and tune to Disc 9 position by pressing the "Disc > " buttons on the remote control.
- Turn the power OFF and insert the Jig A into points A, B and C alternately. Both the tip and the head of Jig A should touch the bottom side of the lifter base and the bottom chassis respectively at all points.



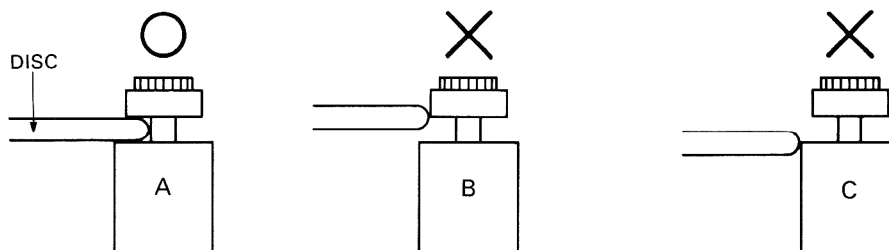
- If either tip or the head of the Jig A does not touch the respective locations, the adjustment will begin with removing all 3 E-rings, ref. No. 245, and 3 extension gears, ref. No. 281.
- Turn the lifter gears, ref. No. 282, either clockwise or counterclockwise until both tip and head are touched by respective locations at all points.
- Replace the E-rings and the extension gears after completion of the adjustment.



2. Lifter Base Height Adjustment



1. Turn the power ON and remove the magazine from the unit.
2. Push SW1 which is located in the magazine compartment with a finger. Do not release until the lifter base completely reaches to the bottom and stop.
3. Press the "Play" button and load a disc into the disc compartment. Disc will begin to play.
4. Press the "Eject" button and the disc will eject.
5. Turn the power OFF when the disc is half way out.
6. Place the Jig B toward the disc and you will note that one of the diagrams listed below will be matched.



JIG B

7. If diagram A is matched, no adjustment is needed.
8. If diagram B is matched, turn the adjusting screw quarter-turn clockwise.
9. If diagram C is matched, turn the adjusting screw quarter-turn counter-clockwise.
10. After the screw adjustment, turn the power ON and remove the disc from the unit. Let the lifter base to elevate to the top and stop.
11. Repeat steps 2–6 above to check if the proper alignment is made. Otherwise, repeat step 8 or 9 until adjustment matches diagram A.
12. After obtaining the correct height adjustment, repeat steps 2–6 one more time for confirmation.

C5-1 ALIGNMENT INSTRUCTION (TYPE C)

C5-1-1 Measuring Instruments and Tools

1. CD adjusting jig YESFZ155
2. Two-channel oscilloscope
3. Low-frequency oscillator
4. Screwdriver for adjusting semi-fixed variable resistors
5. Extension Cord Kit 3 pcs YESFZ155
6. Test Disc (SZZP1014F or SZZP1054C and SZZP1057C)

"Caution-use of controls or adjustments or performance of procedures other than those specified therein may result in hazardous radiation exposure."

C5-1-2 Alignment Procedures and Points

Remove the top case and bottom case, solder the leads to TP and JSW on the back of the circuit board, and follow the sequence below.

Note: Do not attempt alignments when the deck is removed.

- 1) Focus offset alignment VR105 (TP3)
- 2) Tracking offset alignment VR106 (TP1)
- 3) Focus balance alignment VR103 (CN4 for Pin 4 (YEL))
- 4) Tracking balance alignment VR107 (JSW1 both ends of terminals)
- 5) Focus gain alignment VR104 (JSW2)
- 6) Tracking gain alignment VR102 (JSW1)
- 7) Focus retra alignment VR105 (TP3, CN4 for pin 4 (YEL))
- 8) Tracking alignment VR106 (TP1, CN4 for Pin 4 (YEL))

After adjustment the above, disconnect the leads, return the JSW short ties as they were before, and check performance.

Additional Adjustment (PLL)

Remove the deck and connect the PLL with a connecting cable for PLL adjustment.

- ※) Additional PLL adjustment VR401 (Pin ②, IC501)

C5-1-3 Adjusting Methods

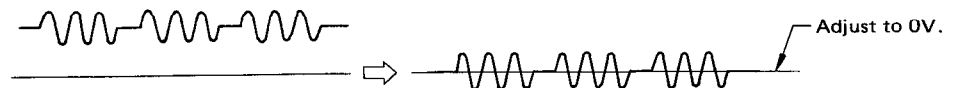
1) Focus offset alignment

- Test point: TP3
- Oscilloscope setting: Volt 100mV Sweep 1ms Input selector DC
- Operation: Open the door without inserting a disc.
- Alignment: Adjust VR105 until the DC voltage is 0V.
- Waveform sample:



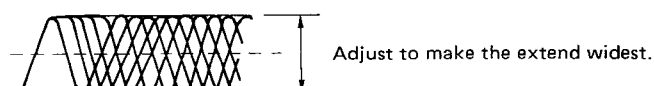
2) Tracking offset alignment

- Test point: TP1
- Oscilloscope setting: Volt 100mV Sweep 1ms Input selector DC
- Operation:
 - a) Insert a disc and play the outermost track.
 - b) During play, hold the disc with fingers to stop the rotation.
- Alignment: Adjust VR106 until the center of the waveform displayed on the scope is 0V.
- Waveform sample:



3) Focus balance alignment (Best eye)

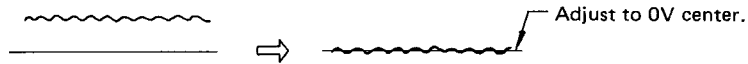
- Test point: Connect the Scope to CN4 for pin 4 (Yellow) and GND.
- Oscilloscope setting: Volt 50–100mV 20mV Sweep 0.5μs Input selector AC
- Operation: Insert disc and play track #1 for alignment, track #6 after the alignment.
- Alignment: RF signal appears. Adjust VR103 until the eye pattern extends widest.
- Waveform sample:



4) Tracking balance alignment

- Test point: JSW1 both ends of terminals, TP1
- CD adjusting jig connection

1) Clip	Chassis
2) Lead TP1 (brown)	TP1
3) TR BAL terminal	Oscilloscope probe
4) Lead JSW IN (green), lead JSW OUT (yellow)	Both ends of JSW1 (Remove short tips)
(Top pin) out, (bottom pin) in when deck at upside down position	(Remove short tips)
- Oscilloscope setting: Volt 100mV Sweep 0.1ms Input selector DC
- Operation: Insert a disc and play track #1. Set the lever switch on the CD adjusting jig to on.
- Alignment: Shift the lever switch to OFF, and adjust VR107 so that the center of the waveform will be 0V with in 15 seconds.
- Waveform sample:



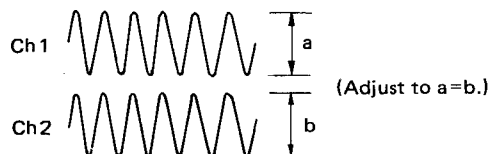
5) Focus gain alignment

- Test point: Both ends of JSW2 (Remove short plug) (Top pin) IN, (bottom pin) OUT when deck at upside down position.
- CD adjusting jig connection

1) Clip	Chassis
2) Lead +10V (orange)	+10V (E of Q109)
3) Lead -10V (red)	-10V (E of Q108)
4) Lead JSW IN (green)	JSW2 IN
5) Lead JSW OUT (yellow)	JSW2 OUT
6) OSC terminal:	Low frequency OSC
7) Oscilloscope Ch1, Ch2 terminals	Oscilloscope probe
- Low frequency oscillator setting

1) Frequency	600Hz ~ 1 kHz
2) Level	150mV p.p.
- Oscilloscope setting (Ch1, Ch2):

Volt 200mV	Sweep 1ms	Input selector AC
--------------------	-------------------	---------------------------
- Operation: Set the lever switch on the CD adjusting to TEST Insert a disc and play.
- Adjust VR104 until the level of both channels at 1kHz are the same.



6) Tracking gain alignment (Use of Adjusting jig)

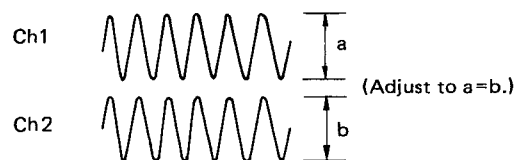
- Test point: Both ends of JSW1 (Remove short tips) (Top pin) OUT (bottom pin) IN
- CD adjusting jig connection:

1) Clip	Chassis
2) Lead +10V (orange)	+10V (E of Q109)
3) Lead -10V (red)	-10V (E of Q108)
4) Lead JSW IN (green)	JSW1 IN
5) Lead JSW OUT (yellow)	JSW1 OUT
6) OSC terminal	Low frequency OSC
7) Monifor1, Monifor2 teminals	Oscilloscope probe Ch1, Ch2
- Low frequency oscillator setting

1) Frequency	1 kHz ~ 1.5 kHz
2) Level	1 50mV p-p
- Oscilloscope setting (Ch1, 2)

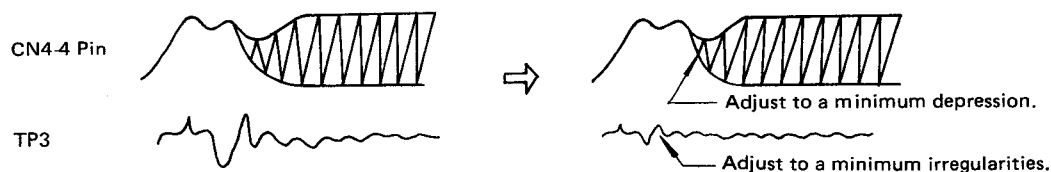
Volt 200mV	Sweep 1ms	Input selector AC
--------------------	-------------------	---------------------------
- Operation: Set the lever switch on the CD adjusting jig to TEST. Insert a disc and play.
- Alignment: Adjust VR102 until the levels of both channels are the same.

- Waveform sample:



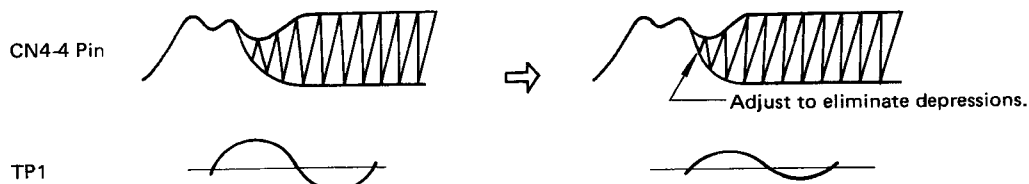
7) Focus retra alignment

- Test point: Connect Pin 4 of CN4 to Ch1; TP3 to Ch2.
- Oscilloscope setting (Ch1, 2): Volt 50mV (CH1), 500mV (CH2) Sweep 0.5ms (Nor.) (CH1)
Input selector AC (CH1) DC (CH2)
- Operation: Insert a test disc and play track #B. Reproduce a black band of 0.5mm.
- Alignment:
- Waveform sample: a) Watch the eye pattern for Ch1, and make fine adjustment of VR105 to a minimum depression.
b) Also make a fine adjustment of VR105 so that FE signal for CH2 will be as flat as possible.



8) Tracking offset alignment

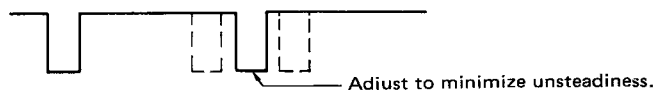
- Test point: Connect the Scope to CN4 for pin 4 (CH1) TPI (CH2)
- Oscilloscope setting (Ch1, 2) Volt 500mV Sweep 0.5ms Input selector AC
- Operation: Insert a test disc and play. Reproduce a black band of 0.5mm.
- Alignment: a) Watch the eye pattern for Ch1, and make fine adjustment of VR106 to a eliminate depression



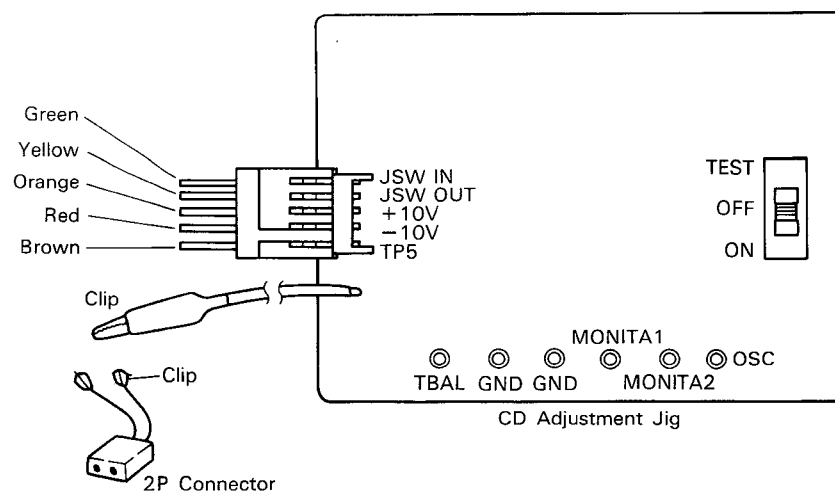
※) Additional alignment PLL adjustment

note: This alignment cannot be made unless the deck is taken out.

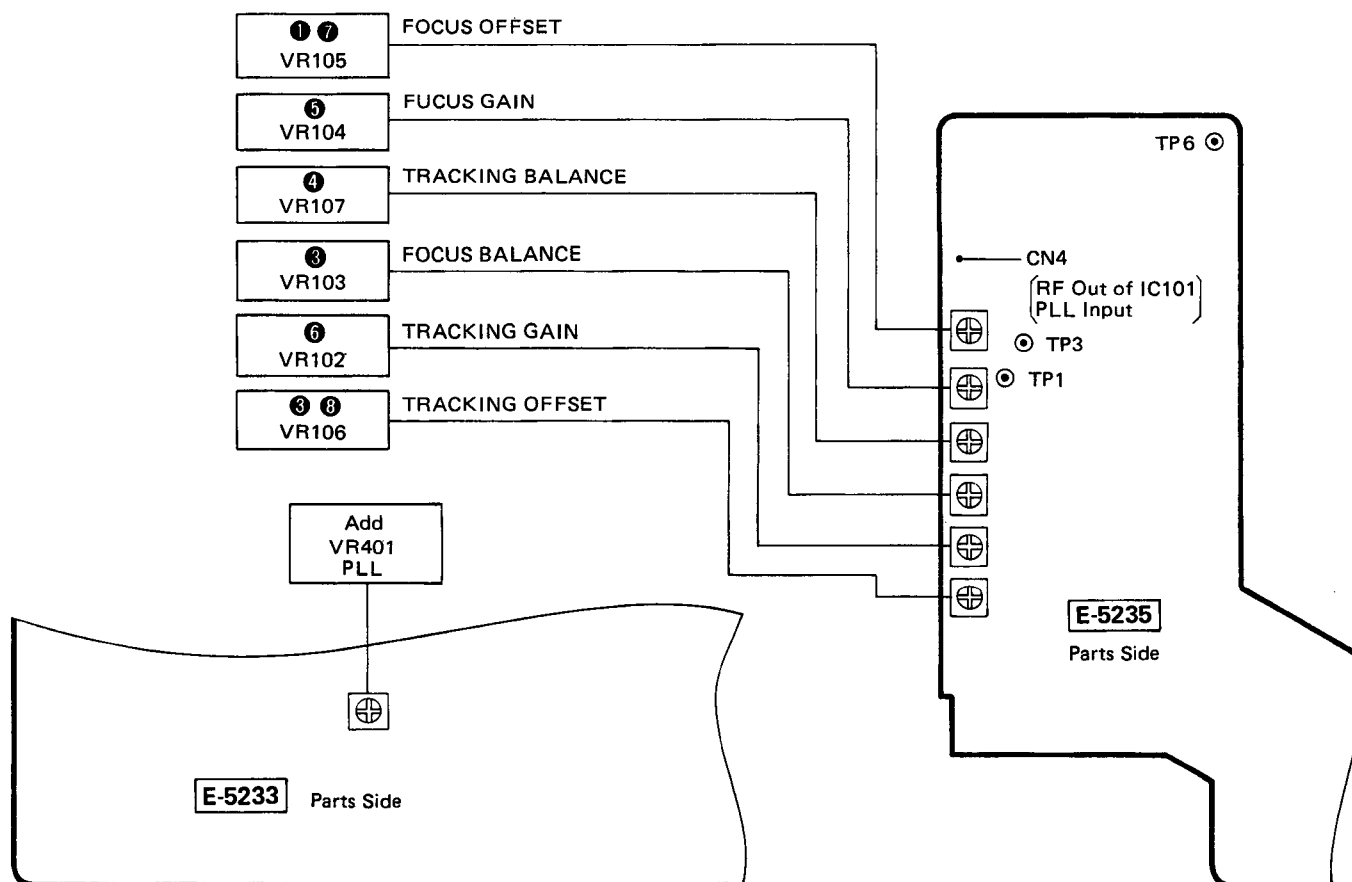
- Test point Pin ② (CLDCK) of LSI (MN6617)
- Oscilloscope setting: Volt 2.0V Sweep 50μs Input selector DC Trig (NOR)
- Operation: Insert a test disc and play. Play track #8. Reproduce a black band of 0.5 mm.
- Alignment: Adjust VR401 until the waveform is steady or as steady as possible.
- Waveform sample: Adjust to minimize unsteadiness.



C5-1-4 CD Adjustment Jig



C5-1-5 Alignment Points



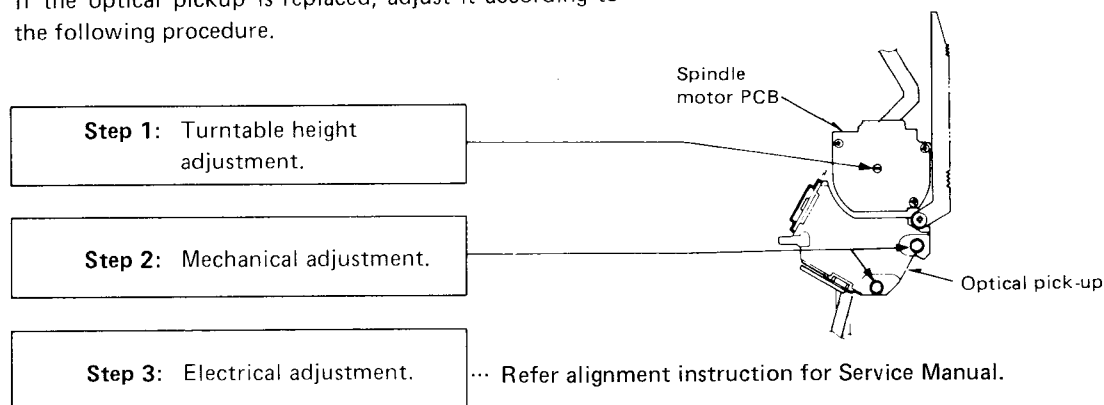
C5-2 OPTICAL PICKUP ADJUSTMENT

C5-2-1 Measuring Instruments and Special Tools

- Two-channel oscilloscope (with trigger) of 30 MHz or over
- Test disc (SZZP1014F) new type CDT-D02
Inspection test disc (SZZP1054C) CDT-D16
Uneven test disc (SZZP1056C) CDT-D18
Inspection test disc (SZZP1054C)
- Hexagonal wrench (2 mm)
- Screw lock paint (RZZ0L01)
- Lead wires
- Extension cord (Jig) [YESFZ154] (Supplied from APD)
- Resistor (4.7 k Ω 1/4W)

C5-2-2 Adjustment Procedure and Adjustment Points

- If the optical pickup is replaced, adjust it according to the following procedure.

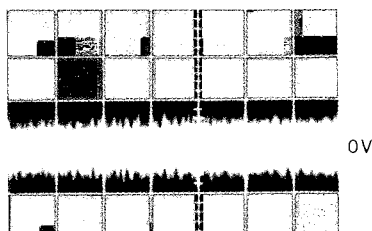
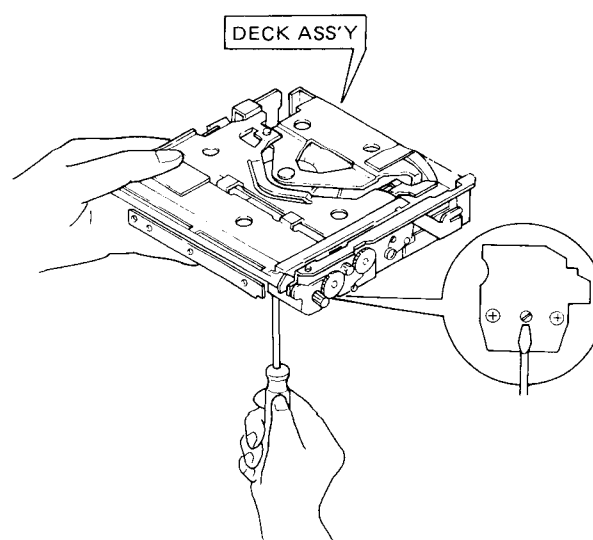
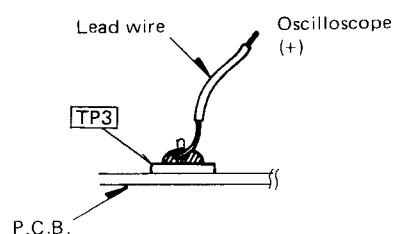


C5-2-3 Turntable Height Adjustment

1. Turn off the power of the player.
2. Solder the lead wires to TP3 (IC101, 37 Pin) of the P.C.B (E-5235) and Ground so that the probe of oscilloscope can be connected.
3. Connect the extension cords between the Deck Ass'y and unit.
4. Connect CH 1 of the oscilloscope to the TP3 (+) and Ground (—).

Oscilloscope setting: VOLT 200 mV
SWEEP 5 msec.
INPUT DC

5. Set the oscilloscope to DC zero balance.
6. Place the **test disc (SZZP1014F or SZZP1054C)** and turn on the power switch of the player.
7. Hold the CD Deck by the hand with the top face up.
8. The player to the play mode.
9. Turn the **adjusting screw** at the bottom of the spindle motor drive P.C.B. with a flat screwdriver so that the waveform is 0 ± 50 mV.
10. Turn off the power switch of the player.
11. After the adjustment, apply **screw lock paint (RZZ0L01)** to the adjusting screw.

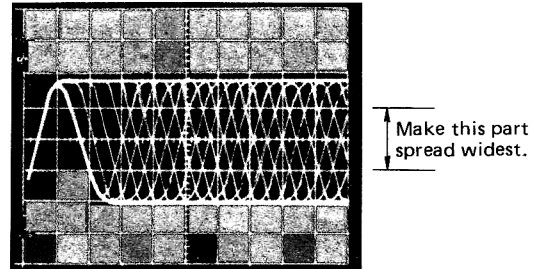
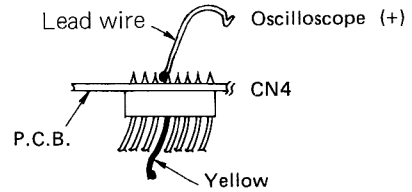


C5-2-4 Mechanical Adjustment

1. Solder a lead wire to Pin 4 of CN4 (Yellow) of the P.C.B.
2. Connect CH1 of the oscilloscope to the lead wire (+) and Ground (-).

Oscilloscope setting: VOLT 200 mV
 SWEEP 0.5 μ sec.
 INPUT AC

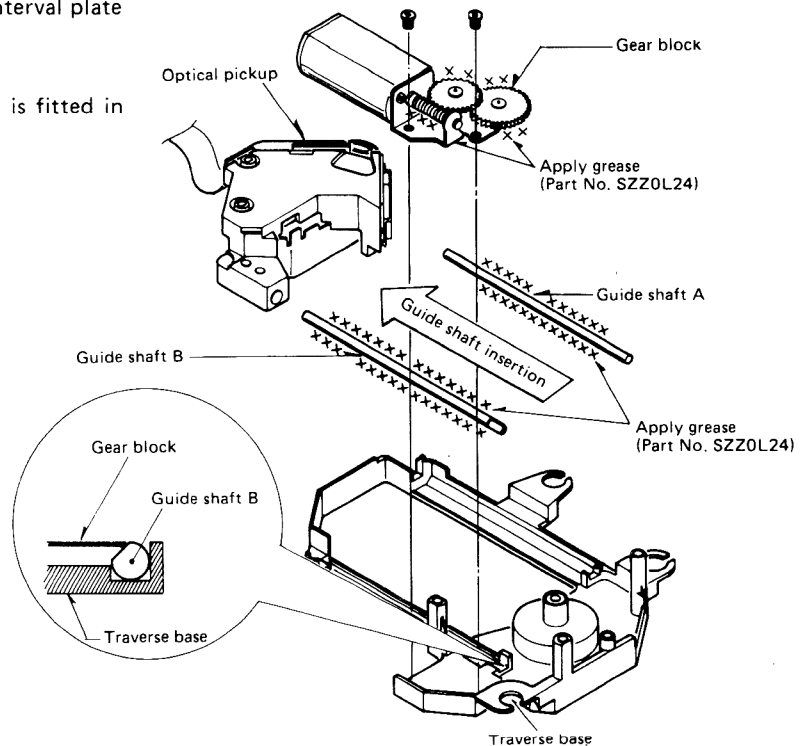
3. Place the uneven test disc and turn on the power switch of the player.
4. Monitoring the RF signal on the oscilloscope, adjust the **two adjusting screws** alternately so that the vertical fluctuation of RF signal is minimized and the eye pattern most open.
5. Turn off the power switch of the player.
6. After the adjustment, apply **screw lock paint (RZZ0L01)** to the two adjusting screws.



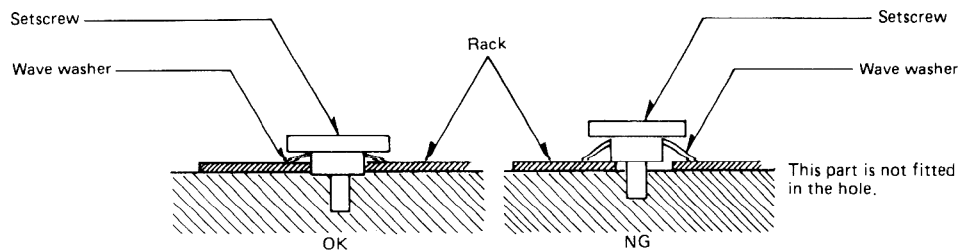
Cautions for assembly

The assembly work can be done by reversing the disassembly procedure. Keep the following points in mind.

1. Be sure to insert the guide shaft in the direction of the arrow.
 If it is inserted in the wrong direction, the interval plate spring may be damaged.
2. Make sure that the D cut off guide shaft B is fitted in the position as shown.



3. Tighten the rack setscrew as shown.



6. TROUBLESHOOTING

A6 TROUBLESHOOTING (TYPE A)

A6-1 SYMPTOM CHECK

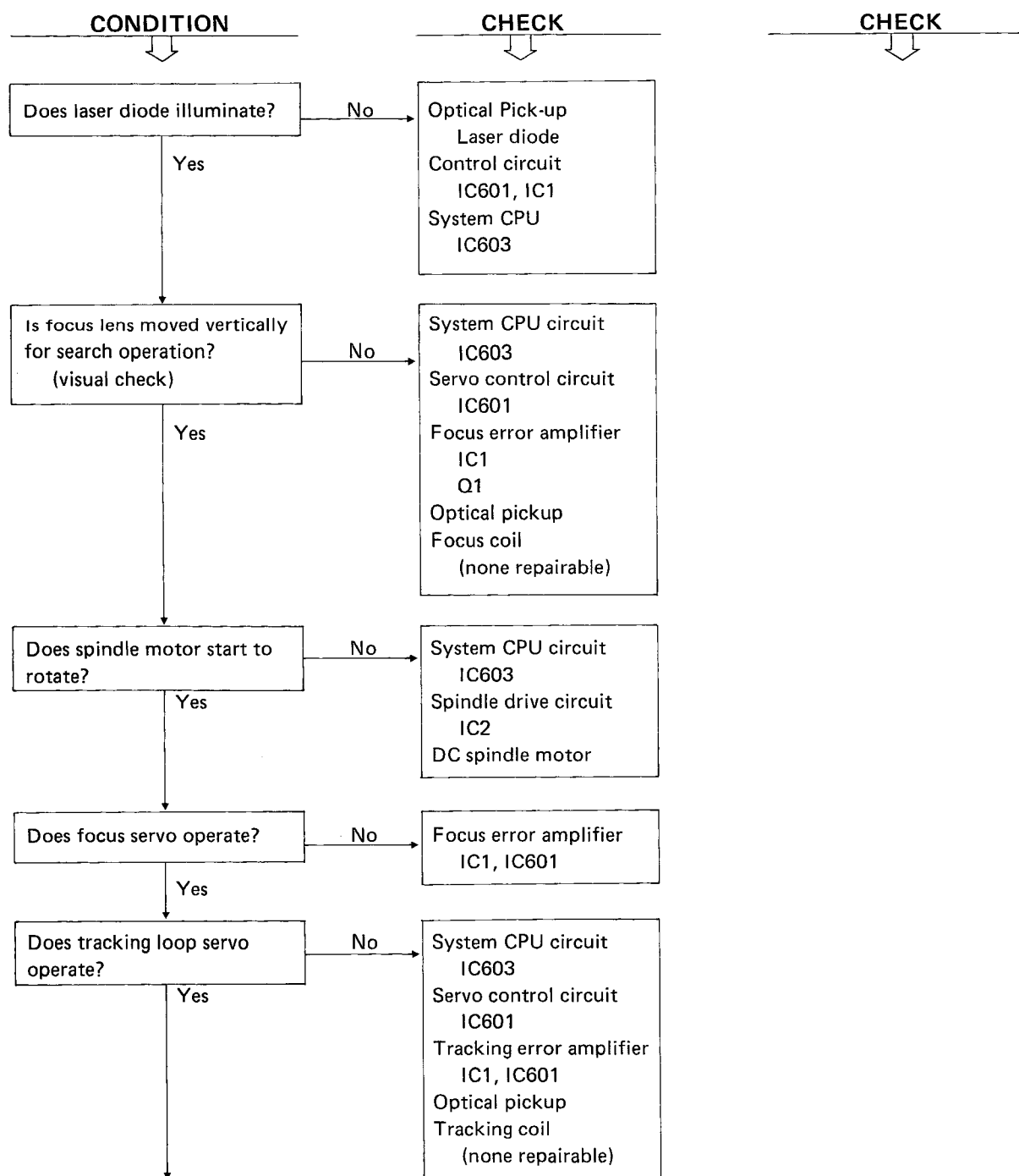
- (1) When traverse motor malfunctions. Check IN SW (H).
Also, when the IN SW is not in changing stage, check the system CPU Pins 18 and 19 (TRV R, TRV F), and Pins 14–17 (CNT1, 2, 3, 4, all H) outputs to IC601, then from IC601 (TRV F/R) to IC602, from IC602 to the traverse motor.
- (2) When the focus lens moves up and down repeatedly, check if the disk is properly set. If the LD ON terminal of Pin 28 IC601 is (H), check the LD terminal of Pin 7 IC1. To see if APC circuit is turned on to intensify the laser.
Check connections between the photo-detector and IC1, check if there is a pulse at the FE terminal of Pin 15 IC1, and a pulse at SENSE terminal of Pin 24 IC601.
- (3) If the disk ejects, when CNT1 is (H), CNT2, 3, and 4 are (L). Check to see if the CLW terminal of Pin 10 IC601, the FPO terminal of Pin 24 IC1, and FD terminal of Pin 10 IC602 generate the triangular wave respectively.
- (4) If the disk ejects at the time when the disk detection ended (CNT1, 2, 3, and 4 (H)), check connection between IC2 and spindle motor, IC2 and IC501 Pin 32 L/R (Clock pulse), Pin 65 EC (Start-up control pulse) and Pin 64 PC (L)).
Also trace and check Pin 23 MLD, Pin 24 MCLK and Pin 25 MDATA (data pulse) of the micro-computer.
- (5) If the disk ejects at the time when CNT1 is (L), CNT2, 3, and 4 are (H), check to see if IC601 Pin 26 FLOCK IS (H → L). Check connection between IC601 and IC1 Pin 32 RFDET (L), Pin 31 BDD and between IC1 Pin 1 AMP1 and optical pickup.

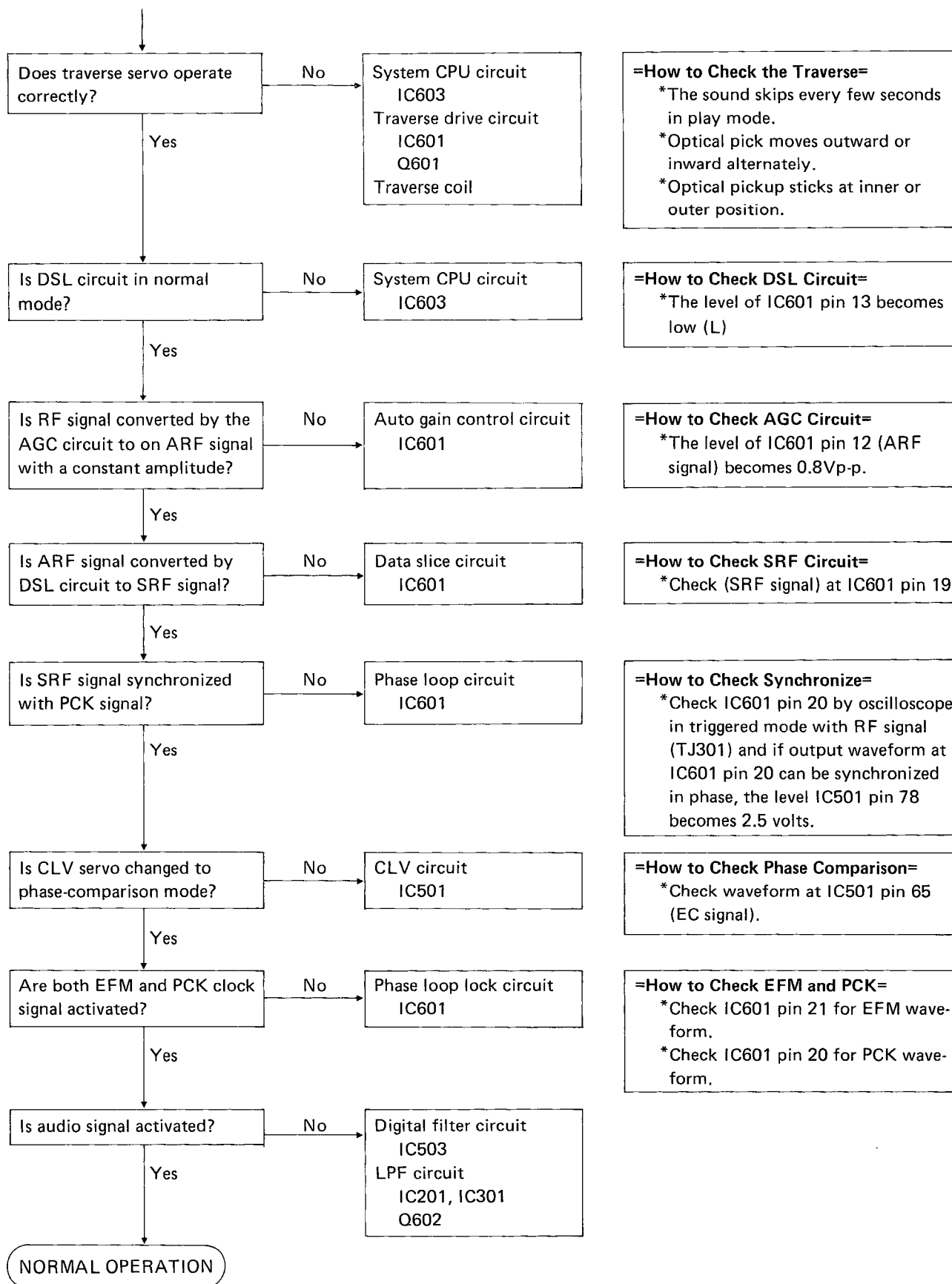
If the disk ejects when CNT1, 2 are (L), CNT3, and 4 are (H),

- 1) check the tracking servo circuit, and check connection between tracking coil and Pin 41 IC601, and Pin 27 VREF (2.5V) of IC601, connection between IC1 and IC601 [IVA (Pin 13) → LSA (Pin 1)] [IVB (14) → LSB (2)] [TPI (17) ← TEO (6)] [TGC (25) (0V) ← VDET (23)] [TPO (22) → BRIN (42)].
 - 2) check the operation from the RF output to the CLV Control.
Also check if there is RF signal at AFR of Pin 35 IC1, at SRF of Pin 19 IC601 and also check TRON (L) Pin 10, STAT (L) Pin 11 and CLVS Pin 77 (L → H) of IC501.
- (6) After CNT1, 2, 3, and 4 became (L), it will not read TOC and the disk ejects, check if there is SUBQ pulse output from Pin 3 IC501, EFM pulse from Pin 21 IC601. Check KICK at Pin 27 IC601 and TVPO output at Pin 39 IC601. If SUBQ is output the AMS (auto-music scan) is operated by the Kick Pulses then read TOC in the micro-computer.

* Is the disk OK? Is the lens dirty?

A6-2 TOC READ OPERATION – PLAY OPERATION FLOW CHART





A6-3 CLEANING PROCEDURE FOR THE FOCUS LENS

Symptoms

The following symptoms may be caused not only by faulty software or hardware but also by the soiled focus lens.

- Not reading TOC
- Not playback
- Skipping sound

Cause

Soiling of the focus lens surfaces with dirt, dust, finger print, and others.

(Note) Clean the lens in the following steps even if it looks to be clean because soiling is not always discernible visually.

Cleaning Steps

(1) Remove the optical pick-up ASSY from unit.

1 Deck ASSY Removal

- Remove the four screws.
- Pull out CN1, CN2 and CN3.
- Pull the deck ass'y upward.

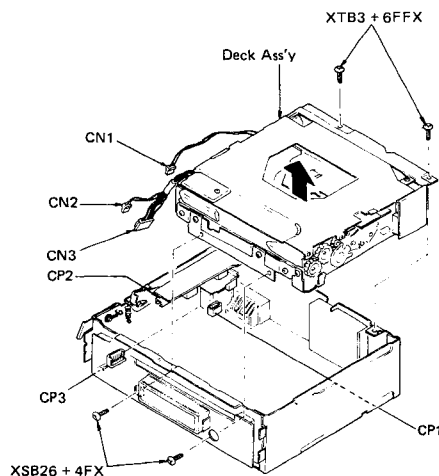


Fig. 1-1

2 Optical Pick-up ASSY Removal

- Pull out the two flexible PCBs from CN1 and CN2.
- Pull out CP1 and CP2.
- Remove the five screws.

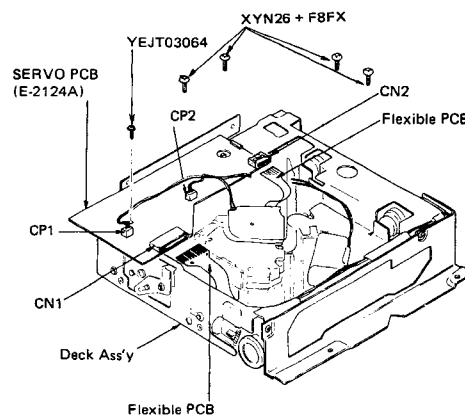
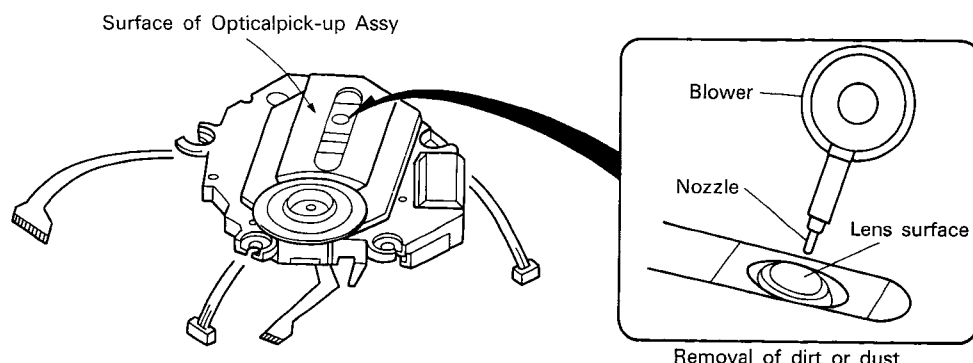


Fig. 1-2

(2) Clean the surface of Optical Pick-up ASSY

(3) Removing method of dirt, dust, or etc., from the focus lens.

- Blow dust, dirt, or etc, off the lens surface by the blower contained in the cleaning kit as shown below.



- (4) Removing method of finger print or the like
- Wet a swab with a little volume of the lens cleaning liquid contained in the cleaning kit. Wipe the surface of the lens with the wet swab by turning the swab.
 - Don't apply a solid portion of the swab to the lens surface again.
- (Note) Cleaning may be made by applying pressure to some degree to the lens.

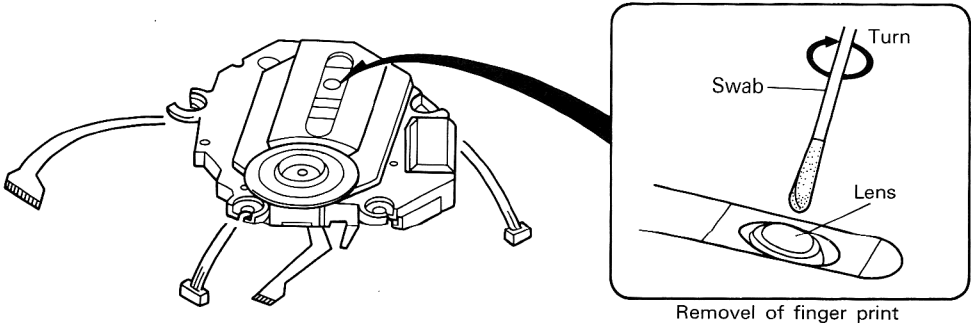


Fig. 1-4

Warning

- Don't apply the cleaning liquid to the lens directly, also don't wet the swab with the cleaning liquid more than necessary, otherwise the liquid may enter the optical pick-up.
- Wipe the lens lightly.
- If the liquid is applied to the lens in excess, wipe off the liquid with a new dry swab.

Kit Item Number: SZZP1038C

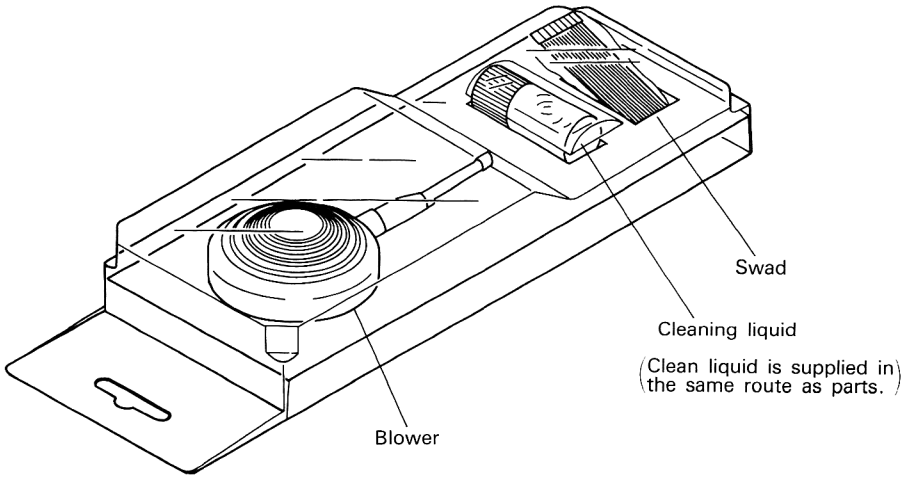


Fig. 1-2

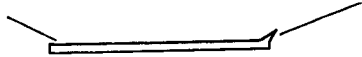
Description	Part number	Kit number
Blower	-----	SZZP1038C
Cleaning liquid	SZZP1050C	
Swab (50 pcs.)		

B6 TROUBLESHOOTING (TYPE B)

B6-1 CAUTION

1. When making any connections, be sure to turn off all power source.
2. This changer is exclusive for 12 cm CD. Therefore, it cannot play 8 cm CD even with some form of adaptor.
3. For enabling the disk moves smoothly in the magazine, check that the CD is free of burr.

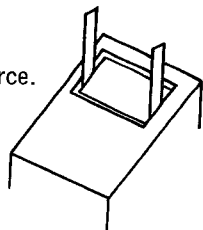
Display surface Burr when disk is taken out from the die.



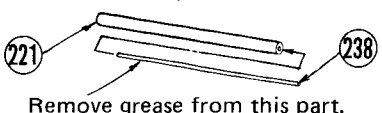
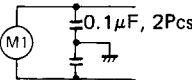
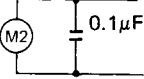
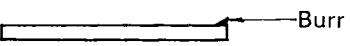
Touch the corner and if the disk has burr,
remove it with emery paper, etc.

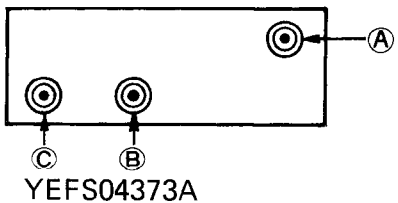
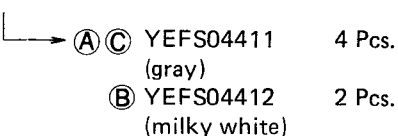
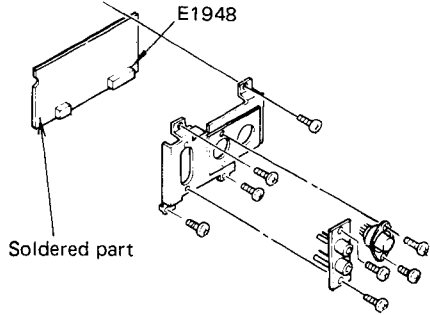
4. When operating the changer main body, be sure to close the cover. If the disk detecting sensor is exposed to the direct sun-light the sensor will malfunction.
5. Sound skipping due to vibration
Since the CD autochanger, is mounted in the trunk, countermeasure for vibration is considered. However, when the car is driving on a very rough road conditions will sometimes cause the player to skip the music. However, the disc and the CD Player will not be damaged by these conditions.
(Example: Pothole, railroad crossing, road paved with stone, etc.)

B6-2 TROUBLESHOOTING

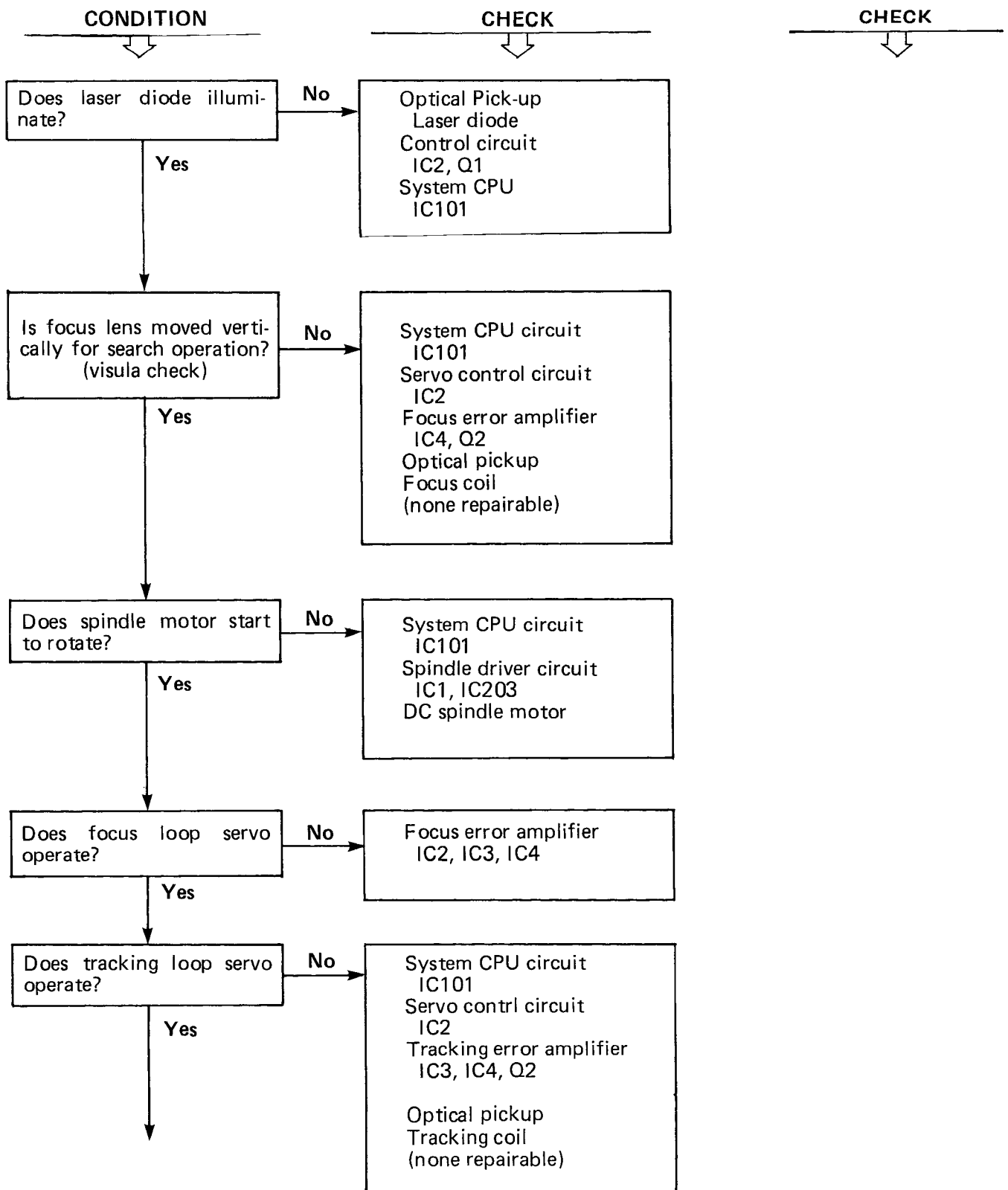
	SYMPTOM	DISPLAY	CAUSE	REMEDY
Disk loading/unloading	Disk magazine won't load.		1. ACC power source is cut off. 2. Back-up power source is cut off. 3. 8P DIN cord of main body is not properly connected.	Check fuse. Check fuse. Check connection.
	System inoperative	E3		Reset power source. 1. Cut ACC power source. 2. Cut back-up power source. (Pull out fuse for back-up power source.) 3. After 10 seconds later, reconnect power sources for ACC and back-up 4. Turn SYSTEM ON and push eject button.  With thin metal plates unlock the claws on both sides and pull out.
CD playing	Selected disk cannot play. (Disk is ejected.)	E1	Disk is dirty or loaded upside down.	Clean disk. Check loaded disk.
		E2	Disk is scarred	Replace the disk with a proper one.
	Some special disk cannot play.	(Example) 6-01 ↓ 8-01	Disk has burr around its outer circumference and which burr prevents smooth playing.	File out the burr of the 7th disk. (Refer to Item 3 of Caution.)
			Disk's pit height is lower.	Compare and check with other disks.
	Selected disk cannot play sometimes.		Disk's positioning is not line-up properly.	Please have the service shop check and adjust the height of the changer.
	Disks No.1 to No.4 cannot play.	(Example) 12-01 ↓ 5-01	Disk detecting sensor is exposed to direct sun-light. Check if the cover of main body is left open.	Close the cover of the changer and then operate. (Refer to Item 4 of Caution.)
	Play by remote control won't function.		Bad batteries in the remote control. used up.	Replace With new batteries.
	No sound only on CD play.		Improper connection of RCA junction cord cord.	1. Between changer main body and controller 2. Between controller and amplifier
	Noise when Disc is changing		1. Change of electric potential ground. 2. Noise is picked up by the cord.	1. Change or add ground points of the set on the amplifier side. 2. Change cords or cords locations.
	Skipping is very frequent.		Loose fitting of the changer main body	Retighten bracket for fitting and screws and nuts.
			Optical lens is dirty.	Have the service shop clean lens.
			Disk itself is dirty or scarred.	Clean disks.

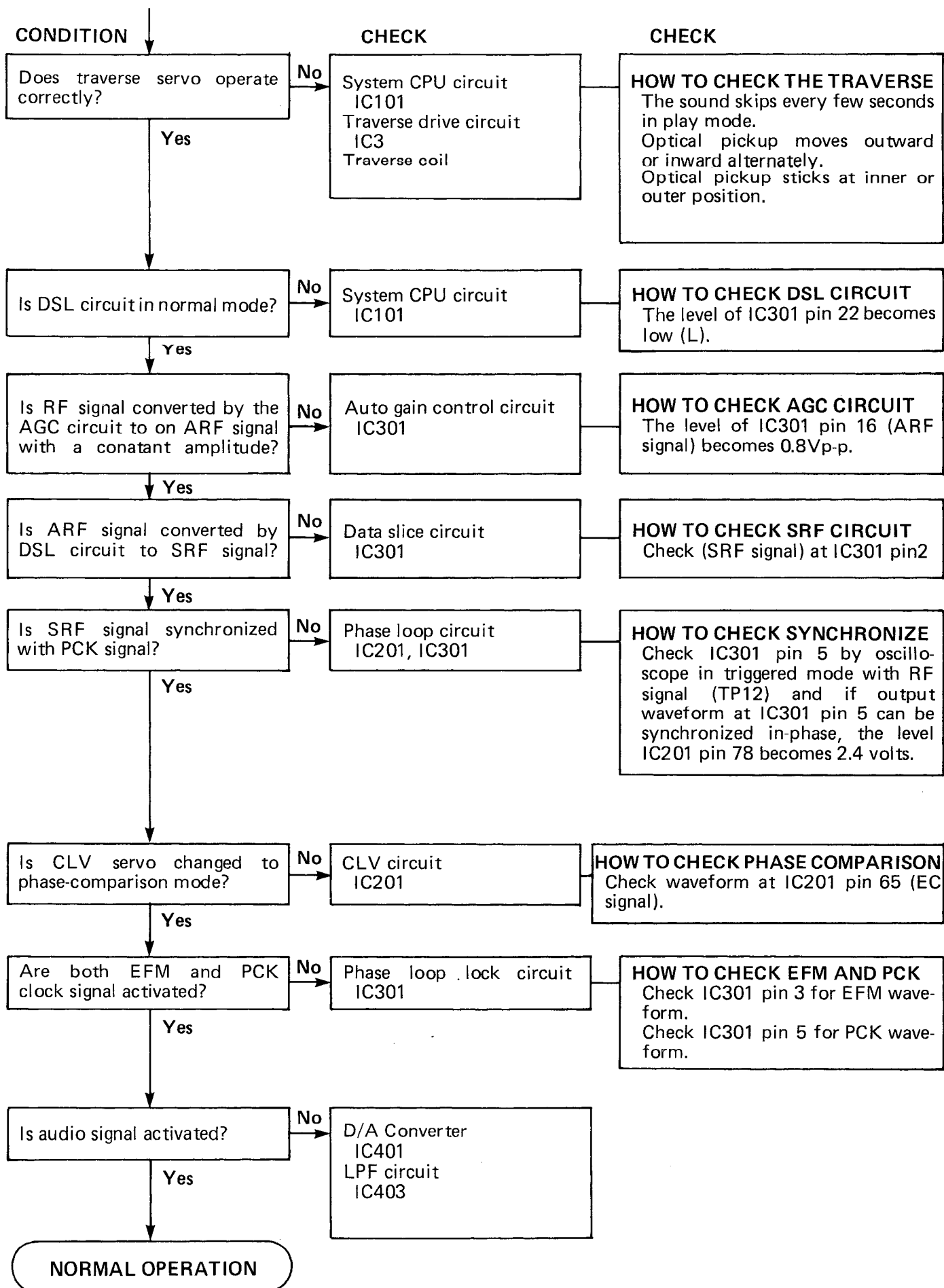
B6-3 EXAMPLES OF CD CHANGER MECHANISM REPAIRS

No.	SYMPTOM	CAUSE	REMEDY
①	Disc won't return to magazine.	Greasy rollers.	Clean rubber rollers by using swab with rubbing alcohol or replace it with roller assembly (YESFX003009). 
②	Magazine won't load.	When the magazine is fully loaded, the loading motor (M1) current may increase and generate noise pulse to CPU IC101.	Add 0.1μF capacitors as showed in diagrams below. M1 : Elevator Motor (241)  M2 : Disc feed motor. (240) 
		SW1 defective. Motor (M1) defective.	Replace defective parts.
③	Disc jams between the rollers.	Lifter base height off specification.	Perform the lifter base and level adjustments.
④	Magazine won't manually eject.	SW101 defective. Motor (M1) defective. IC101 defective.	Check and/or replace defective parts.
⑤	No disc search.	SW2 defective. IC101 defective. IC501 defective. Motor (M1) defectived.	Check and/or replace defective parts.
		No search pulse.	Check and/or replace QON1119 and IC503.
⑥	Disc is stuck on magazine so that disc loading is impossible.	Burrs on the disc.	Remove burrs. 
		Spring preassure inside the magazine is too strong.	Replace with an improved magazine (YEP9FZ1782D).
		Solenoid drive defective.	Check and/or replace Q501 and solenoid.

No.	SYMPTOM	CAUSE	REMEDY
⑦	Disc won't load to deck.	Rollers defective.	Replace roller assembly YESFX003009.
		Motor (M2) defective.	Check and/or replace motor (M2) and IC502.
⑧	Sound skips due to vibration.	Anti-shock system inoperative.	Replace rubber insulators and rubber dampers.  YEFS04373A 
⑨	No read TOC. No playback. Sound skips in normal play condition.	Servo control loops inoperative.	Check CD operation flow chart. Perform optical pickup mechanical alignment perform servo control Perform servo control electrical alignment.
		Soiling of the focus lens surfaces with dirt, dust and others.	Clean focus lens surfaces with lens cleaning kit SZZP1938C.
⑩	One channel out.	Solder joints of pins CN801, CN802 on PCB (E-1948) were cracked due to vibration during installation. 	Resolder the cracked joints.

B6-4 TOC READ OPERATION – PLAY OPERATION FLOW CHART





C6 TROUBLESHOOTING (TYPE C)

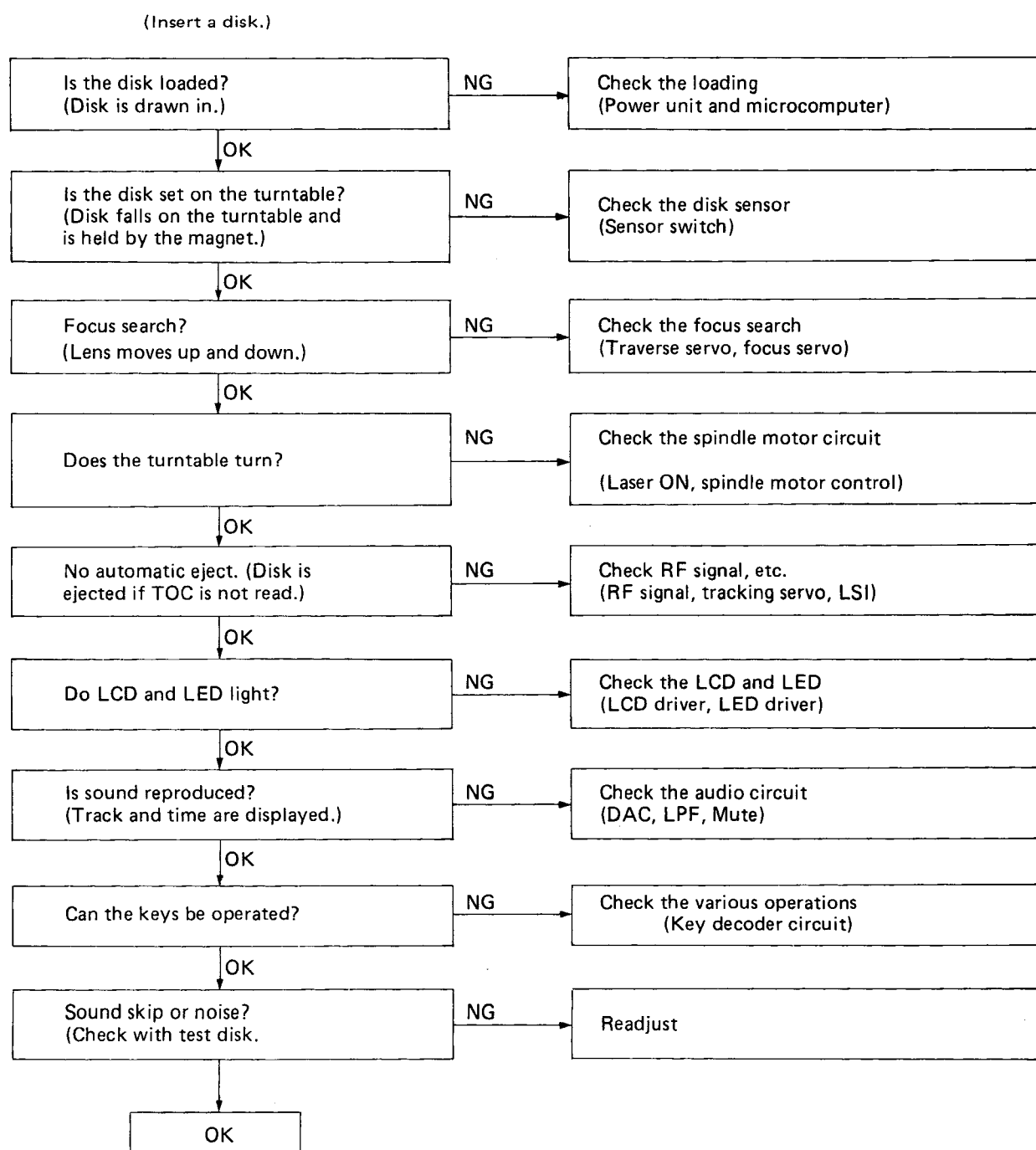
If trouble has occurred, check the symptoms according to the flowchart shown in Section C6-1.

Note: Do not skip steps in the symptom check.

Keep power on and, if the set is disassembled for troubleshooting, make sure that the symptoms are the same as before disassembling the set.

Note: When checking the set with the top cover open, cover the top of the deck to prevent the photo-sensors from wrong operation.

C6-1 SYMPTOM CHECK



C6-2 CD DECK REPAIRS

NO.	SYMPTOM	CAUSE	REMEDY
1	Does not traverse. Does not eject.	Traverse wire disconnected. Improper soldering of traverse wire Foreign substance mixed in optical pickup assy.	Repair by soldering Remove foreign substance.
2	Traverse reverses.	Short due to solder between traverse wires	Correct soldering between traverse wires.
3	Traverse noise	Foreign substance mixed in optical pickup assy. Optical pickup is faulty.	Remove foreign substance. Replace optical pickup assy with a good one.
4	Door does not open. pen.	Door pin is caught. Door shaft is bended. Solenoid is faulty. Improper soldering of solenoid Shutter lever is caught.	Repair mechanically. Repair mechanically. Check and/or replace defective parts. Correct soldering. Repair mechanically.
5	Door does not close.	Door pin is caught. Door shaft is bended. Shutter lever is bended.	Repair mechanically. Check and/or replace defective parts.
6	Disk does not load.	Door switch is bended. Switch 2 is bended. Motor is defective.	Check and/or replace defective parts.
7	Disk does not drop on turntable.	Switch 3 is faulty. Rack arm is caught. Clamp arm is caught.	Check and/or replace defective parts. Repair mechanically. Repair mechanically.
8	Disk surface shakes.	Foreign substance stick to the face of turn table. Optical pickup assy is faulty. Improper adjustment of optical pickup.	Remove foreign substance from turn table. Check and/or replace defective parts. Readjust optical pickup.
9	Turntable does not rotate.	Optical pickup assy flexible PCB is broken. Improper connection of optical pickup assy flexible CN2 to the connector. Foreign substance mixed in optical pickup assy. Optical pickup assy is faulty.	Check and/or replace defective parts. Connect flexible CN2 properly to the connector. Remove foreign substance. Check and/or replace defective parts.
10	Noise when disk loading, dropping and ejecting	Motor gear is broken. Foreign substance is mixed in motor gear. Junction gear is broken. Foreign substance is mixed in junction gear.	Check and/or replace defective parts. Remove foreign substance. Check and/or replace defective parts. Remove foreign substance.
11	Noise when disk is rotating.	Optical pickup assy is faulty. Improper fitting of clamp arm Foreign substance is mixed in.	Check and/or replace defective parts. Correct fitting position of clamp arm. Remove foreign substance.
12	No read TOC No play back Sound skips in normal play condition	Servo control loops inoperative	Check CD operation flow chart. Perform optical pickup mechanical alignment. Perform servo control. Perform servo control electrical alignment..
		Soiling of the focus lens surface with dirt, dust and others	Clean focus lens surfaces with lens cleaning kit SZZP1938C.

C6-3 TOC READ OPERATION – PLAY OPERATION FLOW CHART

